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Systems, methods, and compositions for the generation of water-soluble cannabinoids utilizing fatty-acid binding proteins and fatty-acid binding-like proteins as cannabinoid-carriers

Abstract

The inventive technology includes novel systems, methods, and compositions for the generation of water-soluble short-chain fatty acid phenolic compounds, preferably cannabinoids, terpenes, and other volatile compounds produced in *Cannabis* and other cannabinoid producing plants. In particular, the inventive technology includes novel systems and methods to solubilize short-chain fatty acid phenolic compounds, such as cannabinoids, via binding to a water soluble and readily digested carrier protein.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a national stage application under 35 U.S.C. 371 of PCT Application No. PCT/US20/16642 having an international filing date of Feb. 4, 2020, which designated the United States, which PCT application claimed the benefit of U.S. Application Ser. No. 62/800,692, filed Feb. 4, 2019, both of which are incorporated by reference in their entirety.

SEQUENCE LISTING

(1) The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Aug. 4, 2021, is named 90425-00162-Sequence-Listing-AF.txt, and is 213 Kbytes in size.

(2) This International PCT Application claims the benefit of and priority to U.S. Provisional Application No. 62/800,692 filed Feb. 4, 2019. The entire specification and figures of the above-referenced application is hereby incorporated, in its entirety by reference.

TECHNICAL FIELD

(3) The inventive technology includes novel systems, methods, and compositions for the generation of water-soluble short-chain fatty acid phenolic compounds, preferably cannabinoids, terpenes, and other volatile compounds produced in *Cannabis* and other cannabinoid producing plants. In particular, the inventive technology includes novel systems and methods to solubilize short-chain fatty acid phenolic compounds, such as cannabinoids, via binding to a water soluble and readily digested carrier protein.

BACKGROUND OF THE INVENTION

(4) Cannabinoids are a class of specialized compounds synthesized by *Cannabis*. They are formed by condensation of terpene and phenol precursors. They include these more abundant forms: Δ .sup.9-tetrahydrocannabinol (THC), cannabidiol (CBD), cannabichromene (CBC), and cannabigerol (CBG). Another cannabinoid, cannabinol (CBN), is formed from THC as a degradation product and can be detected in some plant strains. Typically, THC, CBD, CBC, and CBG occur together in different ratios in the various plant strains. These cannabinoids are generally lipophilic, nitrogen-free, mostly phenolic compounds and are derived biogenetically from a monoterpene and phenol, the acid cannabinoids from a monoterpene and phenol carboxylic acid, and have a C21 base. Cannabinoids also find their corresponding carboxylic acids in plant products. In general, the carboxylic acids have the function of a biosynthetic precursor. For example, these compounds arise in vivo from the THC carboxylic acids by decarboxylation the tetrahydrocannabinols Δ .sup.9- and Δ .sup.8-THC and CBD from the associated cannabidiol.

(5) Importantly, cannabinoids are hydrophobic small molecules and, as a result, are highly insoluble. Due to this insolubility, cannabinoids such as THC and CBD may need to be efficiently solubilized to facilitate transport, storage, and adsorption through certain tissues and organs. As described in, U.S. Pat. No. 8,410,064 by Pandya et al., cannabinoids may be subject to cytochrome P450 oxidation and subsequent UDP-glucuronosyltransferase (UGT)-dependent glucuronidation in the body after consumption. The resulting glucuronide of the oxidized cannabinoid is the main metabolite found in urine, and thus, this solubilization process plays a critical role in the metabolic clearance of cannabinoids. In another embodiments outlined in PCT/US18/24409 and PCT/US18/41710 (both of

be incorporated herein in their entirety by reference), by Sayre et al., cannabinoids may be glycosylated in vivo to form water-soluble glycoside and acetylated cannabinoid glycoside compounds. (6) As outlined below, cannabinoids may be solubilized by binding to certain carrier proteins. For example, cannabinoids, and other short-chain fatty acid phenolic compounds, may be transported in biological fluids (such as blood) and tissues (including the intracellular milieu) by these so-called carrier proteins. Generally, the binding to these carrier proteins molecules effectively increases the water-solubility of fatty acids and other lipophilic molecules, thereby facilitating their transport through aqueous environments as well as their transfer across cellular membranes. Human and homologous non-human carrier proteins may offer an opportunity for use in the solubilization of cannabinoids among other compounds. One area where water-soluble cannabinoids has seen renewed interest is in the fields of cannabinoid-infused consumer products. However, the ability to effectively solubilize cannabinoids has limited their applicability. To overcome these limitations, many manufacturers of cannabinoid-infused products have adopted the use of traditional pharmaceutical delivery methods of using nanoemulsions of cannabinoids. This nanoemulsion process essentially coats the cannabinoid in a hydrophilic compound, such as oil or other similar compositions. However, the use of nanoemulsions is limited both technically, and from a safety perspective.

(7) First, a large number of surfactants and cosurfactants are required for nanoemulsion stabilization. Moreover, the stability of nanoemulsions is inherently unstable, and may be disturbed by slight fluctuations in temperature and pH, and is further subject to the “oswald ripening effect” or ORE. ORE describes the process whereby molecules on the surface of particles are more energetically unstable than those within. Therefore, the unstable surface molecules often go into solution shrinking the particle over time and increasing the number of free molecules in solution. When the solution is supersaturated with the molecules of the shrinking particles, those free molecules will redeposit on the larger particles. Thus, small particles decrease in size until they disappear and large particles grow even larger. This shrinking and growing of particles will result in a larger mean diameter of a particle size distribution (PSD). Over time, this causes emulsion instability and eventually phase separation.

(8) Second, nanoemulsions may not be safe for human consumption. For example, nanoemulsions were first developed as a method to deliver small quantities of pharmaceutical compounds having poor solubility. However, the ability to “hide” a compound, such as a cannabinoid, in a nanoemulsion may allow the cannabinoid to be delivered to parts of the body where it was previously prevented from entering, as well as accumulating in tissues and organs where cannabinoids and nanoparticles would not typically be found. Additionally, such nanoemulsions, as well as other water-compatible strategies, do not address one of the major-shortcomings of cannabinoid-infused commercial consumables, namely the strong unpleasant smell and taste. Moreover, such water-compatible strategies deliver inconsistent and delayed cannabinoid uptake in the body which may result in user's ingesting a larger than recommended amount of cannabinoid-infused product, as well as delayed and inconsistent medical and/or psychotropic experiences.

(9) As a result, there is a need for more effective strategies to both solubilize cannabinoids, and other associated compounds, such as terpenes and the like, in a way that is both cost-effective, as well as safe to consumers. In one aspect of the invention, members of the Fatty Acid-Binding Protein (FABP) family may be used in novel systems, methods, and compositions to solubilize cannabinoids and other associated compounds. FABPs generally act as intracellular lipid chaperones and help coordinate lipid responses in cells and are also strongly linked to metabolic and inflammatory pathways. FABPs are abundantly expressed 14-15 kDa proteins that reversibly bind hydrophobic ligands, such as saturated and unsaturated long-chain fatty acids, eicosanoids and other lipids, with high affinity. FABPs are found across species, from *Drosophila melanogaster* and *Caenorhabditis elegans* to mice and humans, demonstrating strong evolutionary conservation. However, little is known about their exact biological functions and mechanisms of action. Studies in cultured cells have suggested potential action of FABPs in fatty-acid import, storage and export as well as cholesterol and phospholipid metabolism. FABPs have also been proposed to sequester and/or distribute ligands to regulate signaling processes and enzyme activities. As will be discussed in more detail below, the current inventive technology overcomes the limitations of traditional cannabinoid emulsion systems while meeting the objectives of a truly effective and scalable cannabinoid production, solubilization, and isolation system.

SUMMARY OF THE INVENTION

(10) Generally, the inventive technology relates to systems, methods and compositions to solubilize short-chain fatty acid phenolic compounds, such as cannabinoids, terpenes and other volatile compounds found in cannabinoid-producing plants such as *Cannabis*. In one aspect, human and homologous non-human FABP proteins, sometimes referred to herein as a carrier, or carrier-proteins, may be used to bind to, and solubilization of one or more cannabinoid compounds. In addition aspects, chimeric FABP proteins may be used to bind to, and solubilization of one or more cannabinoid compounds. In another aspect, rationally engineered FABP proteins having one or more directed mutations may exhibit enhanced binding and solubilization of one or more cannabinoid compounds.

(11) Another aspect of the current invention may include novel methods and compositions for increasing the water solubility of one or more cannabinoid compounds via binding to a select FABP. In this embodiment, FABPs may be utilized to solubilize, transport, and store cannabinoid compounds in in vitro, ex vivo, and in vivo systems. In specific preferred aspects, non-human homologs of FABPs, such as plant FABPs, or engineered FABP may be utilized to solubilize, transport, and store, for example, THC, CBD, and other cannabinoids, terpenoids, and volatile compounds produced in *Cannabis* and other cannabinoid producing plants, or even synthetically generated cannabinoids.

(12) Another aspect of the current invention includes novel methods and compositions for increasing the water solubility of one or more cannabinoid compounds via binding to a select chimeric or genetically modified, sometimes referred to as an engineered, FABP. In this aspect, a novel chimeric FABP construct may be rationally designed from homologs of human FABP to allow for enhanced binding of cannabinoid molecules to a single protein chain. In one specific aspect, a novel chimeric FABP construct may be rationally designed from non-human homologs of FABP to allow for enhanced binding of THC, CBD, or other cannabinoid molecules to a single protein chain. In another aspect, one or more FABPs may be subjected to directed evolutionary process, preferably via in silico random mutagenesis, to identify one or more mutations that may increase the binding affinity, and solubility, of one or more target cannabinoids.

(13) Another aspect of the current invention may include the decrease of aggregation propensity and potential antigenicity for the FABP-carrier proteins that may be specific for short-chain fatty acid phenolic compounds, such as cannabinoids. In this embodiment, mutations on the surface of a carrier protein may be rationally designed to reduce aggregate formation in solution and to reduce any elicited antigenic reactions.

(14) Another aspect of the current invention may include the decrease of aggregation propensity and potential antigenicity for a novel wild-type, or engineered FABP-carrier protein, as discussed below, that may be specific for short-chain fatty acid phenolic compounds, such as cannabinoids. In this embodiment, mutations on the surface of a carrier protein may be rationally designed to reduce aggregate formation in solution and to reduce any elicited antigenic reactions.

(15) Another aspect of the current invention may include systems, methods, and compositions for the solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in cell cultures that express one or more FABP-carrier, or engineered FABP proteins. Exemplary cell cultures may include bacterial, yeast, plant, algae and fungi cell cultures. In another aspect, FABP-carrier, or engineered FABP proteins, may be coupled with secretion signals to allow such proteins to be more easily exported from the cell culture into the surrounding supernatant or media. In this aspect of the invention, a FABP-carrier protein, the terms generally encompassing FABP carrier proteins, or engineered FABP proteins that bind to one or more target compounds, and preferably, cannabinoids, may be exported out of a cell through the action of the secretion signal that may direct posttranslational protein translocation into the endoplasmic reticulum (ER), or in alternative embodiments, a secretion signal that may direct cotranslational translocation across the ER membrane where it may assume its three-dimensional form and bind one or more cannabinoid or other compounds as described herein. In one preferred embodiment, a FABP-carrier protein may be generated in a cell culture, preferably a bacterial, yeast, plant or fungi cell culture, and then be exported out of the cell through natural cellular action, or through the action of the secretion signal where it may assume its three dimensional form and bind one or more cannabinoid or other compounds that may be present, preferably by addition of said compound, such as: a quantity of an isolated cannabinoid; a quantity of a plurality of cannabinoids; or

Cannabis extract, to the culture's supernatant.

(16) In another aspect of the invention, a FABP-carrier protein may be exported out of a cell through the action of the secretion signal after it has assumed a transitory and or final three dimensional form and may further be bound to one or more cannabinoid or other compounds as described herein. In one preferred embodiment, a FABP-carrier protein may be generated in a cell culture, preferably a bacterial, yeast, plant or fungi cell culture, and more preferably a plant suspension culture of a cannabinoid-producing plant such as *Cannabis*, where it may assume a transitory or final three dimensional form and bind one or more cannabinoids or other compounds that may be present or produced in the cell.

(17) Another aspect of the current invention may include systems, methods and compositions for the solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in whole plants and plant cell cultures. In certain embodiments, such plants or cell cultures may be genetically modified to direct cannabinoid synthesis to the cytosol, as opposed to a trichome structure. One or more FABP-carrier proteins may be coupled with a secretion signal, preferable in a plant cell culture, to allow such proteins to be exported from the cell into the surrounding media. Expression of exportable and non-exportable FABP-carrier proteins may be co-expressed with one or more catalase and/or one or more myb transcription factors which may enhance cannabinoid production in a *Cannabis* plant or cell culture.

(18) Another aspect of the current invention may include systems, methods and compositions for the coupled glycosylation and solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in whole cannabinoid-producing plants or cell cultures, preferably a *Cannabis* plant or cell culture. In this embodiment, such *Cannabis* plants or cell cultures may be genetically modified to direct cannabinoid synthesis to the cytosol, as opposed to a trichome structure as generally outlines by the current inventors, Sayre et al. in U.S. Pat. No. 10,378,020, the disclosure of such processes being fully incorporated herein by reference. Such *Cannabis* plant, or *Cannabis* cell culture may be further genetically modified to express one or more heterologous glycosyltransferases having glycosylation activity towards at least one cannabinoid (for example SEQ ID NOs. 32-47, and SEQ ID NOs. 61-62) In additional embodiments, a plant or cell may be further genetically modified to express one or more heterologous glycosyltransferases, wherein in said polynucleotides encoding such glycosyltransferases may be codon-optimized for expression in an exogenous system, such as in yeast (for example SEQ ID NOs. 49-60). In additional embodiments, a heterologous or exogenous, being generally the same term, cytochrome P450 and/or a P450 oxidoreductase may be expressed. In this configuration a heterologous cytochrome P450 (for example SEQ ID NOs. 22-23, and SEQ ID NOs. 26-27) may hydroxylate a cannabinoid to form a hydroxylated cannabinoid and/or oxidize a hydroxylated cannabinoid to form a cannabinoid carboxylic acid. Further, in this embodiment, a heterologous P450 oxidoreductase (for example SEQ ID NOs. 24-25, and SEQ ID NOs. 28-29) may facilitate electron transfer from a nicotinamide adenine dinucleotide phosphate (NADPH) to said cytochrome P450.

(19) As noted above, a heterologous glycosyltransferase may glycosylate a cannabinoid compound and thereby produce a water-soluble cannabinoid glycoside. This glycosylated cannabinoid may bind to a heterologous FABP-carrier also expressed in the *Cannabis* plant or cell that may be coupled with a secretion signal, to allow the carrier proteins to be exported from the cell into the surrounding media. Expression of exportable and non-exportable FABP-carriers may be co-expressed with one or more catalase and/or one or more myb transcription factors. The glycosylated cannabinoids bound to the FABP-carrier, being further coupled with a tag in some embodiments, may be isolated, while in still further embodiments, the FABP-carrier protein may be disrupted by a protease, or other protein disrupting detergent and the like, such that the glycosylated cannabinoid may be released from the FABP-carrier and may be further isolated or reconstituted to their original forms through the action of a glycosidase that may remove the sugar moiety.

(20) Another aspect of the current invention may include systems, methods, and compositions for the coupled glycosylation and solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in non-cannabinoid-producing plants and cell cultures, preferably a tobacco cell culture. In this embodiment, a tobacco cell culture may endogenously express one or more glycosyltransferases having glycosylation activity towards at least one cannabinoid. The tobacco cell culture may optionally be genetically modified to express a heterologous cytochrome P450, and a P450 oxidoreductase. In this configuration a heterologous cytochrome P450 may hydroxylate a cannabinoid

added to a tobacco cell culture for example, to form a hydroxylated cannabinoid and/or oxidizes a hydroxylated cannabinoid to form a cannabinoid carboxylic acid. Further, in this embodiment, a heterologous P450 oxidoreductase may facilitate electron transfer from a nicotinamide adenine dinucleotide phosphate (NADPH) to said cytochrome P450. As noted above, the endogenously expressed heterologous glycosyltransferases (for example, NtGT1, 2, 3, 4 or 5 as identified below) may glycosylate one or more cannabinoids introduced to the tobacco cell culture converting it into a water-soluble cannabinoid-glycoside. This glycosylated cannabinoid may bind to a heterologous FABP-carrier co-expressed or added to the tobacco cell culture. In this aspect, an expression of an exportable FABP-carrier may be co-expressed with one or more catalase and/or one or more myb transcription factors. The glycosylated cannabinoids bound to the FABP-carrier, being further coupled with a tag in some embodiments, may be isolated, while in still further embodiments, the carrier protein may be disrupted by a protease or other protein disrupting detergent and the like such that the glycosylated cannabinoids may be released from the carrier protein and may be further isolated or reconstituted to their original forms through the action of a glycosidase.

(21) Another aspect of the current invention may include systems, methods and compositions for the coupled glycosylation and solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in a cell cultures, preferably a yeast cell culture. In these embodiments, yeast cultures may be genetically modified to biosynthesize one or more cannabinoids. The yeast cell culture may be further genetically modified to express one or more heterologous glycosyltransferases having glycosylation activity towards at least one cannabinoid, as well as in some embodiments, a heterologous cytochrome P450 and/or a P450 oxidoreductase.

(22) As noted above, heterologous glycosyltransferases may glycosylate the cannabinoid making it water-soluble. This glycosylated cannabinoid may bind to a heterologous FABP-carrier protein also expressed in the yeast culture which may further be coupled with a secretion signal, to allow the carrier proteins to be exported from the yeast cell into the surrounding media. Expression of exportable and non-exportable FABP-carrier may be co-expressed with a catalase. The glycosylated cannabinoids bound to the FABP-carrier being further coupled with a tag in some embodiments, may be isolated, while in still further embodiments, the carrier protein may be disrupted by a protease or other protein disrupting detergent and the like such that the glycosylated cannabinoids may be released from the carrier protein and may be further isolated or reconstituted to their original forms through the action of a glycosidase.

(23) Another aspect of the current invention may include systems, methods and compositions for the coupled glycosylation and solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in a cell cultures, preferably yeast, bacteria, fungi or algal cell culture. In these embodiments, a yeast cultures may be genetically modified to express one or more heterologous glycosyltransferases having glycosylation activity towards at least one cannabinoid, as well as in some embodiments, a heterologous cytochrome P450 and/or a P450 oxidoreductase.

(24) As noted above, in one preferred embodiment, a quantity of cannabinoids may be added to the cell culture, and preferably a yeast cell culture, where heterologous glycosyltransferases may glycosylate the cannabinoid making it water-soluble. This glycosylated cannabinoid may bind to a heterologous FABP-carrier co-expressed in the yeast culture which may further be coupled with a secretion signal, to allow the carrier proteins to be exported from the yeast cell into the surrounding media. The glycosylated cannabinoids bound to the FABP-carrier, being further coupled with a tag in some embodiments, may be isolated, while in still further embodiments, the carrier protein may be disrupted by a protease or other protein disrupting detergent and the like such that the glycosylated cannabinoids may be released from the carrier protein and may be further isolated or reconstituted to their original forms through the action of a glycosidase.

(25) Another aspect of the current invention may include one or more heterologous glycosyltransferases coupled with the expression of a FABP-carrier optionally having secretion signal, and in some embodiments a tag, which may be expressed in a plant, yeast or bacterial cell culture. Another aspect of the current invention may include one or more heterologous glycosyltransferases coupled with the addition of a FABP-carrier to a plant, yeast, or bacterial cell culture.

(26) Another aspect of the current invention may include one or more endogenously expressed glycosyltransferases coupled with the expression of a FABP-carrier, and preferable an engineered FABP-

carrier having secretion signal, and in some embodiments a tag, that may be expressed in a plant, yeast or bacterial cell culture. Another aspect of the current invention may include one or more endogenously expressed glycosyltransferases coupled with the addition of a FABP-carrier to a plant cell culture.

(27) Another aspect of the current invention may include the increase of CBD water solubility for transport via binding to a FABP-carrier. In this embodiment, plant or other non-human homologs of FABPs may be utilized to solubilize, transport, and/or store CBD and closely-related cannabinoids. Another aspect of the current invention may include the increase of CBD water solubility for transport via binding to a FABP-carrier. In this aspect, a novel engineered FABP-carrier construct may be rationally designed from homologs of FABPs to allow for enhanced binding of CBD molecules to a single protein chain.

(28) Another aspect of the current invention may include the increase of THC water solubility for transport via binding to a FABP-carrier. In this embodiment, plant or other non-human homologs of FABPs may be utilized to solubilize, transport, and/or store THC and closely-related cannabinoids. Another aspect of the current invention may include the increase of THC water solubility for transport via binding to a FABP-carrier. In this aspect, a novel engineered FABP-carrier construct may be rationally designed from homologs of FABPs to allow for enhanced binding of THC molecules to a single protein chain.

(29) Another aspect of the inventive technology may include polynucleotides encoding one or more engineered FABP carrier proteins having enhanced cannabinoid affinity motifs. Another aspect of the inventive technology may include polynucleotides encoding one or more engineered FABP-carrier proteins being heterologously expressed in a genetically modified microorganism, such as yeast, bacteria, fungi, algae and the like.

(30) Another aspect of the inventive technology may include genetically modified bacteria that express at least one polynucleotide encoding one or more heterologous FABP-carrier, and preferably one or more engineered FABP-carrier proteins having enhanced cannabinoid affinity motifs. Another aspect of the inventive technology may include novel engineered FABP-carrier amino acid sequences having enhanced cannabinoid affinity motifs.

(31) Another aspect of the inventive technology provides for a method of enhancing the solubility and stability of cannabinoids, terpenoids and/or other short-chain fatty acid phenolic compounds utilizing FABP-carrier proteins. In a preferred embodiment, a nucleotide sequence encoding a FABP-carrier protein may be genetically engineered to express a rationally designed FABP-carrier protein having cannabinoid affinity or binding sites having enhanced affinity for cannabinoids such that the engineered FABP-carrier carrier protein may bind cannabinoids with a higher affinity thereby increasing the solubility and stability of the cannabinoid in a solution or other form.

(32) Another aspect of the invention includes compositions of novel engineered FABP-carrier polynucleotides and proteins and their method or manufacture. Another aspect of the invention includes compositions of novel engineered FABP-carrier polynucleotides and proteins and their method or manufacture. Another aspect of the invention involves the identification of FABP-carrier proteins that may have endogenous cannabinoid or other affinity sites. Another aspect of the invention involves the rational design of engineered FABP-carrier proteins have enhanced cannabinoid affinity.

(33) Another aspect of the invention includes compositions of novel consumer products that incorporate one or more solubilized cannabinoids or terpenoids bound to FABP-carrier proteins and/or engineered FABP-carrier proteins.

(34) Additional aspects of the invention may include one or more of the following embodiments:

(35) 1. A method of solubilizing a cannabinoid comprising the steps of: generating a Fatty-Acid Binding Protein (FABP)-carrier protein having affinity towards at least one cannabinoid; and introducing said FABP-carrier protein to said at least one cannabinoid, wherein said FABP-carrier protein binds said at least one cannabinoid to form a water-soluble protein-cannabinoid composition.

(36) 2. The method of embodiment 1, wherein the FABP-carrier protein comprises a FABP-carrier protein having an amino acid sequence selected from the group of consisting of: SEQ ID NOs. 1-2, 21, and 74-86, or a homolog having affinity towards at least one cannabinoid thereof.

(37) 3. The method of embodiment 2, wherein said step of generating a FABP-carrier protein comprises the step of generating a FABP-carrier protein in a protein production system selected from the group

consisting of: a bacterial cell culture; a yeast cell culture; a plant cell culture; a fungi cell culture; an algae cell culture; a bioreactor production system; a plant.

(38) 4. The method of embodiment 3, wherein the FABP-carrier protein is coupled with a secretion signal.

(39) 5. The method of embodiment 4, wherein said secretion signal comprises a secretion signal selected from the group consisting of: SEQ ID NO. 3, SEQ ID NO. 65, SEQ ID NOs. 69-73.

(40) 6. The method of embodiments 3 and 5, wherein the FABP-carrier protein is introduced to said at least one cannabinoid in said protein production system.

(41) 7. The method of embodiment 2, wherein the at least one cannabinoid comprises a cannabinoid selected from the group consisting of: cannabidiol (CBD), cannabidiolic acid (CBDA), Δ^9 -tetrahydrocannabinol (THC), tetrahydrocannabinolic acid (THCA), and (cannabigerolic acid) CBGA).

(42) 8. The method of embodiment 1, and further comprising the step of genetically modifying the FABP-carrier protein form to an engineered FABP-carrier protein having enhanced affinity for at least one cannabinoid, such genetic modification comprising one or more of the following: replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket with side chains pointing towards orientated toward the binding cavity; replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket having a hydrophilic side chain with amino acid residues having a hydrophobic side chain; and replacing one or more small hydrophobic amino acid residues of the FABP-carrier protein cannabinoid binding pocket with larger hydrophobic amino acid residues.

(43) 9. The method of embodiment 8, wherein said engineered FABP-carrier protein comprises an engineered FABP-carrier protein having an amino acid sequence selected from the group of consisting of: SEQ ID NOs. 79-86.

(44) 10. A method of solubilizing a cannabinoid comprising the steps of: genetically modifying a Fatty-Acid Binding Protein (FABP)-carrier protein having affinity towards at least one cannabinoid to produce an engineered FABP-carrier protein having enhanced affinity towards said at least one cannabinoid; introducing said engineered FABP-carrier protein to at least one cannabinoid, wherein said engineered FABP-carrier protein binds said at least one cannabinoid forming a water-soluble protein-cannabinoid composition.

(45) 11. The method of embodiment 10, wherein the engineered FABP-carrier protein comprises an engineered FABP-carrier protein having an amino acid sequence selected from the group of consisting of: SEQ ID NOs. 79-86.

(46) 12. The method of embodiments 10 and 11, wherein said step of genetically modifying the FABP-carrier protein comprises one or more of the following: replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket with side chains pointing towards orientated toward the binding cavity; replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket having a hydrophilic side chain with amino acid residues having a hydrophobic side chain; and replacing one or more small hydrophobic amino acid residues of the FABP-carrier protein cannabinoid binding pocket with larger hydrophobic amino acid residues.

(47) 13. The method of embodiment 11, wherein the engineered FABP-carrier protein is expressed in a protein production system selected from the group consisting of: a bacterial cell culture; a yeast cell culture; a plant cell culture; a fungi cell culture; an algae cell culture; a bioreactor production system; and a plant.

(48) 14. The method of embodiment 13, wherein the engineered FABP-carrier protein is coupled with a secretion signal.

(49) 15. The method of embodiment 14, wherein said secretion signal is selected from the group consisting of: SEQ ID NO. 3, SEQ ID NO. 65, SEQ ID NOs. 69-73.

(50) 16. The method of embodiments 13-15, wherein the engineered FABP-carrier protein is introduced to said at least one cannabinoid in said protein production system.

(51) 17. The method of embodiments 1 and 10, wherein said water-soluble protein-cannabinoid composition undergoes lyophilisation.

(52) 18. The method of embodiment 10, and wherein said at least one cannabinoid comprises at least one cannabinoid selected from the group consisting of: cannabidiol (CBD), cannabidiolic acid (CBDA),

Δ^{sup}.9-tetrahydrocannabinol (THC), tetrahydrocannabinolic acid (THCA), and (cannabigerolic acid) CBGA).

(53) 19. An isolated polynucleotide that encodes one or more amino acid sequences selected from the group of consisting of: SEQ ID NOs. 1-2, 21, and 74-86, or a homolog having affinity towards at least one cannabinoid thereof.

(54) 20. The polynucleotide of embodiment 19, wherein said polynucleotide is operably linked to a promotor forming an expression vector.

(55) 21. The polynucleotide of embodiment 19, wherein said polynucleotide is codon optimized for expression in a microorganism, or plant cell, and is further operably linked to a promotor forming an expression vector.

(56) 22. A genetically modified organism expressing at least one of the expression vectors of embodiments 20 and 21

(57) 23. The genetically modified organism of embodiment 22, wherein said genetically modified organism is selected from the group consisting of: a genetically modified bacterial cell a genetically modified yeast cell, a genetically modified plant cell, a genetically modified fungi cell, a genetically modified algae cell, and a genetically modified plant.

(58) 24. A method of solubilizing a cannabinoid comprising the steps of: establishing a cell culture of genetically modified yeast, plant, or bacteria cells that express a nucleotide sequence encoding a heterologous Fatty-Acid Binding Protein (FABP)-carrier protein operably linked to a promotor wherein said heterologous FABP-carrier protein exhibits activity towards one or more cannabinoids; introducing one or more cannabinoids to the genetically modified yeast or bacteria cell culture; and wherein said FABP-carrier protein binds said one or more cannabinoids to form a water-soluble protein-cannabinoid composition.

(59) 25. The method of embodiment 24, wherein the step of introducing comprises the step of introducing one or more cannabinoids to a genetically modified yeast, plant, or bacteria cell culture in a fermenter or suspension cell culture.

(60) 26. The method of embodiment 24, wherein the step of introducing comprises the step of biosynthesizing one or more cannabinoids in a genetically modified yeast or bacteria cell culture wherein said heterologous FABP-carrier protein binds said one or more biosynthesized cannabinoids to form a water-soluble protein-cannabinoid composition.

(61) 27. The method of embodiment 24, wherein said heterologous FABP-carrier protein comprises a heterologous FABP-carrier protein having an amino acid sequence selected from the group of consisting of: SEQ ID NOs. 1-2, 21, and 74-86, or a homolog having affinity towards at least one cannabinoid thereof.

(62) 28. The method of embodiments 24 and 27, wherein said heterologous FABP-carrier protein is coupled with a tag.

(63) 29. The method of embodiments 24 and 27, wherein said heterologous FABP-carrier protein is coupled with a secretion signal.

(64) 30. The method of embodiment 29, wherein said secretion signal comprises a secretion signal selected from the group consisting of: SEQ ID NO. 3, SEQ ID NO. 65, SEQ ID NOs. 69-73.

(65) 31. The method of embodiment 24, wherein the at least one cannabinoid comprises a cannabinoid selected from the group consisting of: cannabidiol (CBD), cannabidiolic acid (CBDA), Δ^{sup}.9-tetrahydrocannabinol (THC), tetrahydrocannabinolic acid (THCA), and (cannabigerolic acid) CBGA).

(66) 32. The method of embodiment 24, and further comprising the of step of genetically modifying the FABP-carrier protein form an engineered FABP-carrier protein having enhanced affinity for at least one cannabinoid, such genetic modification comprising one or more of the following: replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket with side chains pointing towards orientated toward the binding cavity; replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket having a hydrophilic side chain with amino acid residues having a hydrophobic side chain; and replacing one or more small hydrophobic amino acid residues of the FABP-carrier protein cannabinoid binding pocket with larger hydrophobic amino acid residues.

(67) 33. The method of embodiment 32, wherein said engineered FABP-carrier protein comprises an engineered FABP-carrier protein having an amino acid sequence selected from the group of consisting

of: SEQ ID NOs. 79-86.

(68) 34. The FABP-carrier protein of embodiments 1, 2, 10-11, 19, 24, and 27, wherein said FABP-carrier protein is further genetically modified to decrease aggregation propensity and potential antigenicity.

(69) 35. The FABP-carrier protein of embodiments 34, wherein said FABP-carrier protein comprises a FABP-carrier protein having one or more of the following mutations: V58E, V42R.

(70) 36. The water-soluble protein-cannabinoid composition of any of the embodiments above wherein said water-soluble protein-cannabinoid composition is introduced to a consumer product meant for human-consumption, or a pharmaceutical composition for administration of a therapeutically effective dose to a subject in need thereof; or a prodrug for administration of a therapeutically effective dose to a subject in need thereof.

(71) 37. An exportable FABP-carrier protein having a first domain having affinity for at least one cannabinoid, and a second domain comprising a eukaryotic or prokaryotic secretion signal.

(72) 38. The exportable FABP-carrier protein of embodiment 37, wherein said first domain is selected from the group of amino acid sequences identified as SEQ ID NO. 1-2, 21, and 74-86

(73) 39. The exportable FABP-carrier protein of embodiment 37, wherein said second domain is selected from the group of amino acid sequences identified as: SEQ ID NO. 3, SEQ ID NO. 65, SEQ ID NOs. 69-73.

(74) 40. The exportable polypeptide of embodiments 37-39, wherein said exportable FABP-carrier protein binds at least one cannabinoid.

(75) 41. The exportable polypeptide of embodiments 37-39, wherein said exportable FABP-carrier protein binds at least one cannabinoid introduced, or biosynthesized in a cell culture.

(76) 42. The exportable polypeptide of embodiment 41, wherein said cell culture comprises a cell culture selected from the group consisting of: a bacterial cell culture, a yeast cell culture, a plant cell culture.

(77) 43. A genetically modified *Cannabis* plant expressing a nucleotide sequence operably linked to a promoter encoding at least one FABP-carrier protein.

(78) 44. The *Cannabis* plant of embodiment 43 and wherein said FABP-carrier protein comprises a FABP-carrier protein selected from the group consisting of: an amino acid sequence according to SEQ ID NOs. 1-2, 21, and 74-86.

(79) 45. The *Cannabis* plant of embodiments 43 and 44, and further comprising the step of expressing a nucleotide sequence operably linked to a promoter encoding one or more cannabinoid synthases having its trichome targeting sequence disrupted or removed.

(80) 46. The *Cannabis* plant of embodiment 45, wherein one or more cannabinoid synthase genes has been disrupted or knocked out.

(81) 47. The *Cannabis* plant of embodiment 46, wherein said one or more cannabinoid synthases having its trichome targeting sequence disrupted or removed is selected from the group consisting of: the nucleotide sequence identified as: SEQ ID NO. 13, SEQ ID NO. 14, and SEQ ID NO. 15.

(82) 48. The *Cannabis* plant of embodiment 44, and further comprising the step of expressing at least one myb transcription factor.

(83) 49. The *Cannabis* plant of embodiment 48, wherein said at least one myb transcription factor is selected from the group consisting of: SEQ ID NOs. 16-20.

(84) 50. The *Cannabis* plant of embodiment 44, and further comprising the step of expressing at least one catalase.

(85) 51. The *Cannabis* plant of embodiment 50, wherein said at least one catalase is selected from the group consisting of: SEQ ID NOs. 6-10.

(86) 52. The *Cannabis* plant of embodiment 44, and further comprising the step of expressing at least one heterologous glycosyltransferase,

(87) 53. The *Cannabis* plant of embodiment 52, wherein said at least one at least one heterologous glycosyltransferase is selected from the group consisting of: 32-47, and SEQ ID NOs. 61-62. Additional aspects of the invention may be evident from the specification and figures below.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1. (A) an exemplary cannabinoid (CBD) molecule (purple spheres) is bound to the predicted micromolar affinity site in *Bos taurus* FABP1 (white cartoons). (B) an exemplary cannabinoid (THC) molecule (purple spheres) is bound to the predicted micromolar affinity site in *Mus musculus* FABP1 (white cartoons).
- (2) FIG. 2. Vector map of modified pET24a(+).
- (3) FIG. 3 Small-scale of protein expression from 10 ml of culture for WT and engineered FABPs. (A) FABP (from the plant *Rhodamnia argentea*), FABP1 (from pig) and FABP7 (from chicken) Wild type (no modification, natural protein). (B) Engineered FABP from the same organism described above with mutations in binding sites using smina (sm) method of binding prediction of affinity binding. (C) Engineered FABP using rosetta (ro) method for prediction of affinity binding. All wild type FABPs expressed as soluble protein, the protein is found in supernatant (soluble).
- (4) FIG. 4. FABP binding to cannabinoids by ANS displacement by ligands. (A) Wild type FABP from chicken indicating no binding to either CBD or THC. (B) Chicken FABP engineered by smima showing no binding to THC or CBD. (C) Chicken FABP engineered by rosetta showing binding to CBD but not THC.
- (5) FIG. 5. Binding of the plant wild type (A) and engineered (B and C) FABP to cannabinoids by ANS displacement. Both wild-type and engineered proteins bind to CBD and THC.
- (6) FIG. 6. Binding of pig wild type (A) and engineered (B and C) FABP to CBD and THC by ANS displacement.
- (7) FIG. 7. Summary of ANS-displacement by ligand, showing the relative change in fluorescence. 2 uM of protein in 50 mM Tris-HCl buffer was labelled with 20 uM ANS. 33 uM of CBD, THC and Arachidonic acid (AAc) was used as ligand diluted in 10% of MeOH.
- (8) FIG. 8. Structural models of engineered variants of (A) plant FABP (images encompass modified amino acid residues for SEQ ID NOs. 85-86) and (B) horse FABP1 (images encompass modified amino acid residues for both SEQ ID NOs. 83-84) with predicted enhanced binding affinity to CBD relative to the WT proteins. Protein backbones are shown in green or blue cartoons, while bound CBD is shown with magenta sticks. The mutations in each protein that were introduced by the in silico directed evolution procedure are shown as red spheres. Note that all these mutations, which are predicted to help improve each protein's binding affinity to CBD, occur within the binding cavity.

DETAILED DESCRIPTION OF THE INVENTION

- (9) In certain embodiments, the invention may include the use of a FABP as a carrier protein, generally referred to as a FABP-carrier, to solubilize cannabinoids, terpenes/terpenoids, and other short-chain fatty acid phenolic compounds.
- (10) In one embodiment, the invention may include systems, methods and compositions to solubilize cannabinoids, terpenes/terpenoids, and other short-chain fatty acid phenolic compounds utilizing FABPs an, preferably non-human homologs of FABPs to solubilize cannabinoids and other compounds generally identified herein. In this embodiment, the use of FABPs compositions to solubilize cannabinoids may facilitate the transport, and/or storage of cannabinoids in in vitro, and in vivo systems, as well as their use in commercial products where enhanced solubility may enhance the product's characteristics or price.
- (11) In another embodiment, the current invention may include the rational design of a novel chimeric FABP construct, preferably based on non-human homologs of FABPs, to increase cannabinoid water solubility via binding. In a preferred embodiment, a human homolog of hepatic FABP1 (SEQ ID NO. 21), for example as identified in SEQ ID NO. 1-2, or a homolog thereof, may be used to solubilize cannabinoids and other compounds in both in vitro and in vivo systems generally identified herein. Additional embodiments may include the generation of genetically modified or engineered FABPs that may be used to solubilize cannabinoids.
- (12) In a preferred embodiment, such site-directed mutations may be rationally designed such that one or more mutations may be made near a cannabinoid, or other binding site. Such rationally designed mutations may modulate the compounds binding affinity with the FABP-carrier protein. In a preferred embodiment, rationally designed mutations may increase its strength of binding with a cannabinoid, terpene, or other short-chain fatty acid phenolic compound. In some further embodiments, rationally

designed mutations may enhance binding affinity for the FABP-carrier protein that is compound specific. In this embodiment, mutations at and/or near the binding site may be rationally designed to increase its strength of binding with THC, CBD or other cannabinoids as identified herein.

(13) In one example, the directed evolution algorithm may identify amino acid substitution mutations that aid in enhancing the binding affinity to one or more target cannabinoids, such as CBD based quantitatively on the Metropolis criterion. As generally shown in FIG. 8, which shows t 1 three-dimensional models of the FABP-carriers according to SEQ ID NOs. 84-85, and 85-86, the specifically identified mutations sites followed a pattern. In these examples, the set of residues at, or near the cannabinoid binding pocket and with side chains pointing towards this cavity were targeted for mutations. Thus, when looking at any two variants (that started from the same WT parent), there is an expected higher chance of seeing a mutation occur at a particular position. For instance, both chicken FABP7 variants (SEQ ID NOs. 79 & 80) have mutations at positions 34 and 54, and both pig FABP1 variants (SEQ ID NOs. 81 & 82) have mutations at positions 100 and 102. Such mutations may induce conformational or other changes that enhance the affinity of the cannabinoid binding pocket on the respective FABP-carrier proteins.

(14) Such mutations may induce conformational or other changes that enhance the affinity of the cannabinoid binding pocket on the respective FABP-carrier proteins for a specific ligand cannabinoid, such as CBD. Since CBD is highly nonpolar or hydrophobic, as generally shown in SEQ ID NOs. 79-86, the specific mutations may: i) replace a residue with a hydrophilic side chain to another residue with a hydrophobic side chain; or ii) replace a small hydrophobic residue with a bulkier/larger hydrophobic residue. These mutations may enhance the overall hydrophobicity of the cannabinoid binding pocket, thereby making it more attractive to the ligand, for example CBD, for binding. In one example, the FABP-carrier identified a chicken FABP7 (SEQ ID NO. 74) for instance, the wild-type Gly at position 34 is mutated to either Ala or Phe in the variants (case B), and the wild-type Thr at position 54 is replaced with either Met or Ile in the variants (case A). In another example, the FABP-carrier identified a pig FABP1 (SEQ ID NO. 75), the wild-type Ser at position 100 is mutated to Ala in both variants and the wild-type Thr at position 102 is replaced with either Phe or Ile in the variants.

(15) In another example, as noted in the SEQUENCING LISTING below, amino acid residues that have been modified from their wild type versions are underlined (see for example the amino acid sequences according to SEQ ID NO. 79-86, compared to wild type amino acid sequences according to SEQ ID NO. 74-78). For example, in one embodiment, the amino acid sequence for a wild type FABP-carrier having affinity for one or more cannabinoids from the plant *Rhodamnia argentea*, being identified as SEQ ID NO. 77, may be engineered to form an engineered FABP-carrier protein through rationally designed mutations derived from the in silico directed evolution of the proteins binding site such that the rationally designed proteins may exhibit enhanced binding affinity for the FABP-carrier protein. For example, in one embodiment, specific rationally designed mutations identified in the FABP-carrier according to SEQ ID NO 85 may include: A34M, G35M, P39A, T41V, F53I, T55V, L77F, Q95I, R107A, V117F, W119E, and R128A. In another example, specific rationally designed mutations identified in the FABP-carrier according to SEQ ID NO. 86 may include: T41I, F53I, T55M, I64V, N74A, R76A, 84GA, Q95I, Q96N, K104T, and A128A.

(16) In certain embodiment, the invention may include an engineered FABP-carrier protein having one or more mutations that enhance the overall hydrophobicity of the cannabinoid binding pocket, thereby making it more attractive to the cannabinoid ligand. In certain embodiment, the invention may include an engineered FABP-carrier protein having at least one mutation that: (i) that replace a residue with a hydrophilic side chain to another residue with a hydrophobic side chain thereby enhancing the overall hydrophobicity of the cannabinoid binding pocket, thereby making it more attractive to the cannabinoid ligand; and/or that replace a small hydrophobic residue with a bulkier/larger hydrophobic residue thereby enhancing the overall hydrophobicity of the cannabinoid binding pocket, thereby making it more attractive to a cannabinoid ligand. As noted above, in certain embodiments the invention may include an engineered FABP-carrier protein having at least one of the substitution mutations identified in SEQ ID NO. 79-86. In alternative embodiments, invention may include an engineered FABP-carrier protein according to SEQ ID NO. 79-86.

(17) In one embodiment, the current invention may include the increase of THC water solubility for

transport via binding to a non-human homolog of an FABP. In a preferred embodiment, the FABP1 (hepatic FABP) homolog from *Mus musculus* may be used due to its sequence homology with *Homo sapiens* FABP1 (high sequence identity of around 84%) and the presence of a site with predicted micromolar affinity for binding the cannabinoid THC. The protein sequence for *Mus musculus* FABP1 is identified as SEQ ID NO. 1. The left panel of FIG. 1 above demonstrates a structural model of a THC molecule bound to *Mus musculus* FABP1.

(18) In another embodiment, the current invention may include the rational design of mutations to the binding site of an FABP-carrier protein to enhance its binding affinity for THC or other related cannabinoids. In one preferred embodiment, these mutations may be designed into wild type FABP-carriers identified by amino acid SEQ ID NO. 1-2, 21, and 74-78, using a combination of in vitro, in vivo studies as well as bioinformatics approaches such as computational docking, binding affinity estimation, and molecular dynamics simulations. Such bioinformatics applications may be further employed to identify additional potential FABP carrier proteins, as well as direct specific point-mutations to modulate or enhance cannabinoid binding affinity.

(19) In another embodiment, the current invention may include the increase of CBD water solubility for transport via binding to a non-human homolog of an FABP. In a preferred embodiment, the FABP1 (hepatic FABP) homolog from *Bos taurus* may be used due to its sequence homology with *Homo sapiens* FABP1 (high sequence identity of around 81%) and the presence of a site with predicted micromolar affinity for binding CBD. The protein sequence for *Bos taurus* FABP1 is identified as amino acid sequence SEQ ID NO. 2. The right panel of FIG. 1 above demonstrates a structural model of a CBD molecule bound to *Bos taurus* FABP1.

(20) In another embodiment, the current invention may include the increase of CBD water solubility for transport via binding to a non-human homolog of an FABP selected from the amino acid sequences according to SEQ ID NO. 1-2, 21, 74-86. In one preferred embodiment, the aforementioned sequences may be coupled with a tag, and/or secretion signal.

(21) In another embodiment, the current invention may include the rational design of mutations at and/or near the binding site of an FABP-based carrier protein to enhance its binding affinity for CBD. In a preferred embodiment, these mutations will be designed into the FABP1 homolog identified as amino acid sequence SEQ ID NO. 1-2, 21, and 74-78 using a combination of in vitro, in vivo studies as well as bioinformatics approaches such as computational docking, binding affinity estimation, and molecular dynamics simulations. Such bioinformatics applications may be further employed to identify additional potential FABP carrier proteins, as well as direct specific point-mutations to modulate or enhance cannabinoid binding affinity.

(22) In another embodiment, the current invention may include the rational design of mutations on the surface of a FABP-carrier protein to decrease its aggregation propensity and potential antigenicity. In a preferred embodiment, one or more of the mutations V58E, or V42R, may be introduced into one or more of the FABP-carriers identified herein. In a preferred embodiment, one or more of the mutations V58E, or V42R, may be introduced into one or more of the FABP-carriers according to SEQ ID NOs. 21, 75-76, 78, and 81-84. Each mutation may reduce the tendency of this protein to form aggregates in solution, without destabilizing its conformational stability. Preliminary analyses of linear epitopes for this FABP1 homolog suggest a high false positive prediction rate, based on a comparison with corresponding predictions for *Homo sapiens* FABP1 (SEQ ID NO. 21).

(23) The above FABP-carrier proteins are provided as exemplary embodiments only. Nor are they limiting as to the number of punitive cannabinoid, or other short-fatty-acid phenolic compound affinity sites that may be engineered in a FABP-carrier. Consideration of which may include the desired type of short-fatty-acid phenolic compound to be bound by the FABP-carrier, as well as steric considerations resulting from the addition of such modified affinity motifs presented in the three-dimensional folded protein. Naturally, certain modifications may be made to a FABP-carrier that may alter the affinity strength of one or more existing cannabinoid affinity sites.

(24) In another embodiment, the current invention may include the rational design of mutations on the surface of a cannabinoid-specific FABP-carrier protein that may decrease its aggregation propensity and potential antigenicity. In a preferred embodiment, the mutation V58K may be introduced into the FABP-carrier which may further include SEQ ID NOs. 21, 75-76, 78, and 81-84, or a homolog

thereof. Such a reduction may reduce the tendency of a FABP-carrier to form aggregates in solution, without destabilizing its conformational stability. Analyses of the linear epitopes for this modified FABP-carrier suggest a high false positive prediction rate, based on a comparison with corresponding predictions for *Homo sapiens* FABP1 (SEQ ID NO. 21).

(25) In another embodiment, the current invention may include the increase of cannabinoid water solubility for transport via binding to an artificial or non-human homolog of FABP. In this embodiment of the current invention may include the increase of CBD water solubility for transport via binding to a non-human homolog of FABP.

(26) In another embodiment, a FABP-carrier protein may be identified an engineered FABP-carrier protein having at least one modified cannabinoid-affinity sites or motifs, such as a THC, or CBD-affinity site or motif, when the protein is in its folded three-dimensional configuration. In one embodiment, a cannabinoid affinity from an engineered FABP-carrier may be modified to have increased or decreased cannabinoid affinity. In one preferred embodiment, a FABP-carrier protein, preferably a non-human homolog such as identified in SEQ ID NO. 74-86, or a homolog thereof may be identified and established as an engineered FABP-carrier protein. This engineered FABP-carrier protein may be genetically modified to construct a FABP-chimera or fusion protein having one or more modified cannabinoid-affinity sites or motifs having, preferable enhanced cannabinoid affinity. For example, as shown in FIG. 1, a FABP-carrier may have a micromolar affinity for a cannabinoid, while an engineered FABP-carrier may be engineered to have a nanomolar or greater affinity for cannabinoids.

(27) As one of ordinary skill in the art would recognize, a ligand, such as a cannabinoid, or other short-chain fatty acid phenolic compound, with nanomolar (nM) dissociation constant may bind more tightly to a particular protein than a ligand with micromolar (μ M) dissociation constant. As a result, in certain embodiments of the inventive technology, customized FABP-carrier proteins, and in particular engineered FABP-carrier proteins may be generated that have a customized dissociation constant. This customized dissociation constant may be engineered according to the specifications of a particular application. For example, in one application a FABP-carrier may be engineered to be a fusion, or chimeric protein having one or more cannabinoid affinity sites having nanomolar (nM) or greater dissociation constant. Such FABP-carrier may be useful for long-term storage of cannabinoids in solution, or for applications inclusion in various commercial and other consumer products where the FABP-carrier may be exposed to artificial, or natural environmental conditions or other chemical processes that might degrade the protein structure and prematurely released the cannabinoid.

Alternatively, in one application a FABP-carrier may be engineered to have one or more a cannabinoid affinity sites having micromolar (μ M) dissociation constant. Such FABP-carrier may allow for one or more cannabinoid compounds to be more easily release from the FABP-carrier. In one preferred embodiment, a FABP-carrier may include one or more a cannabinoid affinity sites having a macro- or micromolar (μ M) dissociation that may allow for greater release, as compared for example to a nanomolar (nM) dissociation, and bioavailability of the cannabinoid upon consumption.

(28) As noted above, cannabinoid producing strains of *Cannabis*, as well as other plants may be utilized with the inventive technology. In certain preferred embodiments, *Cannabis* plant material may be harvested and undergo cannabinoid extraction through one or more of the methods generally known in the art. These extracted cannabinoids, terpenoids and other short chain fatty acid phenolic compounds, may be introduced to a quantity of FABP-carrier proteins to be solubilized as described herein.

(29) In one embodiment, cells of a microorganism, such as yeast or bacteria, may be transformed with artificially created expression vectors encoding one or more FABP-carrier proteins. In this preferred embodiment, the nucleotide sequences encoding a FABP-carrier protein may be codon optimized for exogenous expression. Additional embodiments may include genetic control elements such as promoters and/or enhancers as well as post-transcriptional regulatory elements that may also be expressed in the genetically engineered microorganisms, such as yeast or bacteria, such that the presence, quantity and activity of any FABP-carrier proteins present in a culture may be controlled. In a preferred embodiment, the microorganism, such as yeast or bacteria strain, may be further modified to generate high-levels of FABP-carrier proteins. In a preferred embodiment, the bacterial strain may include a high expression strain of bacteria, such as *E. coli* strain BL21(DE3) for optimal protein expression. In another preferred embodiment, the yeast strain may include genetically modified yeast cells selected from the group

consisting of: genetically modified *Pichia pastoris* cells, genetically modified *Saccharomyces cerevisiae* cells, and/or genetically modified *Kluyveromyces marxianus* cells.

(30) As noted above, in one embodiment the inventive technology may include individual expression of one or more FABP-carrier proteins each having a selected molecular tag. In a preferred embodiment, an FABP-carrier-protein, for example engineered from the amino acid sequences SEQ ID NO. 1-2, 21, and 74-86, or other homologous sequences, may each be configured to contain a poly-His or His-6 tag, which may be used later for protein purification. In this embodiment, the expressed FABP-carrier-protein may be detected and purified because the string of histidine residues bind to several types of immobilized metal ions, including nickel, cobalt and copper, under appropriate buffer conditions.

(31) In one embodiment of the inventive technology, a cell culture, such as a plant, yeast or bacterial culture, may be genetically modified to express a tagged heterologous FABP-carrier and may further be allowed to grow to a desired level of cell or optical density, or in other instances until a desired level of FABP-carrier proteins have accumulated in the cultured cells and/or media, for example through the addition of a secretion signal that directs the FABP-carrier to be exported from the cell. All, or a portion of the accumulated FABP-carrier proteins may then be harvested from the culture and/or media, which in a preferred embodiment may be an industrial-scale fermenter or other apparatus suitable for the large-scale culturing of microorganisms. Optionally, the harvested cells may be lysed such that the accumulated FABP-carrier proteins may be released to the surrounding lysate. Additional steps may include treating this lysate. Examples of such treatment may include filtering, centrifugation or screening to remove extraneous cellular material as well as chemical treatments to improve later FABP-carrier yields.

(32) The FABP-carrier may be further isolated and purified. In one preferred embodiment, the cell lysate or media may be processed utilizing affinity chromatography or other purification methods. In this preferred embodiment, an affinity column having a ligand configured to bind with one or more of the tags coupled with the FABP-carrier protein, for example, a poly-His or His-6 tag, among others, may be immobilized or coupled to a solid support. The lysate may then be passed over the column such that the tagged FABP-carrier, having specific binding affinity to the ligand becomes bound and immobilized. In some embodiments, non-binding and non-specific binding proteins that may have been present in the lysate may be removed. Finally, the FABP-carrier may be eluted or displaced from the affinity column by, for example, a corresponding protein, tag or other compound that may displace or disrupt the tag-ligand bond. The eluted FABP-carrier proteins may be collected and further purified or processed. Notably, in other embodiments, FABP-carrier proteins may be commercially obtained or synthesized and used consistent with the embodiments described herein.

(33) All FABP and FABP-carrier amino sequences described herein include homologs of said sequences which may have between 75-99% homology, while retaining specific activity towards one or more short-chain fatty-acid phenolic compound, and preferably one or more cannabinoid compounds.

(34) One embodiment of the inventive technology includes the generation of novel genetically modified short-chain fatty-acid phenolic compound-carrier proteins that may have enhanced affinity for short-chain fatty-acid phenolic compounds, and preferably a terpenoid or cannabinoid from *Cannabis*. In one preferred embodiment, the inventive technology includes the generation of novel genetically modified cannabinoid-carrier FABPs engineered from one or more wild type FABP-carriers, for example SEQ ID NO. 1-2, and 74-78, or homologs thereof that may have enhanced affinity for cannabinoids.

(35) Another embodiment of the inventive technology provides for systems and methods of high-capacity cannabinoid solubilization. In this preferred embodiment, a polynucleotide configured to express one or more FABP-carrier proteins, for example the amino acid sequences SEQ ID NO. 1-2, 74-86, or a homolog thereof, which may optionally be coupled with a tag for purification or isolation purposes, and may be operably linked to a promoter forming an expression vector. This expression vector may be used to transform a microorganism which may express one or more tagged FABP-carrier proteins, which may be further isolated, preferably through affinity purification. The isolated tagged FABP-carrier may be placed into a bio-reactor or other suitable environment where they may be introduced to one or more cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds. The tagged FABP-carrier proteins may solubilize the cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds through affinity binding to one or more affinity site. The

solubilized cannabinoids may be isolated and used for commercial, pharmaceutical and other applications as generally described herein.

(36) Another embodiment of the invention provides for methods of masking the typical unpleasant small and taste of cannabinoid-infused commercial products and beverages. For example, in this embodiment a FABP-carrier, for example as identified in the amino acid sequences 1-2, 21, and 74-86, may bind to one or more cannabinoids and allow it to be solubilized. In this solubilized state, the carrier protein allows for the masking of the cannabinoids natural small and taste. Moreover, in additional embodiments, a FABP-carrier may bind to, and solubilize one or more terpenes, the compounds in *Cannabis* primarily responsible for its distinctive smell. In this manner, the invention may generate cannabinoid-infused commercial products, such as consumables and beverages that eliminate, mask or ameliorate the undesired smell and taste of the cannabinoid and terpene compounds.

(37) Another embodiment of the invention provides for methods of generating solubilized cannabinoids, terpenes and other short-chain fatty-acid phenolic compounds that may have a more rapid metabolic uptake or bioavailability upon ingestion. In this embodiment, a FABP-carrier, such as an engineered FABP-carrier, may bind to one or more cannabinoids and/or terpenes and allow it to be solubilized such that upon ingestion, it may be more readily taken up by the body, for example, through the association with the carrier protein. This embodiment may allow for not only a more rapid uptake of the target compound, but allow for consistent consumer experiences, as well as facilitating safe and effective consumer-controlled dosing of cannabinoids and other compounds. This embodiment may further allow for lower amounts of cannabinoid and terpenes to be used in infused consumables and beverages as a result of this improved bioavailability. For example, absent this enhance bioavailability of the solubilized cannabinoids and terpenes, a large portion of the compounds may not be efficiently taken up by the body and may be eventually eliminated through natural chemical degradation or other strategies to metabolically clear the compounds from the body.

(38) Another embodiment of the invention provides for methods of generating precise doses and/or formulations and/or ratios of cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds. In a preferred embodiment, a polynucleotide may be generated that is configured to express one or more FABP-carrier proteins configured to have binding affinity motifs that selectively bind an individual or class of cannabinoid, terpenoids, and/or other short-chain fatty-acid phenolic compounds. Again, this selective FABP-carrier-protein may be coupled with a tag for purification or isolation purposes and may be operably linked to a promoter forming an expression vector. This expression vector may be used to transform a microorganism, such as bacteria, yeast, fungi, or algae, which may express the tagged selective FABP-carrier protein which may be further isolated, preferably through affinity purification. The isolated selective FABP-carrier-protein may be placed into a bio-reactor or other suitable environment where they may be introduced to one or more cannabinoid, terpenoids, and/or other short-chain fatty-acid phenolic compounds. The FABP-carrier may selectively solubilize a quantity of cannabinoid, terpenoids, and/or other short-chain fatty-acid phenolic compounds, consistent with its endogenous and/or engineered affinity sites. The solubilized cannabinoid, terpenoids, and/or other short-chain fatty-acid phenolic compounds may be used for commercial, pharmaceutical, and other applications as generally described herein.

(39) Another aspect of the invention provides for methods of generating precise mixed doses, ratios, and/or formulations of cannabinoids, terpenoids, and/or other short-chain fatty acid phenolic compounds. In a preferred embodiment, a first polynucleotide may be generated that is configured to express a FABP-carrier configured to have a selective binding affinity motif(s) that selectively bind an individual or class of cannabinoid, terpenoid, and/or other short-chain fatty-acid phenolic compound. An additional polynucleotide may be generated that is configured to express a FABP-carrier configured to have a cannabinoid binding affinity motif(s) that selectively bind a different individual or class of cannabinoid, terpenoid, and/or other short-chain fatty-acid phenolic compound. Both selective FABP-carrier proteins may be coupled with a tag for purification or isolation purposes and may be incorporated into one or more expression vectors being operably linked to a promoter. Such expression vector(s) may be used to transform a microorganism, such as bacteria, yeast, fungi, or algae, which may express the tagged selective FABP-carrier proteins which may be further isolated, preferably through affinity purification. The isolated selective FABP-carrier proteins may be placed into a bio-reactor or other suitable

environment where they may be introduced to one or more cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds. The first FABP-carrier protein may selectively solubilize a quantity of individual or class of cannabinoid, terpenoid, and/or other short-chain fatty-acid phenolic compound consistent with the number and type of its endogenous and/or engineered affinity sites. The additional FABP-carrier protein may selectively solubilize a quantity of a separate individual or class of cannabinoid, terpenoid, and/or other short-chain fatty-acid phenolic compound consistent with the number and type of its endogenous and/or engineered affinity sites. The solubilized cannabinoid, terpenoids, and/or other short-chain fatty-acid phenolic compounds may be used for commercial, pharmaceutical, and other applications as generally described herein.

(40) Another aspect of the invention may include in vitro systems and methods to solubilize cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds. In a preferred embodiment, FABP-carrier proteins, for example the proteins identified according to SEQ ID NO. 1-2, 21, and 74-86, may be artificially synthesized in vitro and then placed into a bio-reactor, or other suitable environment, where they may be introduced to one or more cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds. The FABP-carrier proteins carrier proteins may solubilize the cannabinoids, terpenoids, and/or other short-chain fatty acid phenolic compounds as generally described herein.

(41) Another embodiment of the inventive technology provides for direct systems and methods of high-capacity cannabinoid solubilization. In this preferred embodiment, a polynucleotide configured to express one or more FABP-carriers, for example SEQ ID NO. 1-2, 21, and 74-86, may be coupled with a tag for purification or isolation purposes. This polynucleotide may be operably linked to a promoter forming an expression vector. This expression vector may be used to transform a microorganism which may be grown in an industrial scale fermenter or other like apparatus known in the art for high-level protein production. While in culture, the genetically modified microorganism may express one or more tagged FABP-carrier proteins. Short-chain fatty-acid phenolic compounds, such as cannabinoids, terpenes, and other volatiles may be extracted from cannabinoid-producing plants or artificially biosynthesized and added to the cell culture. The isolated FABP-carrier proteins produced in culture may be introduced to one or more cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds in the culture. In one embodiment, one or more FABP-carrier proteins produced in culture may be coupled with a secretion signal, for example as identified in SEQ ID NO. 3 or 65, so as to be able to be more easily exported to the cultures media or supernatant, preferably in a yeast culture. The FABP-carrier proteins may bind to, and solubilize one or more cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds in the culture's supernatant or media. The tagged FABP-carrier proteins, and their bound compounds, may be isolated utilizing affinity chromatography or other purification methods.

(42) Another embodiment of the inventive technology provides for direct systems and methods of high-capacity cannabinoid solubilization. In this preferred embodiment, a polynucleotide configured to express one or more FABP-carrier proteins may be coupled with a tag for purification or isolation purposes. Such polynucleotide may be operably linked to a promoter forming an expression vector. This expression vector may be used to transform a bacterium which may be grown in an industrial scale fermenter or other like apparatus known in the art for high-level protein production. While in culture, the genetically modified bacteria may express one or more tagged FABP-carriers, for example one or more of the amino acid sequences according to SEQ ID NO. 1-2, 74-86, that may also be coupled with a secretion signal. Short-chain fatty-acid phenolic compounds, such as cannabinoids, terpenes, and other volatiles, may be extracted from cannabinoid-producing plants or artificially biosynthesized and added to the cell culture. The isolated FABP-carrier proteins produced in culture may be introduced to one or more cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds in the culture. The FABP-carrier proteins may bind to and solubilize one or more cannabinoids, terpenoids, and/or other short-chain fatty-acid phenolic compounds. The tagged FABP-carrier proteins carrier proteins, and their bound compounds, may be isolated utilizing affinity chromatography or other purification methods.

(43) Another embodiment of the inventive technology provides for systems and methods of high-capacity cannabinoid solubilization coupled with cannabinoid biosynthesis in microorganisms genetically engineered to produce cannabinoids. Implementing cannabinoid biosynthesis strategies

proposed by: Carvalho A, et al.; US Pat. App. No. US20180371507, by Paulos et al.; and WO2017139496, by Hussain et al.; (all of which are incorporated herein by reference) for the generation of cannabinoids in microorganisms such as yeast, fungi, and bacteria, in one embodiment the inventive technology may include systems and methods for solubilization of cannabinoids produced in non-cannabinoid producing microorganisms or artificial chemically-synthesized cannabinoids.

(44) In one embodiment, one or more metabolic pathways for cannabinoid biosynthesis may be reconstructed in microorganisms, such as bacteria, fungi, algae, or yeast. Such pathways may be reconstructed through the expression of a plurality of heterologous genes necessary for the biosynthesis of precursor and cannabinoid compounds. In one preferred embodiment, a microorganism, such as bacteria, yeast, or fungi, may be genetically engineered to produce one or more cannabinoids, terpenes, or other short-chain fatty acid phenolic compounds. The microorganism may be further genetically modified to express a polynucleotide encoding a cannabinoid-carrier FABPs or a homolog thereof, such as those identified in SEQ ID NO. 1-2, 74-86 or more preferably an engineered FABP-carrier protein, such as those identified in SEQ ID NO. 79-86, or other homolog thereof. In one preferred embodiment, a FABP-carrier protein may bind to and solubilize one or more exogenously biosynthesized cannabinoids. This FABP-carrier protein may be tagged to facilitate isolation and purification as generally described herein and may further be coupled with a secretion signal as discussed below.

(45) In another aspect of the invention, a FABP-carrier protein may be exported out of a cell through the natural action of the cell, gradient movement, or diffusion, as well as the action of the secretion signal, where it may bind to one or more cannabinoid or other compounds located externally to a cell. In one preferred embodiment, a FABP-carrier protein may be generated in a cell culture, preferably a bacterial, yeast, plant or fungi cell culture, and more preferably a plant suspension culture of a cannabinoid-producing plant such as *Cannabis*, where it may be exported out of the cell and bind one or more cannabinoid or other compounds that may be present in the external cellular environment. In additional embodiments, a FABP-carrier protein may bind one or more cannabinoid or other compounds that may be present in the intracellular environment.

(46) In another aspect of the invention, a FABP-carrier protein having a secretion signal may be expressed in a genetically modified yeast culture and exported out of a cell through the action of the secretion signal, or other natural action of the yeast cell. In one preferred embodiment, a heterologous polynucleotide may express one or more exportable FABP-carrier proteins having a secretion signal. In one embodiment, a secretion signal may direct posttranslational protein translocation into the endoplasmic reticulum (ER). In additional embodiments, a secretion signal may direct cotranslational translocation of the carrier protein across the ER membrane. Notably, protein translocation is the process by which peptides are transported across a membrane bilayer. Translocation of proteins across the membrane of the ER is known to occur in one of two ways: cotranslationally, in which translocation is concurrent with peptide synthesis by the ribosome, or posttranslationally, in which the protein is first synthesized in the cytosol and later is transported into the ER.

(47) In eukaryotic organisms such as yeast, proteins that are targeted for translocation across the ER membrane have a distinctive amino-terminal signal sequence, such as the amino acid sequence identified in SEQ ID NO. 65, which is recognized by the signal recognition particle (SRP). The SRP in eukaryotes is a large ribonucleoprotein which, when bound to the ribosome and the signal sequence of the nascent peptide, is able to arrest protein translation by blocking tRNA entry. The ribosome is targeted to the ER membrane through a series of interactions, starting with the binding of the SRP by the SRP receptor. The signal sequence of the nascent peptide chain is then transferred to the protein channel, Sec61. The binding of SRP to its receptor causes the SRP to dissociate from the ribosome, and the SRP and SRP receptor also dissociate from each other following GTP hydrolysis. As the SRP and SRP receptor dissociate from the ribosome, the ribosome is able to bind directly Sec61.

(48) The Sec61 translocation channel (known as SecY in prokaryotes) is a highly conserved heterotrimeric complex composed of α -, β - and γ -subunits. The pore of the channel, formed by the α -subunit, is blocked by a short helical segment which may become unstructured during the beginning of protein translocation, allowing the peptide to pass through the channel. The signal sequence of the nascent peptide intercalates into the walls of the channel, through a side opening known as the lateral gate. During translocation, the signal sequence is cleaved by a signal peptide peptidase, freeing the

amino terminus of the growing peptide.

(49) During cotranslational translocation in eukaryotes, the ribosome provides the motive power that pushes the growing peptide into the ER lumen. During posttranslational translocation, additional proteins are necessary to ensure that the peptide moves unidirectionally into the ER membrane. In eukaryotes, posttranslational translocation requires the Sec62/Sec63 complex and the chaperone protein BiP. BiP is a member of the Hsp70 family of ATPases, a group which is characterized as having an N-terminal nucleotide-binding domain (NBD), and a C-terminal substrate-binding domain (SBD) which binds to peptides. The nucleotide binding state of the NBD determines whether the SBD can bind to a substrate peptide, in this case a FABP-carrier or engineered FABP-carrier. While the NBD is bound to ATP, the SBD is in an open state, allowing for peptide release, while in the ADP state, the SBD is closed and peptide-bound. The primary role of the membrane protein complex Sec62/Sec63 is to activate the ATPase activity of BiP via a J-domain located on the lumen-facing portion of Sec63. The SBD of BiP binds non-specifically to the peptide as it enters the ER lumen, and keeps the peptide from sliding backwards in a ratchet-type mechanism.

(50) Again, in one preferred embodiment, a FABP-carrier may be modified to include at least one secretion signal that may facilitate vesicle transport of the protein out of the cell, preferably a yeast cell. In one embodiment, a FABP-carrier may be modified to include a secretion signal which directs posttranslational protein translocation into the ER. In one preferred embodiment, a secretion signal which directs posttranslational protein translocation into the ER may be identified in amino acid SEQ ID NO. 3 (see below) which encodes an N-terminal secretion signal from α -factor mating pheromone in *S. cerevisiae*. The secretion signal is made up of a 19 amino acid 'presequence' which directs posttranslational protein translocation into the ER, and a 66-amino acid 'pro region' mediating receptor-dependent packaging into ER-derived COPII transport vesicles.

(51) SEQ ID NO. 3:

(52) TABLE-US-00001 MRFPSIFTAVLFAASSALAAPVNTTTEDETAQIPA
EAVIGYSDLEGDFDVAVLPFSNSTNNGLLFINTTI ASIAAKEEGVSLEKR

(53) In another embodiment, a FABP-carrier may be modified to include a secretion signal which directs cotranslational translocation across the ER membrane. In one preferred embodiment, an enhanced secretion signal which directs cotranslational translocation across the ER membrane may be identified in amino acid sequence of SEQ ID NO. 65, where the 19 amino acid 'presequence' is replaced with the enhance 'presequence' with the Ost1 (OST=oligosaccharyltransferase) signal sequence identified by amino acid SEQ ID NO. 66.

(54) TABLE-US-00002 MRQVWFSWIVGLFLCFFNVSSA

(55) In this preferred embodiment, an enhanced secretion signal may be identified as SEQ ID NO. 65:

(56) TABLE-US-00003 MRQVWFSWIVGLFLCFFNVSSAAPVNTTTEDETA
QIPAEAVIGYSDLEGDFDVAVLPFSNSTNNGLLF INTTASIAAKEEGVSLEKR

(57) Again, in a preferred embodiment, one or more of the FABP-carrier proteins identified herein may be modified and expressed, preferably in a yeast cell, to include a secretion signal which directs posttranslational protein translocation into the ER, such signal preferably being SEQ ID NO. 32 which encode an N-terminal secretion signal from α -factor mating pheromone in *S. cerevisiae*. Such exportable FABP-carrier carrier proteins may include exemplary amino acid sequence identified as SEQ ID NO. 4-5, and amino acid sequences 1-2, 21, and 74-86 coupled with the N-terminal secretion signal according to SEQ ID NO. 3, may bind to, and solubilize one or more cannabinoids located in the cell, or more preferably they may solubilize one or more cannabinoids outside in the cell, such as cannabinoids added to a cell culture supernatant. The exportable FABP-carrier proteins, having solubilized one or more target cannabinoids or other compounds identified herein may be further isolated as generally described herein.

(58) In another embodiment, a FABP-carrier protein may be modified and expressed, preferably in a yeast cell, to include an enhanced secretion signal which directs cotranslational translocation across the ER membrane, such signal preferably being SEQ ID NO. 65 which include the Ost1 signal sequence identified as amino acid sequence SEQ ID NO. 66 coupled with the 66-amino acid 'pro region' of the N-terminal secretion signal from α -factor mating pheromone in *S. cerevisiae*. Such enhanced exportable FABP-carrier proteins may include exemplary amino acid sequence identified as SEQ ID NO. 67-68, and amino acid sequences 1-2, 21, and 74-86 coupled with the enhanced N-terminal secretion signal

according to SEQ ID NO. 65, may bind to, and solubilize one or more cannabinoids located in the cell, or more preferably one or more cannabinoids located outside in the cell, such as cannabinoids added to a cell culture supernatant. The exportable FABP-carrier, having solubilized one or more target cannabinoids or other compound identified herein, may be further isolated as generally described herein. (59) Additional embodiments also feature a method for producing FABP-carrier polypeptides. The method includes culturing a recombinant bacterium cell in a culture medium under conditions that allow the FABP-carrier polypeptides to be secreted into the culture medium, the recombinant bacterium cell comprising at least one exogenous nucleic acid, the exogenous nucleic acid comprising first and second nucleic acid sequences, wherein the first nucleic acid sequence encodes a signal peptide and the second nucleic acid sequence encodes a FABP-carrier carrier polypeptides, wherein the first and second nucleic acid sequences are operably linked to produce a fusion polypeptide comprising the signal peptide and the FABP-carrier polypeptide, and wherein upon secretion of the fusion polypeptide from the cell into the culture medium, the signal peptide may be removed from the -containing polypeptide. The method further can include isolating the FABP-carrier polypeptides from the culture medium.

(60) In another aspect of the invention, a FABP-carrier protein may be exported out of a bacterial cell through the action of a secretion signal where the FABP-carrier protein may be secreted in an unfolded conformation and bind to one or more cannabinoid or other compounds located externally to a cell. In one preferred embodiment, a FABP-carrier protein may be generated in a cell culture, preferably a bacterial cell culture, where it may be exported out of the cell and bind one or more cannabinoid or other compounds that may be present in the external cellular environment. In this embodiment, a FABP-carrier protein may be coupled with a secretion signal that may direct the carrier protein to be secreted from a bacterium through a SEC-mediated secretion pathway.

(61) Notably, in bacteria, translated peptides may be actively translocated post-translationally through a SecY channel by a protein called SecA. SecA is composed of a nucleotide-binding domain, a polypeptide crosslinking domain, and helical wing and scaffold domains. During translocation, a region of the helical scaffold domain forms a two-finger helix which inserts into the cytoplasmic side of the SecY channel, thereby pushing the translocating carrier peptide through. A tyrosine found on the tip of the two-finger helix plays a critical role in translocation, and is thought to make direct contact with the translocating peptide. The polypeptide crosslinking domain (PPXD) forms a clamp which may open as the translocating peptide is being pushed into the SecY channel by the two-finger helix, and close as the two-finger helix resets to its "up" position. The conformational changes of SecA are powered by its nuclease activity, with one ATP being hydrolyzed during each cycle. This SEC system secretes proteins having a consensus signal peptide that is similar to, but distinct from, that of the Tat system as described below. The Sec signal sequence lacks an N-terminal consecutive-arginine sequence and has a relatively hydrophobic central region and a relatively short signal sequence compared with that of Tat. Exemplary Sec signal sequences may be identified as SEQ ID NO. 69.

(62) Again, in one preferred embodiment, a FABP-carrier may be modified to include at least one Sec-mediated secretion signal that may facilitate translocation of transport of the unfolded carrier protein out of a bacterial cell via a Sec-secretion pathway. In one embodiment, a FABP-carrier may be modified to include a secretion signal which directs posttranslational protein translocation. In one preferred embodiment, a secretion signal which directs posttranslational protein translocation may be identified in amino acid SEQ ID NO. 69 which encodes an exemplary Sec-signal sequence from *E. coli* L-asparaginase II.

(63) Again, in a preferred embodiment, one or more of the FABP-carrier carrier proteins may be selected from SEQ ID NO. 1-2, 21, and 74-86, and may be modified and expressed, preferably in a bacterial cell, to include a secretion signal which directs posttranslational protein translocation of the unfolded protein, such signal preferably being SEQ ID NO. 70, or homologous or similar Tat-secretion signal sequence, which may encode an exemplary Sec-secretion signal sequence. Such exportable FABP-carrier proteins may be translocated from a bacterial cell to the external environment where they may come into contact with, bind to, and solubilize one or more cannabinoids located outside in the cell, such as cannabinoids added to a cell culture supernatant.

(64) In another aspect of the invention, a FABP-carrier protein may be exported out of a bacterial cell through the action of a secretion signal where the FABP-carrier protein may assume its folded three-

dimensional configuration prior to secretion. In this embodiment, a FABP-carrier protein may bind to one or more cannabinoid or other compounds located internally or externally to the cell. In one preferred embodiment, a FABP-carrier protein may be generated in a cell culture, preferably a bacterial cell culture, where it may be exported out of the cell and into the external cellular environment. In this embodiment, a FABP-carrier protein may be coupled with a secretion signal that may direct the carrier protein to be secreted from a bacterium through a TAT-mediated secretion pathway.

(65) Unlike the Sec system, the Tat system is involved in the transport of pre-folded protein substrates. Proteins are targeted to the Tat pathway by possession of N-terminal tripartite signal peptides. The signal peptides include a conserved twin-arginine motif in the N-region of Tat signal peptide. The motif has been defined as R-R-x- Φ - Φ , where Φ represents a hydrophobic amino acid. In *E. coli* the Tat pathway comprises the three-membrane protein TatA, TatB and TatC. A fourth protein TatE forms a minor component of the Tat machinery and has a similar function to TatA. Because of the ability to secrete pre-folded protein substrates, the Tat pathway may be especially suited for secreting a high level of heterologous FABP-carrier proteins. Estimates of Tat substrates in organisms other than *Bacillus subtilis* and *E. coli* have been based predominantly in in silico analysis of genome sequences using programs trained to recognize specific features of tat targeting sequences. An exemplary Tat signal sequences may be identified as SEQ ID NO. 70.

(66) Again, in one preferred embodiment, a FABP-carrier may be modified to include at least one Tat-mediated secretion signal that may facilitate translocation of transport of the folded carrier protein out of a bacterial cell. In one embodiment, a FABP-carrier may be modified to include a secretion signal which directs posttranslational protein translocation via a Tet-secretion pathway. In one preferred embodiment, a secretion signal which directs posttranslational protein translocation may be identified in amino acid SEQ ID NO. 70 or homologous or similar Sec-secretion signal sequence which encodes an exemplary tat signal peptide for *E. coli* strain k12 periplasmic nitrate reductase.

(67) Again, in a preferred embodiment, one or more of the FABP-carrier proteins may be selected from SEQ ID NO. 1-2, 21, and 74-86 and may be modified and expressed, preferably in a bacterial cell, to include a secretion signal which directs posttranslational protein translocation of the folded protein via a Tet-secretion pathway, such signal preferably being SEQ ID NO. 70 or homologous or similar Sec-secretion signal sequence. Such exportable FABP-carrier proteins may be translocated from a bacterial cell already having one or more bound cannabinoids, or other compounds. In alternative embodiments, an exportable FABP-carrier protein may be translocated from a bacterial cell where it may come into contact with, bind to, and solubilize one or more cannabinoids located outside in the cell, such as cannabinoids added to a cell culture supernatant. The exportable FABP-carrier proteins, having solubilized one or more target cannabinoids or other compounds identified herein may be further isolated.

(68) In another embodiment, the invention includes a recombinant plant, or plant cell producing FABP-carrier proteins. The plant or plant cell can include at least one heterologous nucleic acid encoding a FABP-carrier, wherein the plant or plant cell is from a species of *Cannabis*. The plant or plant cell can include at least one heterologous nucleic acid encoding a FABP-carrier, wherein the plant or plant cell is from a species of *Nicotiana*. The plant or plant cell can include at least one heterologous nucleic acid encoding a FABP-carrier, wherein the plant or plant cell is from a species other than *Nicotiana*. The heterologous nucleic acid further can include a regulatory control element such as a promoter (e.g., a tissue-specific promoter such as leaves, roots, stems, or seeds).

(69) A polypeptide can be expressed in monocot plants and/or dicot plants. Techniques for introducing nucleic acids into plants are known in the art, and include, without limitation, *Agrobacterium*-mediated transformation, viral vector-mediated transformation, electroporation, and particle gun transformation (also referred to as biolistic transformation). See, for example, U.S. Pat. Nos. 5,538,880; 5,204,253; 6,329,571; and 6,013,863; Richards et al., Plant Cell. Rep. 20:48-20 54 (2001); Somleva et al., Crop Sci. 42:2080-2087 (2002); Sinagawa-Garcia et al., Plant Mol Biol (2009) 70:487-498; and Lutz et al., Plant Physiol., 2007, Vol. 145, pp. 1201-1210. In some instances, intergenic transformation of plastids can be used as a method of introducing a polynucleotide into a plant cell. In some instances, the method of introduction of a polynucleotide into a plant comprises chloroplast transformation. In some instances, the leaves and/or stems can be the target tissue of the introduced polynucleotide. If a cell or cultured tissue is

used as the recipient tissue for transformation, plants can be regenerated from transformed cultures if desired, by techniques known to those skilled in the art.

(70) Other suitable methods for introduce polynucleotides include electroporation of protoplasts, polyethylene glycol-mediated delivery of naked DNA into plant protoplasts, direct gene transformation through imbibition (e.g., introducing a polynucleotide to a dehydrated plant), transformation into protoplasts (which can comprise transferring a polynucleotide through osmotic or electric shocks), chemical transformation (which can comprise the use of a polybrene-spermidine composition), microinjection, pollen-tube pathway transformation (which can comprise delivery of a polynucleotide to the plant ovule), transformation via liposomes, shoot apex method of transformation (which can comprise introduction of a polynucleotide into the shoot and regeneration of the shoot), sonication-assisted *Agrobacterium* transformation (SAAT) method of transformation, infiltration (which can comprise a floral dip, or injection by syringe into a particular part of the plant (e.g., leaf)), silicon-carbide mediated transformation (SCMT) (which can comprise the addition of silicon carbide fibers to plant tissue and the polynucleotide of interest), electroporation, and electrophoresis. Such expression may be from transient or stable transformations.

(71) Additional embodiments also feature a method for producing FABP-carrier polypeptides in plants and preferably a plant cell in culture. The method includes culturing a recombinant plant cell in a culture medium under conditions that allow the FABP-carrier polypeptides to be secreted into the culture medium, the recombinant plant cell comprising at least one heterologous nucleic acid, the heterologous nucleic acid encoding a FABP-carrier polypeptide, wherein the nucleic acid sequence is operably linked to a promoter and optionally may further produce a fusion polypeptide comprising the signal peptide and the FABP-carrier polypeptide, and wherein upon secretion of the polypeptide from the plant cell into the culture medium, the signal peptide may be removed from the FABP-carrier polypeptide. The method further can include isolating the FABP-carrier polypeptides from the culture medium.

(72) In another aspect of the invention, a FABP-carrier protein may be exported out of a plant cell through the action of a secretion signal where the FABP-carrier protein may be secreted via a plant protein secretion pathway. In a preferred embodiment, FABP-carrier may be coupled with an N-terminal signal peptide which may direct their translocation to the extracellular region via the Endoplasmic Reticulum-Golgi apparatus and the subsequent endomembrane system. In one preferred embodiment, a FABP-carrier protein may be generated in a plant, and preferably a plant cell culture, where it may be exported out of the cell and bind one or more cannabinoid or other compounds that may be present in the external cellular environment. In this embodiment, a FABP-carrier protein may be coupled with a secretion signal that may direct the carrier protein to be secreted from a plant cell via the Endoplasmic Reticulum-Golgi apparatus and the subsequent endomembrane system.

(73) Again, in one preferred embodiment, a FABP-carrier may be modified to include at least one plant secretion signal that may facilitate translocation of transport of the protein out of a plant cell. In one embodiment, a FABP-carrier may be modified to include a secretion signal which directs. In one preferred embodiment, a secretion signal which directs protein translocation from a plant cell may be identified in amino acid SEQ ID NO. 71, which encodes an exemplary secretion signal from an extracellular *Arabidopsis* protease Ara12 (At5g67360). Additional examples include the amino acid SEQ ID NO. 72, which encodes an exemplary secretion signal from a barley (*Hordeum vulgare*) alpha amylase. Still further examples include the amino acid SEQ ID NO. 73, which encodes an exemplary secretion signal from a rice a-Amylase.

(74) Again, in a preferred embodiment, one or more of the FABP-carrier carrier proteins may be selected from SEQ ID NO. 1-2, 21, and 74-86, and may be further modified and expressed, preferably in a plant cell, to include a secretion signal which directs protein translocation out of the plant cell, such signal preferably being SEQ ID NO. 71, 72-74. Such exportable FABP-carrier proteins may be translocated from a plant cell already having one or more bound cannabinoids, or other compounds. In alternative embodiments, an exportable FABP-carrier protein may be translocated from a plant cell where it may come into contact with, bind to, and solubilize one or more cannabinoids located outside in the cell, such as cannabinoids added to a cell culture supernatant. The exportable FABP-carrier protein, having solubilized one or more target cannabinoids or other compounds identified herein, may be further isolated.

(75) In another embodiment, one or more of the FABP-carrier proteins may be secreted from a plant cell in culture using the Hydroxyproline-Glycosylation (Hyp-Glyco) technology. In this embodiment, one or more of the FABP-carrier proteins may be selected from SEQ ID NO. 1-2, 21, and 74-86 and may be modified and expressed, preferably in a plant cell and further fused with Hyp-rich repetitive peptide (HypRP) tag that directs extensive Hyp-O-glycosylation in plant cells resulting in arabinogalactan polysaccharides populating this repetitive peptide fusion facilitating the secretion of the expressed protein from cultured plant cells.

(76) In certain embodiments, a catalase enzyme may be co-expressed with cannabinoid biosynthesis genes and FABP-carrier proteins, as well as optionally FABP-transporters or other genes that may reduce cannabinoid biosynthesis toxicity and/or facilitate transport of the solubilized cannabinoids through or out of the cell. In one embodiment a heterologous catalase is selected from the group consisting of: the amino acid sequence SEQ ID NO. 6, the amino acid sequence SEQ ID NO. 7, the amino acid sequence SEQ ID NO. 8, the amino acid sequence SEQ ID NO. 9, the amino acid sequence SEQ ID NO. 10 and a sequence having at least 80% homology to amino acid sequence SEQ ID NO. 6, SEQ ID NO. 7, SEQ ID NO. 8, SEQ ID NO. 9 and SEQ ID NO. 10.

(77) Another embodiment of the inventive technology provides for systems and methods of high-capacity cannabinoid solubilization coupled with cannabinoid biosynthesis in cannabinoid producing plants or plants engineered to produce cannabinoids. In this preferred embodiment, cannabinoid biosynthesis may be redirected from the plant's trichome to be localized in the plant cell's cytosol. In certain embodiments, a cytosolic cannabinoid production system may be established as directed in PCT/US18/24409 and PCT/US18/41710, both by the present inventors Sayre et al. (These applications are both incorporated by reference with respect to their disclosure related to cytosolic cannabinoid production and/or modification in whole, and plant cell systems).

(78) In one embodiment, a cytosolic cannabinoid production and solubilization system may include the in vivo creation of one or more recombinant proteins that may allow cannabinoid biosynthesis to be localized to the cytosol where one or more heterologous FABP-carrier proteins may also be expressed and present in the cytosol. This inventive feature allows not only higher levels of cannabinoid production and accumulation, but efficient production of cannabinoids in suspension cell cultures. Even more importantly, this inventive feature allows cannabinoid production and accumulation without a trichome structure in whole plants, allowing cells that would not traditionally produce cannabinoids, such as cells in *Cannabis* leaves and stalks, to become cannabinoid-producing cells

(79) More specifically, in this preferred embodiment, one or more cannabinoid synthases may be modified to remove all or part of an N-terminal extracellular trichome targeting. An exemplary N-terminal trichome targeting sequence for THCA synthase is identified as SEQ ID NO. 11, while an N-terminal trichome targeting sequence for CBDA synthase is identified as SEQ ID NO. 12. Co-expression with this cytosolic-targeted synthase (for example SEQ ID NOs. 13-15) with a heterologous FABP-carrier protein, and preferably an engineered FABP-carrier protein, may allow the localization of cannabinoid synthesis, accumulation and solubilization to the cytosol. The FABP cannabinoid carrier proteins may be later isolated with their bound cannabinoid molecules through a water-based extraction process due to their solubility, as opposed to traditional chemical or super-critical CO₂ extractions methods. In this way, a cell culture of a cannabinoid producing plant may form a continuous production platform for solubilized cannabinoids.

(80) As noted below, in certain embodiments cannabinoid biosynthesis may be coupled with cannabinoid glycosylation in a cell cytosol. For example, in one preferred embodiment a cytosol-target glycosyltransferase (for example SEQ ID NOs. 63-64) may be expressed in a cell, preferably a cannabinoid producing cell, and even more preferably a *Cannabis* cell.

(81) Such cytosolic target enzymes may be co-expressed with heterologous catalase and cannabinoid transporters or other genes that may reduce cannabinoid biosynthesis toxicity and/or facilitate transport through or out of the cell. In one embodiment a heterologous catalase is selected from the group consisting of: the amino acid sequence SEQ ID NO. 6, the amino acid sequence SEQ ID NO. 7, the amino acid sequence SEQ ID NO. 8, the amino acid sequence SEQ ID NO. 9, the amino acid sequence SEQ ID NO. 10 and a sequence having at least 80% homology to amino acid sequence SEQ ID NO. 6, SEQ ID NO. 7, SEQ ID NO. 8, SEQ ID NO. 9 and SEQ ID NO. 10.

(82) Such cytosolic target enzymes may also be co-expressed with one or more myb transcriptions factors that may enhance metabolite flux through the cannabinoid biosynthetic pathway which may increase cannabinoid production. In one embodiment a myb transcription factor may be endogenous to *Cannabis*, or an ortholog thereof. Examples of endogenous myb transcription factor may include SEQ ID NO. 16 and 17, or orthologs thereof. In one embodiment a myb transcription factor may be heterologous to *Cannabis*. A heterologous myb transcription factor may be selected from the group consisting of a nucleotide sequence that expresses: amino acid sequence SEQ ID NO. 18, amino acid sequence SEQ ID NO. 19, amino acid sequence SEQ ID NO. 20.

(83) In an alternative embodiment, isolated heterologous FABP-carrier proteins may be added to a cell culture of a cannabinoid-producing plant, preferably a *Cannabis* suspension cell culture, having a cytosolic cannabinoid production system. In this preferred embodiment, one or more cannabinoid may be produced in the cytosol and transported into the surrounding culture media through passive or active transport mechanisms. Once the cannabinoids have been transported to the surrounding culture media, a quantity of FABP-carrier proteins may be added to the media and bind to and solubilize one or more cannabinoids. This media may then be removed and replenished, such that the solubilized cannabinoids bound to FABP-carrier proteins may be further isolated from the media as generally described herein. In one embodiment, the FABP-carrier proteins may be later isolated with their bound cannabinoid molecules through a water-based extraction process due to their solubility, as opposed to traditional chemical or super-critical CO₂ extractions methods. In this way, a cell culture of a cannabinoid producing plant may form a continuous production platform for solubilized cannabinoids.

(84) Another embodiment of the invention may include the generation of an expression vector comprising this polynucleotide, namely a cannabinoid synthase lacking an N-terminal extracellular trichome targeting sequence and a heterologous FABP-carrier gene, for example a nucleotide sequence that encodes one or more of the amino acid sequences identified in SEQ ID NOs. 1-2, 21, and 74-86, as well as optionally one or more secretion signals as discussed herein, operably linked to a promoter. This expression vector may be used to create a genetically altered plant or parts thereof and its progeny comprising this polynucleotide operably linked to a promoter, wherein said plant or parts thereof and its progeny produce said proteins. For example, seeds and pollen contain this expression vector, a genetically altered plant cell comprising this expression vector such that said plant cell produces said chimeric protein. Another embodiment comprises a tissue culture comprising a plurality of the genetically altered plant cells having this expression vector.

(85) One preferred embodiment of the invention may include a genetically altered cannabinoid-producing plant or cell expressing a cytosolic-targeted cannabinoid synthase protein having a cannabinoid synthase N-terminal extracellular targeting sequence (SEQ ID NOs. 11-12) inactivated or removed. In one embodiment, a cytosolic targeted THCA synthase (ctTHCAs) may be identified as SEQ ID NO. 13, while in another embodiment, cytosolic targeted CBDA synthase (cytCBDAs) are identified as SEQ ID NO. 14-15, respectively. Such cytosolic-targeted cannabinoid synthase proteins may be operably linked to a promoter. Another embodiment provides a method for constructing a genetically altered plant or part thereof having solubilization of cannabinoids in the plant's cytosol compared to a non-genetically altered plant or part thereof, the method comprising the steps of: introducing a polynucleotide encoding a cannabinoid synthase into a plant or part thereof to provide a genetically altered plant or part thereof, wherein the cannabinoid synthase N-terminal extracellular targeting sequence has been disrupted or removed and further expressing a polynucleotide encoding a cannabinoid-carrier FABPs, such as those identified in SEQ ID NO. 1-2, 21, and 74-86, or a homolog thereof.

(86) Notably, in a preferred embodiment, one or more endogenous cannabinoid synthase genes may be disrupted and/or knocked out and replaced with cytosolic-targeted cannabinoid synthase proteins as described herein. The disrupted endogenous cannabinoid synthase gene(s) may be the same or different than the expressed cytosolic-targeted cannabinoid synthase protein. Methods of disrupting or knocking-out a gene are known in the art and could be accomplished by one of ordinary skill without undue experimentation.

(87) In another embodiment, one or more endogenous cannabinoid synthase genes may be disrupted and/or knocked out in a *Cannabis* plant or suspension cell culture wherein one or more cannabinoid

synthase genes has been disrupted and/or knocked out is selected from the group consisting of: a CBG synthase gene; a THCA synthase, a CBDA synthase, and a CBCA synthase. In this embodiment, the *Cannabis* plant or suspension cell culture may express a polynucleotide encoding one or more cannabinoid synthases having its trichome targeting sequence disrupted and/or removed which may be selected from the group consisting of: a CBG synthase gene having its trichome targeting sequence disrupted and/or removed; a THCA synthase having its trichome targeting sequence disrupted and/or removed; a CBDA synthase having its trichome targeting sequence disrupted and/or removed; and a CBCA synthase having its trichome targeting sequence disrupted and/or removed.

(88) The current invention may further include systems, methods and compositions for the solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in cell cultures. Exemplary cell cultures may include bacterial, yeast, plant, algae and fungi cell cultures. FABP-carrier may be coupled with secretion signals to allow such proteins to be exported from the cell culture into the surrounding media. In this embodiment, a FABP-carrier protein may be engineered to include a secretion signal that may allow it to be exported from a cell. In one preferred embodiment, one or more of sequences identified as SEQ ID NO. 1-19, and 74-86, may be coupled with a secretion signal. In one preferred embodiment, one or more of sequences identified as SEQ ID NO. 1-2, 21, and 74-86, may be coupled with the N-terminal secretion signal identified in SEQ ID NO. 3 or SEQ ID NO. 65. One exemplary exportable FABP-carrier protein may include SEQ ID NO. 4 or SEQ ID NO. 5 or a FABP-carrier protein engineered from SEQ ID NO. 4 or SEQ ID NO. 5 or may be coupled with the secretion signal identified as amino acid sequence SEQ ID NO. 66, to form an enhanced exportable a FABP-carrier protein being identified herein as SEQ ID NO. 67 and SEQ ID NO. 68. Naturally, such examples are meant to be illustrative of the type and number of exportable FABP-carrier and FABP-carrier proteins within the scope of the current invention.

(89) Another aspect of the current invention may include systems, methods and compositions for the solubilization of cannabinoids, terpenoids and other short-chain fatty acid phenolic compounds in whole plants and plant cell cultures. In certain embodiments, such plants or cell cultures may be genetically modified to direct cannabinoid synthesis to the cytosol, as opposed to a trichome structure. Further, FABP-carrier proteins may be coupled with a secretion signal, preferable in a plant cell culture, to allow such proteins to be exported from the cell into the surrounding media. Expression of exportable and non-exportable FABP-carrier proteins may be co-expressed with one or more catalase and/or myb transcription factors

(90) Another embodiment of the inventive technology may include the generation of a powder containing one or more solubilized cannabinoids. In one preferred embodiment, cannabinoids, terpenes, and other short-chain fatty acid phenolic compounds may be solubilized by association with a FABP-carrier protein. FABP-carrier proteins, having solubilized a quantity of cannabinoids, may undergo lyophilisation, to form a FABP-carrier protein powder containing the solubilized cannabinoids. In a preferred embodiment, a FABP-carrier protein may solubilize a quantity of cannabinoids through one of the methods generally described herein and then may further undergo lyophilisation, to form a FABP-carrier protein powder containing the solubilized cannabinoids. This powder may have enhanced properties, such as enhanced cannabinoid affinity to provide greater retention and shelf-life to the cannabinoids in the powdered composition. Additionally, this cannabinoid infused powder may be reintroduced to a liquid such that the cannabinoids are re-dissolved in the liquid. This powder may be used, for example, by consumers that wish to add a quantity of one or more cannabinoids to a beverage or other consumable product. It may also be used for pharmaceutical preparations and for proper cannabinoid dosing. This type of soluble cannabinoid-infused powder may be used as a food additive, or even coupled with flavoring agents to be used as a beverage additive. The presence of the FABP-carrier proteins, as well as the enhanced cannabinoid affinity and binding capacity, may allow less powder to be used to achieve an equivalent dose, whether in a pharmaceutical or consumer beverage/consumable product.

(91) Other embodiments may allow for the creation of high-concentration solutions of solubilized cannabinoids bound to FABP-carrier proteins. Such solutions may allow a user to generate liquid-based food and beverage additives of varying concentrations. Such solutions may further allow a user to generate liquid-based food and beverage additives of varying types of cannabinoids or combinations of

cannabinoids and/or terpenes and the like. Due to the enhanced characteristics of certain FABP-carriers, in particular the enhanced ability to bind and solubilize greater numbers of individual cannabinoid molecules per protein chain, such solutions may achieve higher than normal concentrations of solubilized cannabinoids. Also, due to the enhanced affinity characteristics of certain FABP-carriers, liquid solutions having solubilized cannabinoids may achieve a longer-shelf life.

(92) In another embodiment, the inventive technology may include novel systems, methods and compositions to decrease potential antigenicity for the FABP-carrier proteins. In one preferred embodiment, the recognition sequences of one or more FABP-carrier, or preferably FABP-carrier proteins that correspond to the formation of one or more post-translational glycosylation sites or motifs may be disrupted. In this embodiment, site-directed mutagenesis of recognition sequences that allow for post-translational glycosylation for the sequences identified as SEQ ID NO. 1-2, 21, and 74-86, or a homolog thereof may be accomplished. The removal of such glycosylation sites in a FABP-carrier may result in decreased antigenicity.

(93) The inventive technology may further include novel engineered FABP-carrier proteins that may exhibit enhance cannabinoid affinity and/or novel affinity motifs allowing for a greater of number of individual cannabinoid compounds to be bound and solubilized at one time. In one preferred embodiment, the invention may include a pharmaceutical composition as active ingredient an effective amount or dose of one or more FABP-carrier proteins coupled with one or more cannabinoids, terpenes or other short-chain fatty acid phenolic compounds. In some instances, the active ingredient may be provided together with pharmaceutically tolerable adjuvants and/or excipients in the pharmaceutical composition. Such pharmaceutical composition may optionally be in combination with one or more further active ingredients. In one embodiment, one of the aforementioned FABP-carrier proteins coupled with one or more cannabinoids, terpenes or other short-chain fatty acid phenolic compounds may act as a prodrug. The term “prodrug” is taken to mean FABP-carrier proteins according to the invention which have been modified by, for example, the additional of novel cannabinoid affinity sites, or enhanced cannabinoid affinity sites, and wherein the FABP-carrier proteins are sterically modified in the organism to released the bound cannabinoid or other compound to form the therapeutically effective dose according to the invention. The terms “effective amount” or “effective dose” or “dose” are interchangeably used herein and denote an amount of the pharmaceutical compound having a prophylactically or therapeutically relevant effect on a disease or pathological conditions, i.e. which causes in a tissue, system, animal or human a biological or medical response which is sought or desired, for example, by a researcher or physician. Pharmaceutical formulations can be administered in the form of dosage units which comprise a predetermined amount of active ingredient per dosage unit. The concentration of the prophylactically or therapeutically active ingredient in the formulation may vary from about 0.1 to 100 wt %. Preferably, the compound of formula (I) or the pharmaceutically acceptable salts thereof are administered in doses of approximately 0.5 to 1000 mg, more preferably between 1 and 700 mg, and most preferably 5 and 100 mg per dose unit. Generally, such a dose range is appropriate for total daily incorporation. In other terms, the daily dose is preferably between approximately 0.02 and 100 mg/kg of body weight. The specific dose for each patient depends, however, on a wide variety of factors as already described in the present specification (e.g. depending on the condition treated, the method of administration and the age, weight and condition of the patient). Preferred dosage unit formulations are those which comprise a daily dose or part-dose, as indicated above, or a corresponding fraction thereof of an active ingredient. Furthermore, pharmaceutical formulations of this type can be prepared using a process which is generally known in the pharmaceutical art.

(94) As noted above, the present invention allows the scaled production of water-soluble or solubilized cannabinoids (the terms being generally used to denote a cannabinoid or other compound, such as a terpene or short-chain fatty acid phenolic compound that is water-soluble or may be dissolved in water). Because of this solubility, the invention allows for the addition of such solubilized cannabinoid to a variety of compositions without requiring oils and/or emulsions that are generally required to maintain the generally hydrophobic cannabinoid compounds in suspension. As a result, the present invention may allow for the production of a variety of compositions for the food and beverage industry, as well as pharmaceutical applications that do not required oils or emulsion suspensions and the like.

(95) In one embodiment, the invention may include aqueous compositions containing one or more

solubilized cannabinoids that may be introduced to a food or beverage. In a preferred embodiment, the invention may include an aqueous solution containing one or more solubilized cannabinoids. In this embodiment, one or more cannabinoids, terpenes, or other short-chain fatty acid phenolic compounds may be solubilized through binding to a FABP-carrier protein. Here, the solubilized cannabinoids may be generated in vivo as generally described herein, or in vitro. In additional embodiments, the solubilized cannabinoid may be an isolated non-psychoactive, such as CBD and the like. Such selection of one or more cannabinoids may be due to specific affinity specificities in a FABP-carrier protein for one cannabinoid over another. Moreover, in this embodiment, the aqueous solution may contain one or more of the following: saline, purified water, propylene glycol, deionized water, and/or an alcohol such as ethanol, as well as a pH buffer that may allow the aqueous solution to be maintained at a pH below 7.4. Additional embodiments may include the addition of an acid or base, such as formic acid, or ammonium hydroxide.

(96) In another embodiment, the invention may include a consumable food additive having at least one FABP-carrier solubilized cannabinoid. In this embodiment, one or more cannabinoids, terpenes or other short-chain fatty acid phenolic compounds may be solubilized through binding to a FABP-carrier protein. Here, the solubilized cannabinoids may be generated in vivo as generally described herein, or in vitro. This consumable food additive may further include one or more food additive polysaccharides, such as dextrin and/or maltodextrin, as well as an emulsifier. Example emulsifiers may include, but not be limited to: gum arabic, modified starch, pectin, xanthan gum, gum ghatti, gum tragacanth, fenugreek gum, mesquite gum, mono-glycerides and di-glycerides of long chain fatty acids, sucrose monoesters, sorbitan esters, polyethoxylated glycerols, stearic acid, palmitic acid, mono-glycerides, di-glycerides, propylene glycol esters, lecithin, lactylated mono- and di-glycerides, propylene glycol monoesters, polyglycerol esters, diacetylated tartaric acid esters of mono- and di-glycerides, citric acid esters of monoglycerides, stearyl-2-lactylates, polysorbates, succinylated monoglycerides, acetylated monoglycerides, ethoxylated monoglycerides, quillaia, whey protein isolate, casein, soy protein, vegetable protein, pullulan, sodium alginate, guar gum, locust bean gum, tragacanth gum, tamarind gum, carrageenan, furcellaran, Gellan gum, psyllium, curdlan, konjac mannan, agar, and cellulose derivatives, or combinations thereof.

(97) The consumable food additive of the invention may be a homogenous composition and may further comprise a flavoring agent. Exemplary flavoring agents may include: sucrose (sugar), glucose, fructose, sorbitol, mannitol, corn syrup, high fructose corn syrup, saccharin, aspartame, sucralose, acesulfame potassium (acesulfame-K), neotame. The consumable food additive of the invention may also contain one or more coloring agents. Exemplary coloring agents may include: FD&C Blue Nos. 1 and 2, FD&C Green No. 3, FD&C Red Nos. 3 and 40, FD&C Yellow Nos. 5 and 6, Orange B, Citrus Red No. 2, annatto extract, beta-carotene, grape skin extract, cochineal extract or carmine, paprika oleoresin, caramel color, fruit and vegetable juices, saffron, Monosodium glutamate (MSG), hydrolyzed soy protein, autolyzed yeast extract, disodium guanylate or inosinate. In one embodiment, this powdered lyophilized FABP-carrier protein, having solubilized a quantity of cannabinoids, may be a food additive. In certain preferred embodiments, one or more flavoring agents may be added to a quantity of powdered or lyophilized FABP-carrier proteins having solubilized a quantity of cannabinoids.

(98) The consumable food additive of the invention may also contain one or more surfactants, such as glycerol monostearate and polysorbate 80. The consumable food additive of the invention may also contain one or more preservatives. Exemplary preservatives may include ascorbic acid, citric acid, sodium benzoate, calcium propionate, sodium erythorbate, sodium nitrite, calcium sorbate, potassium sorbate, BHA, BHT, EDTA, or tocopherols. The consumable food additive of the invention may also contain one or more nutrient supplements, such as: thiamine hydrochloride, riboflavin, niacin, niacinamide, folate or folic acid, beta carotene, potassium iodide, iron or ferrous sulfate, alpha tocopherols, ascorbic acid, Vitamin D, amino acids, multi-vitamin, fish oil, co-enzyme Q-10, and calcium.

(99) In one embodiment, the invention may include a consumable fluid containing at least one FABP-carrier solubilized cannabinoid, terpenoid, or other short chain fatty acid phenolic compound. In one preferred embodiment, this consumable fluid may be added to a drink or beverage to infuse it with the solubilized cannabinoid generated through binding to a FABP-carrier protein, in an in vitro, ex vivo, or in

vivo system as generally described, or through an in vitro process. The consumable fluid may include a food additive polysaccharide such as maltodextrin and/or dextrin, which may further be in an aqueous form and/or solution. For example, in one embodiment, an aqueous maltodextrin solution may include a quantity of sorbic acid and an acidifying agent to provide a food grade aqueous solution of maltodextrin having a pH of 2-4 and a sorbic acid content of 0.02-0.1% by weight.

(100) In certain embodiments, the consumable fluid may include water, as well as an alcoholic beverage; a non-alcoholic beverage, a noncarbonated beverage, a carbonated beverage, a cola, a root beer, a fruit-flavored beverage, a citrus-flavored beverage, a fruit juice, a fruit-containing beverage, a vegetable juice, a vegetable containing beverage, a tea, a coffee, a dairy beverage, a protein containing beverage, a shake, a sports drink, an energy drink, and a flavored water. The consumable fluid may further include at least one additional ingredient, including but not limited to: xanthan gum, cellulose gum, whey protein hydrolysate, ascorbic acid, citric acid, malic acid, sodium benzoate, sodium citrate, sugar, phosphoric acid, and water. In certain embodiments, the consumable fluid of the invention may be generated by addition of a quantity of solubilized cannabinoid in powder or liquid form as generally described herein to an existing consumable fluid, such as a branded beverage or drink.

(101) In one embodiment, the invention may include a consumable gel having at least one FABP-carrier solubilized cannabinoid and gelatin in an aqueous solution. In a preferred embodiment, the consumable gel may include a one or more cannabinoids, terpenes or other short-chain fatty acid phenolic compounds solubilized through binding to a FABP-carrier protein. Here, the solubilized cannabinoids may be generated in vivo as generally described herein, or in vitro.

(102) Additional embodiments may include a liquid composition having at least one cannabinoid solubilized by a FABP-carrier protein, in a first quantity of water; and at least one of: xanthan gum, cellulose gum, whey protein hydrolysate, ascorbic acid, citric acid, malic acid, sodium benzoate, sodium citrate, sugar, phosphoric acid, and/or a sugar alcohol. In one preferred embodiment, the composition may further include a quantity of ethanol. Here, the amount of solubilized cannabinoids may include: less than 10 mass % water; more than 95 mass % water; about 0.1 mg to about 1000 mg of the solubilized cannabinoid; about 0.1 mg to about 500 mg of the solubilized cannabinoid; about 0.1 mg to about 200 mg of the solubilized cannabinoid; about 0.1 mg to about 100 mg of the solubilized cannabinoid; about 0.1 mg to about 10 mg of the solubilized cannabinoid; about 0.5 mg to about 5 mg of the solubilized cannabinoid; about 1 mg/kg to 5 mg/kg (body weight) in a human of the solubilized cannabinoid.

(103) In alternative embodiments, the composition may include at least one cannabinoid solubilized by a FABP-carrier protein, in the range of 50 mg/L to 300 mg/L; at least one FABP-carrier solubilized cannabinoid in the range of 50 mg/L to 100 mg/L; at least one FABP-carrier solubilized cannabinoid in the range of 50 mg/L to 500 mg/L; at least one FABP-carrier solubilized cannabinoid over 500 mg/L; at least one FABP-carrier solubilized cannabinoid under 50 mg/L. Additional embodiments may include one or more of the following additional components: a flavoring agent; a coloring agent; and/or caffeine.

(104) In one embodiment, the invention may include a liquid composition having at least one cannabinoid solubilized by a FABP-carrier protein, being solubilized in said first quantity of water and a first quantity of ethanol in a liquid state. In a preferred embodiment, a first quantity of ethanol in a liquid state may be between 1% to 20% weight by volume of the liquid composition. In this embodiment, a solubilized cannabinoid may include a cannabinoid solubilized by a FABP-carrier protein, a terpenoid/terpene solubilized by a FABP-carrier protein, or a mixture of both. Such solubilized cannabinoids may be generated in an in vivo, ex vivo, and/or in vitro system as herein identified. In a preferred embodiment, the ethanol or ethyl alcohol component may be up to about ninety-nine point nine-five percent (99.95%) by weight and the solubilized cannabinoid about zero point zero five percent (0.05%) by weight.

(105) Examples of the preferred embodiment may include liquid ethyl alcohol compositions having at least one cannabinoid solubilized by a FABP-carrier protein, wherein said ethyl alcohol has a proof greater than 100, and/or less than 100. Additional examples of a liquid composition containing ethyl alcohol and at least one cannabinoid solubilized by a FABP-carrier protein, may include, beer, wine and/or distilled spirits.

(106) Additional embodiments of the invention may include a chewing gum composition having a first

quantity of at least one cannabinoid solubilized by a FABP-carrier protein. In a preferred embodiment, a chewing gum composition may further include a gum base comprising a buffering agent selected from the group consisting of acetates, glycinates, phosphates, carbonates, glycerophosphates, citrates, borates, and mixtures thereof. Additional components may include at least one sweetening agent and at least one flavoring agent. As noted above, in a preferred embodiment, at least one cannabinoid solubilized by a FABP-carrier protein, may be generated in vitro, or in vivo respectively.

(107) In one embodiment, the chewing gum composition described above may include: 0.01 to 1% by weight of at least one FABP-carrier solubilized cannabinoid; 25 to 85% by weight of a gum base; 10 to 35% by weight of at least one sweetening agent; and 1 to 10% by weight of a flavoring agent.

(108) Here, such flavoring agents may include: menthol flavor, eucalyptus, mint flavor and/or L-menthol. Sweetening agents may include one or more of the following: xylitol, sorbitol, isomalt, aspartame, sucralose, acesulfame potassium, and saccharin. Additional preferred embodiment may include a chewing gum having a pharmaceutically acceptable excipient selected from the group consisting of: fillers, disintegrants, binders, lubricants, and antioxidants. The chewing gum composition may further be non-disintegrating and also include one or more coloring and/or flavoring agents.

(109) The invention may further include a composition for a cannabinoid infused solution comprising essentially of: water and/or purified water, at least one cannabinoid solubilized by a FABP-carrier protein and at least one flavoring agent. A FABP-carrier solubilized cannabinoid infused solution of the invention may further include a sweetener selected from the group consisting of: glucose, sucrose, invert sugar, corn syrup, stevia extract powder, stevioside, steviol, aspartame, saccharin, saccharin salts, sucralose, potassium acetosulfam, sorbitol, xylitol, mannitol, erythritol, lactitol, alitame, miraculin, monellin, and thaumatin or a combination of the same. Additional components of the solubilized cannabinoid infused solution may include, but not be limited to: sodium chloride, sodium chloride solution, glycerin, a coloring agent, and a demulcent. As to this last potential component, in certain embodiments, a demulcent may include: pectin, glycerin, honey, methylcellulose, and/or propylene glycol. As noted above, in a preferred embodiment, a FABP-carrier solubilized cannabinoid may include at least one FABP-carrier solubilized cannabinoid wherein such solubilized cannabinoids may be generated in vivo, ex vivo, and/or in vitro respectively.

(110) The invention may further include a composition for a FABP-carrier solubilized cannabinoid infused anesthetic solution having water, or purified water, at least one FABP-carrier solubilized cannabinoid, and at least one oral anesthetic. In a preferred embodiment, an anesthetic may include benzocaine, and/or phenol in a quantity of between 0.1% to 15% volume by weight.

(111) Additional embodiments may include a FABP-carrier solubilized cannabinoid infused anesthetic solution having a sweetener which may be selected from the group consisting of: glucose, sucrose, invert sugar, corn syrup, stevia extract powder, stevioside, steviol, aspartame, saccharin, saccharin salts, sucralose, potassium acetosulfam, sorbitol, xylitol, mannitol, erythritol, lactitol, alitame, miraculin, monellin, and thaumatin or a combination of the same. Additional components of a FABP-carrier solubilized cannabinoid infused solution may include, but not be limited to: sodium chloride, sodium chloride solution, glycerin, a coloring agent, and a demulcent. In a preferred embodiment, a demulcent may be selected from the group consisting of: pectin, glycerin, honey, methylcellulose, and propylene glycol. As noted above, in a preferred embodiment, a FABP-carrier solubilized cannabinoid may include at least one cannabinoid solubilized by a FABP-carrier protein. In this embodiment, such solubilized cannabinoids may have been generated in vivo, ex vivo, and/or in vitro respectively.

(112) The invention may further include a composition for a hard lozenge for rapid delivery of solubilized cannabinoids through the oral mucosa. In this embodiment, such a hard lozenge composition may include: a crystalized sugar base and at least one FABP-carrier solubilized cannabinoid, wherein the hard lozenge has moisture content between 0.1 to 2%. In this embodiment, the solubilized cannabinoid may be added to the sugar base when it is in a liquefied form and prior to the evaporation of the majority of water content. Such a hard lozenge may further be referred to as a candy.

(113) In a preferred embodiment, a crystalized sugar base may be formed from one or more of the following: sucrose, invert sugar, corn syrup, and isomalt or a combination of the same. Additional components may include at least one acidulant. Examples of acidulants may include, but not be limited to: citric acid, tartaric acid, fumaric acid, and malic acid. Additional components may include at least one

pH adjustor. Examples of pH adjustors may include, but not be limited to: calcium carbonate, sodium bicarbonate, and magnesium trisilicate.

(114) In another preferred embodiment, the composition may include at least one anesthetic. Example of such anesthetics may include benzocaine, and phenol. In this embodiment, first quantity of anesthetic may be between 1 mg to 15 mg per lozenge. Additional embodiments may include a quantity of menthol. In this embodiment, such a quantity of menthol may be between 1 mg to 20 mg. The hard lozenge composition may also include a demulcent, for example: pectin, glycerin, honey, methylcellulose, propylene glycol, and glycerin. In this embodiment, a demulcent may be in a quantity between 1 mg to 10 mg. As noted above, in a preferred embodiment, a FABP-carrier solubilized cannabinoid may include at least one cannabinoid solubilized by a FABP-carrier protein. In this embodiment, such solubilized cannabinoid may have been generated in vivo and/or in vitro respectively.

(115) The invention may include a chewable lozenge for rapid delivery of solubilized cannabinoids through the oral mucosa. In a preferred embodiment, the compositions may include: a glycerinated gelatin base, at least one sweetener, and at least one FABP-carrier solubilized cannabinoid dissolved in a first quantity of water. In this embodiment, a sweetener may include a sweetener selected from the group consisting of: glucose, sucrose, invert sugar, corn syrup, stevia extract powder, stevioside, steviol, aspartame, saccharin, saccharin salts, sucralose, potassium acetosulfam, sorbitol, xylitol, mannitol, erythritol, lactitol, alitame, miraculin, monellin, and thaumatin or a combination of the same.

(116) Additional components may include at least one acidulant. Examples of acidulants may include, but not be limited to: citric acid, tartaric acid, fumaric acid, and malic acid. Additional components may include at least one pH adjustor. Examples of pH adjustors may include, but not be limited to: calcium carbonate, sodium bicarbonate, and magnesium trisilicate.

(117) In another preferred embodiment, the composition may include at least one anesthetic. Example of such anesthetics may include benzocaine and phenol. In this embodiment, first quantity of anesthetic may be between 1 mg to 15 mg per lozenge. Additional embodiments may include a quantity of menthol. In this embodiment, such a quantity of menthol may be between 1 mg to 20 mg. The chewable lozenge composition may also include a demulcent, for example: pectin, glycerin, honey, methylcellulose, propylene glycol, and glycerin. In this embodiment, a demulcent may be in a quantity between 1 mg to 10 mg. As noted above, in a preferred embodiment, a FABP-carrier solubilized cannabinoid may include at least one cannabinoid solubilized by a FABP-carrier protein. In this embodiment, such solubilized cannabinoid may be generated in vivo, ex vivo, or in vitro respectively.

(118) The invention may include a soft lozenge for rapid delivery of solubilized cannabinoids through the oral mucosa. In a preferred embodiment, the compositions may include: a polyethylene glycol base, at least one sweetener, and at least one FABP-carrier solubilized cannabinoid, referring generally to a cannabinoid solubilized by a FABP-carrier, dissolved in a first quantity of water. In this embodiment, a sweetener may include sweetener selected from the group consisting of: glucose, sucrose, invert sugar, corn syrup, stevia extract powder, stevioside, steviol, aspartame, saccharin, saccharin salts, sucralose, potassium acetosulfam, sorbitol, xylitol, mannitol, erythritol, lactitol, alitame, miraculin, monellin, and thaumatin or a combination of the same. Additional components may include at least one acidulant. Examples of acidulants may include, but not be limited to: citric acid, tartaric acid, fumaric acid, and malic acid. Additional components may include at least one pH adjustor. Examples of pH adjustors may include, but not be limited to: calcium carbonate, sodium bicarbonate, and magnesium trisilicate.

(119) In another preferred embodiment, the composition may include at least one anesthetic. Example of such anesthetics may include benzocaine and phenol. In this embodiment, first quantity of anesthetic may be between 1 mg to 15 mg per lozenge. Additional embodiments may include a quantity of menthol. In this embodiment, such a quantity of menthol may be between 1 mg to 20 mg. The soft lozenge composition may also include a demulcent, for example: pectin, glycerin, honey, methylcellulose, propylene glycol, and glycerin. In this embodiment, a demulcent may be in a quantity between 1 mg to 10 mg. As noted above, in a preferred embodiment, a FABP-carrier solubilized cannabinoid may include at least one cannabinoid solubilized by a FABP-carrier protein. In this embodiment, such solubilized cannabinoid may be generated in vivo, ex vivo, or in vitro respectively.

(120) In another embodiment, the invention may include a tablet or capsule consisting essentially of a solubilized cannabinoid and a pharmaceutically acceptable excipient. Examples may include solid, semi-

solid, and aqueous excipients such as: maltodextrin, whey protein isolate, xanthan gum, guar gum, diglycerides, monoglycerides, carboxymethyl cellulose, glycerin, gelatin, polyethylene glycol and water-based excipients. In this embodiment, the cannabinoid solubilized by a FABP-carrier protein, may have an improved shelf-life, composition stability, and bioavailability upon injection.

(121) In a preferred embodiment, a solubilized cannabinoid may include at least one cannabinoid solubilized by a FABP-carrier protein. In this embodiment, such solubilized cannabinoids may be generated in vivo, ex vivo, or in vitro respectively. Examples of such in vivo systems being generally described herein, including in plant, as well as cell culture systems including *Cannabis* cell culture, tobacco cell culture, bacterial cell cultures, fungal cell cultures, and yeast cell culture systems. In one embodiment, a tablet or capsule may include an amount of solubilized cannabinoid of 5 milligrams or less. Alternative embodiments may include an amount of solubilized cannabinoid between 5 milligrams and 200 milligrams. Still other embodiments may include a tablet or capsule having an amount of solubilized cannabinoid that is more than 200 milligrams. Still other embodiments may include a tablet or capsule having an amount of solubilized cannabinoid that is more than 500 milligrams.

(122) The invention may further include a method of manufacturing and packaging a solubilized cannabinoid dosage, consisting of the following steps: 1) preparing a fill solution with a desired concentration of one or more FABP-carrier solubilized cannabinoid in a liquid carrier wherein said cannabinoid is dissolved in said liquid carrier; 2) encapsulating said fill solution in capsules; 3) packaging said capsules in a closed packaging system; and 4) removing atmospheric air from the capsules. In one embodiment, the step of removing atmospheric air consists of purging the packaging system with an inert gas, such as, for example, nitrogen gas, such that said packaging system provides a room temperature stable product. In one preferred embodiment, the packaging system may include a plaster package, which may be constructed of material that minimizes exposure to moisture and air.

(123) In one embodiment, a preferred liquid carrier may include a water-based carrier, such as for example an aqueous sodium chloride solution. In a preferred embodiment, a solubilized cannabinoid may include at least one cannabinoid solubilized by a FABP-carrier protein. In this embodiment, such solubilized cannabinoids may be generated in vivo, ex vivo, or in vitro respectively. In one embodiment, a desired solubilized cannabinoid concentration may be about 1-10% w/w, while in other embodiments it may be about 1.5-6.5% w/w. Alternative embodiments may include an amount of solubilized cannabinoid between 5 milligrams and 200 milligrams. Still, other embodiments may include a tablet or capsule having amount of solubilized cannabinoid that is more than 200 milligrams. Other embodiments may include a tablet or capsule having an amount of solubilized cannabinoid that is more than 500 milligrams.

(124) The invention may include an oral pharmaceutical solution, such as a sub-lingual spray having solubilized cannabinoids and a liquid carrier. One embodiment may include a solubilized cannabinoid, 30-33% w/w water, about 50% w/w alcohol, 0.01% w/w butylated hydroxyanisole (BHA) or 0.1% w/w ethylenediaminetetraacetic acid (EDTA) and 5-21% w/w co-solvent, having a combined total of 100%, wherein said co-solvent is selected from the group consisting of propylene glycol, polyethylene glycol, and combinations thereof, and wherein said solubilized cannabinoid is at least one cannabinoid solubilized by a FABP-carrier protein. In an alternative embodiment, such a oral pharmaceutical solution may consist essentially of 0.1 to 5% w/w of said solubilized cannabinoid, about 50% w/w alcohol, 5.5% w/w propylene glycol, 12% w/w polyethylene glycol and 30-33% w/w water. In a preferred composition, the alcohol component may be ethanol.

(125) The invention may include an oral pharmaceutical solution, such as a sublingual spray, consisting essentially of about 0.1% to 1% w/w solubilized cannabinoids, about 50% w/w alcohol, 5.5% w/w propylene glycol, 12% w/w polyethylene glycol, 30-33% w/w water, 0.01% w/w butylated hydroxyanisole, having a combined total of 100%, and wherein said solubilized cannabinoid is at least one cannabinoid solubilized by a FABP-carrier protein, that may be further generated in vivo, ex vivo, or in vitro respectively. In an alternative embodiment, such a oral pharmaceutical solution may consist essentially of 0.54% w/w solubilized cannabinoid, 31.9% w/w water, 12% w/w polyethylene glycol 400, 5.5% w/w propylene glycol, 0.01% w/w butylated hydroxyanisole, 0.05% w/w sucralose, and 50% w/w alcohol, wherein the a the alcohol components may be ethanol.

(126) The invention may include a solution for nasal and/or sublingual administration of a solubilized

cannabinoid including: 1) an excipient of propylene glycol, ethanol anhydrous, or a mixture of both; and 2) a solubilized cannabinoid which may include at least one cannabinoid solubilized by a FABP-carrier protein, that may be further generated in vivo, ex vivo, or in vitro respectively. In a preferred embodiment, the composition may further include a topical decongestant, which may include phenylephrine hydrochloride, Oxymetazoline hydrochloride, and Xylometazoline in certain preferred embodiments. The composition may further include an antihistamine, and/or a steroid. Preferably, the steroid component is a corticosteroid selected from the group consisting of: neclomethasone dipropionate, budesonide, ciclesonide, flunisolide, fluticasone furoate, fluticasone propionate, mometasone, and triamcinolone acetonide. In alternative embodiments, the solution for nasal and/or sublingual administration of a solubilized cannabinoid may further comprise at least one of the following: benzalkonium chloride solution, benzyl alcohol, boric acid, purified water, sodium borate, polysorbate 80, phenylethyl alcohol, microcrystalline cellulose, carboxymethylcellulose sodium, dextrose, dipasic, sodium phosphate, edetate disodium, monobasic sodium phosphate, and propylene glycol.

(127) The invention may further include an aqueous solution for nasal and/or sublingual administration of a solubilized cannabinoid comprising: a water and/or saline solution; and a solubilized cannabinoid which may include at least one cannabinoid solubilized by a FABP-carrier protein that may be further generated in vivo, ex vivo, or in vitro, respectively. In a preferred embodiment, the composition may further include a topical decongestant, which may include phenylephrine hydrochloride, Oxymetazoline hydrochloride, and Xylometazoline in certain preferred embodiments. The composition may further include an antihistamine and/or a steroid. Preferably, the steroid component is a corticosteroid selected from the group consisting of: neclomethasone dipropionate, budesonide, ciclesonide, flunisolide, fluticasone furoate, fluticasone propionate, mometasone, and triamcinolone acetonide. In alternative embodiments, the aqueous solution may further comprise at least one of the following: benzalkonium chloride solution, benzyl alcohol, boric acid, purified water, sodium borate, polysorbate 80, phenylethyl alcohol, microcrystalline cellulose, carboxymethylcellulose sodium, dextrose, dipasic, sodium phosphate, edetate disodium, monobasic sodium phosphate, or propylene glycol.

(128) The invention may include a topical formulation for the transdermal delivery of solubilized cannabinoids. In a preferred embodiment, a topical formulation for the transdermal delivery of solubilized cannabinoids which may include at least one cannabinoid solubilized by a FABP-carrier protein, and a pharmaceutically acceptable excipient. The solubilized cannabinoids may be generated in vivo, ex vivo, or in vitro respectively. Preferably a pharmaceutically acceptable excipient may include one or more: gels, ointments, cataplasms, poultices, pastes, creams, lotions, plasters and jellies or even polyethylene glycol. Additional embodiments may further include one or more of the following components: a quantity of capsaicin; a quantity of benzocaine; a quantity of lidocaine; a quantity of camphor; a quantity of benzoin resin; a quantity of methylsalicylate; a quantity of triethanolamine salicylate; a quantity of hydrocortisone; or a quantity of salicylic acid.

(129) The invention may include a gel for transdermal administration of a solubilized cannabinoid which may include at least one cannabinoid solubilized by a FABP-carrier protein, and which may be generated in vivo, ex vivo, or in vitro. In this embodiment, the mixture preferably contains from 15% to about 90% ethanol, about 10% to about 60% buffered aqueous solution or water, about 0.1 to about 25% propylene glycol, from about 0.1 to about 20% of a gelling agent, from about 0.1 to about 20% of a base, from about 0.1 to about 20% of an absorption enhancer and from about 1% to about 25% polyethylene glycol, and a solubilized cannabinoid as generally described herein.

(130) In another embodiment, the invention may further include a transdermal composition having a pharmaceutically effective amount of a solubilized cannabinoid for delivery of the cannabinoid to the bloodstream of a user. This transdermal composition may include a pharmaceutically acceptable excipient and at least one FABP-carrier solubilized cannabinoid, which may include at least one cannabinoid solubilized by a FABP-carrier protein, and which may be generated in vivo, ex vivo, or in vitro, wherein the solubilized cannabinoid is capable of diffusing from the composition into the bloodstream of the user. In a preferred embodiment, a pharmaceutically acceptable excipient to create a transdermal dosage form selected from the group consisting of: gels, ointments, cataplasms, poultices, pastes, creams, lotions, plasters and jellies. The transdermal composition may further include one or

more surfactants. In one preferred embodiment, the surfactant may include a surfactant-lecithin organogel, which may further be present in an amount of between about 95% and about 98% w/w. In an alternative embodiment, a surfactant-lecithin organogel comprises lecithin and PPG-2 myristyl ether propionate and/or high molecular weight polyacrylic acid polymers. The transdermal composition may further include a quantity of isopropyl myristate.

(131) The invention may further include transdermal composition having one or more permeation enhancers to facilitate transfer of the solubilized cannabinoid across a dermal layer. In a preferred embodiment, a permeation enhancer may include one or more of the following: propylene glycol monolaurate, diethylene glycol monoethyl ether, an oleoyl macrogolglyceride, a caprylocaproyl macrogolglyceride, and an oleyl alcohol.

(132) The invention may also include a liquid cannabinoid liniment composition consisting of water, isopropyl alcohol solution, and a solubilized cannabinoid, which may include at least one cannabinoid solubilized by a FABP-carrier protein, and which may be generated in vivo, ex vivo, or in vitro. This liquid cannabinoid liniment composition may further include approximately 97.5% to about 99.5% by weight of 70% isopropyl alcohol solution and from about 0.5% to about 2.5% by weight of a solubilized cannabinoid mixture.

(133) Based on the improved solubility and other physical properties, as well as cost advantages, improved cannabinoid affinity and capacity, extended shelf-life, and scalability of the inventions in vivo, ex vivo, or in vitro solubilized cannabinoid production platform, the invention may include one or more commercial infusions. For example, commercially available products, such as a lip balm, soap, shampoos, lotions, creams, and cosmetics may be infused with one or more solubilized cannabinoids.

(134) The invention may further include a novel composition that may be used to supplement a cigarette or other tobacco-based product. In this embodiment, the composition may include at least one FABP-carrier solubilized cannabinoid in a powder as already described, or dissolved in an aqueous solution. This aqueous solution may be introduced to a tobacco product, such as a cigarette and/or a tobacco leaf such that the aqueous solution may evaporate generating a cigarette and/or a tobacco leaf that contains the aforementioned solubilized cannabinoid(s), which may further have been generated in vivo as generally described herein.

(135) In one embodiment, the invention may include one or more methods of treating a medical condition in a mammal. In this embodiment, the novel method may include of administering a therapeutically effective amount of a solubilized cannabinoid, such as an in vivo, ex vivo, or in vitro cannabinoid solubilized by a FABP-carrier protein, wherein the medical condition is selected from the group consisting of: obesity, post-traumatic stress syndrome, anorexia, nausea, emesis, pain, wasting syndrome, HIV-wasting, chemotherapy induced nausea and vomiting, alcohol use disorders, anti-tumor, amyotrophic lateral sclerosis, glioblastoma multiforme, glioma, increased intraocular pressure, glaucoma, *Cannabis* use disorders, Tourette's syndrome, dystonia, multiple sclerosis, inflammatory bowel disorders, arthritis, dermatitis, Rheumatoid arthritis, systemic lupus erythematosus, anti-inflammatory, anti-convulsant, anti-psychotic, anti-oxidant, neuroprotective, anti-cancer, immunomodulatory effects, peripheral neuropathic pain, neuropathic pain associated with post-herpetic neuralgia, diabetic neuropathy, shingles, burns, actinic keratosis, oral cavity sores and ulcers, post-episiotomy pain, psoriasis, pruritis, contact dermatitis, eczema, bullous dermatitis herpetiformis, exfoliative dermatitis, mycosis fungoides, pemphigus, severe erythema multiforme (e.g., Stevens-Johnson syndrome), seborrheic dermatitis, ankylosing spondylitis, psoriatic arthritis, Reiter's syndrome, gout, chondrocalcinosis, joint pain secondary to dysmenorrhea, fibromyalgia, musculoskeletal pain, neuropathic-postoperative complications, polymyositis, acute nonspecific tenosynovitis, bursitis, epicondylitis, post-traumatic osteoarthritis, synovitis, and juvenile rheumatoid arthritis. In a preferred embodiment, the pharmaceutical composition may be administered by a route selected from the group consisting of: transdermal, topical, oral, buccal, sublingual, intra-venous, intra-muscular, vaginal, rectal, ocular, nasal and follicular. The amount of solubilized cannabinoids may be a therapeutically effective amount, which may be determined by the patient's age, weight, medical condition cannabinoid-delivered, route of delivery, and the like. In one embodiment, a therapeutically effective amount may be 50 mg or less of a solubilized cannabinoid. In another embodiment, a therapeutically effective amount may be 50 mg or more of a solubilized cannabinoid.

(136) It should be noted that for any of the above composition, unless otherwise stated, an effective amount of solubilized cannabinoids may include amounts between: 0.01 mg to 0.1 mg; 0.01 mg to 0.5 mg; 0.01 mg to 1 mg; 0.01 mg to 5 mg; 0.01 mg to 10 mg; 0.01 mg to 25 mg; 0.01 mg to 50 mg; 0.01 mg to 75 mg; 0.01 mg to 100 mg; 0.01 mg to 125 mg; 0.01 mg to 150 mg; 0.01 mg to 175 mg; 0.01 mg to 200 mg; 0.01 mg to 225 mg; 0.01 mg to 250 mg; 0.01 mg to 275 mg; 0.01 mg to 300 mg; 0.01 mg to 325 mg; 0.01 mg to 350 mg; 0.01 mg to 375 mg; 0.01 mg to 400 mg; 0.01 mg to 425 mg; 0.01 mg to 450 mg; 0.01 mg to 475 mg; 0.01 mg to 500 mg; 0.01 mg to 525 mg; 0.01 mg to 550 mg; 0.01 mg to 575 mg; 0.01 mg to 600 mg; 0.01 mg to 625 mg; 0.01 mg to 650 mg; 0.01 mg to 675 mg; 0.01 mg to 700 mg; 0.01 mg to 725 mg; 0.01 mg to 750 mg; 0.01 mg to 775 mg; 0.01 mg to 800 mg; 0.01 mg to 825 mg; 0.01 mg to 850 mg; 0.01 mg to 875 mg; 0.01 mg to 900 mg; 0.01 mg to 925 mg; 0.01 mg to 950 mg; 0.01 mg to 975 mg; 0.01 mg to 1000 mg; 0.01 mg to 2000 mg; 0.01 mg to 3000 mg; 0.01 mg to 4000 mg; 0.01 mg to 5000 mg; 0.01 mg to 0.1 mg/kg; 0.01 mg to 0.5 mg/kg; 0.01 mg to 1 mg/kg; 0.01 mg to 5 mg/kg; 0.01 mg to 10 mg/kg; 0.01 mg to 25 mg/kg; 0.01 mg to 50 mg/kg; 0.01 mg to 75 mg/kg; and 0.01 mg to 100 mg/kg.

(137) The FABP solubilized cannabinoid compounds of the present invention are useful for a variety of therapeutic applications. For example, the compounds are useful for treating or alleviating symptoms of diseases and disorders involving CB1 and CB2 receptors, including appetite loss, nausea and vomiting, pain, multiple sclerosis and epilepsy. For example, they may be used to treat pain (i.e. as analgesics) in a variety of applications including but not limited to pain management. In additional embodiments, such solubilized cannabinoids may be used as an appetite suppressant. Additional embodiments may include administering the FABP solubilized cannabinoid compounds of the invention.

(138) By “treating,” the present inventors mean that the compound is administered in order to alleviate symptoms of the disease or disorder being treated. Those of skill in the art will recognize that the symptoms of the disease or disorder that is treated may be completely eliminated or may simply be lessened. Further, the compounds may be administered in combination with other drugs or treatment modalities, such as with chemotherapy or other cancer-fighting drugs.

(139) Implementation may generally involve identifying patients suffering from the indicated disorders and administering the compounds of the present invention in an acceptable form by an appropriate route. The exact dosage to be administered may vary depending on the age, gender, weight, and overall health status of the individual patient, as well as the precise etiology of the disease. However, in general, for administration in mammals (e.g. humans), dosages in the range of from about 0.01 to about 300 mg of compound per kg of body weight per 24 hr., and more preferably about 0.01 to about 100 mg of compound per kg of body weight per 24 hr., may be effective.

(140) Administration may be oral or parenteral, including intravenously, intramuscularly, subcutaneously, intradermal injection, intraperitoneal injection, etc., or by other routes (e.g. transdermal, sublingual, oral, rectal and buccal delivery, inhalation of an aerosol, etc.). In a preferred embodiment of the invention, the solubilized cannabinoid are provided orally or intravenously.

(141) The compounds, for example a cannabinoid solubilized by a FABP-carrier, may be administered in the pure form or in a pharmaceutically acceptable formulation including suitable elixirs, binders, and the like (generally referred to as a “secondary carrier”) or as pharmaceutically acceptable salts (e.g. alkali metal salts such as sodium, potassium, calcium or lithium salts, ammonium, etc.) or other complexes. It should be understood that the pharmaceutically acceptable formulations include liquid and solid materials conventionally utilized to prepare both injectable dosage forms and solid dosage forms such as tablets and capsules and aerosolized dosage forms. In addition, the compounds may be formulated with aqueous or oil based vehicles. Water may be used as the carrier for the preparation of compositions (e.g. injectable compositions), which may also include conventional buffers and agents to render the composition isotonic. Other potential additives and other materials (preferably those which are generally regarded as safe [GRAS]) include: colorants; flavorings; surfactants (TWEEN, oleic acid, etc.); solvents, stabilizers, elixirs, and binders or encapsulants (lactose, liposomes, etc). Solid diluents and excipients include lactose, starch, conventional disintegrating agents, coatings and the like. Preservatives such as methyl paraben or benzalkonium chloride may also be used. Depending on the formulation, it is expected that the active composition will consist of about 1% to about 99% of the composition and the secondary carrier will constitute about 1% to about 99% of the composition. The pharmaceutical compositions of

the present invention may include any suitable pharmaceutically acceptable additives or adjuncts to the extent that they do not hinder or interfere with the therapeutic effect of the active compound.

(142) The administration of the solubilized compounds of the present invention may be intermittent, bolus dose, or at a gradual or continuous, constant, or controlled rate to a patient. In addition, the time of day and the number of times per day that the pharmaceutical formulation is administered may vary and are best determined by a skilled practitioner such as a physician. Further, the effective dose can vary depending upon factors such as the mode of delivery, gender, age, and other conditions of the patient, as well as the extent or progression of the disease. The compounds may be provided alone, in a mixture containing two or more of the compounds, or in combination with other medications or treatment modalities.

(143) Notably, as used herein, “FABP,” or “FABP-carrier” or “FABP-carrier protein” explicitly includes any FABPs that have affinity for a cannabinoid, terpene or other short-chain fatty acid phenolic compounds. Additionally, “FABP-carrier” may be generically used to explicitly describe FABP-like proteins, regardless of family or classification, that exhibit an affinity for a cannabinoid, terpene or other short-chain fatty acid phenolic compounds. The term “FABP-carrier” explicitly encompasses FABP-like proteins, FABP homologs, and FABP orthologs having affinity for a cannabinoid, terpene or other short-chain fatty acid phenolic compounds. As used herein a “FABP,” or “FABP-carrier” or “FABP-carrier protein” specifically encompasses wild-type, as well as “engineered” or “engineered FABP-carrier” or “engineered FABP-carrier proteins.” As used herein, a “engineered” or “engineered FABP-carrier” or “engineered FABP-carrier protein” may be a FABP-carrier protein that may bind or, to have some affinity for a target compound, such as a cannabinoid, that may be further modified to have enhanced affinity, or additional affinity sites for the target compound, such as a mutation that may modulate a characteristic of that protein, such as cannabinoid binding characteristic, or antigenicity characteristic, or for example may be modified to form a chimera or fusions protein through the addition of a secretion signal. In one embodiment, as “engineered” may be established through bioinformatics approaches such as computational docking, binding affinity estimation, and molecular dynamics simulations. Additional embodiments may include the site-directed mutations of an “engineered,” “engineered FABP-carrier,” or “engineered carrier protein” may also be determined by through bioinformatics approaches such as computational docking, binding affinity estimation, and molecular dynamics simulations as well as in high-throughput binding assays and the like. According to some embodiments of the present invention, a suitable computational platform for executing the method presented herein, in one embodiment may include the generation of an engineered FABP-carrier, may include the Rosetta™ software suite platform, publically available from the “Rosetta@home” at the Baker laboratory, University of Washington, U.S.A. Briefly, Rosetta™ is a molecular modeling software package for understanding protein structures, protein design, protein docking, protein-DNA and protein-protein interactions. The Rosetta software contains multiple functional modules, including RosettaAbinitio, RosettaDesign, RosettaDock, RosettaAntibody, RosettaFragments, RosettaNMR, RosettaDNA, RosettaRNA, RosettaLigand, RosettaSymmetry, and more. According to some embodiments of the present invention, a suitable computational platform for executing the method presented herein, is the SMINA software suite platform, which generally encompasses a version of AutoDock Vina specially optimized to support high-throughput scoring and user-specified custom scoring functions. Additional embodiment may further include software suites for AutoDock Vina, and idock.

(144) As used herein, the term “enhanced” as used to described the binding of a cannabinoid to a FABP-carrier, or engineered FABP-carrier means having a binding affinity greater than wild-type binding affinity.

(145) As used herein, a “cannabinoid” is a chemical compound (such as cannabinol, THC or cannabidiol) that is found in the plant species *Cannabis* among others like: *Echinacea*; *Acmella Oleracea*; *Helichrysum umbraculigerum*; *Radula marginata* (Liverwort) and *Theobroma cacao*, and metabolites and synthetic analogues thereof that may or may not have psychoactive properties. Cannabinoids therefore include (without limitation) compounds (such as THC) that have high affinity for the cannabinoid receptor (for example $K_i < 250$ nM), and compounds that do not have significant affinity for the cannabinoid receptor (such as cannabidiol, CBD). Cannabinoids also include compounds that have a characteristic dibenzopyran ring structure (of the type seen in THC) and cannabinoids which do not

possess a pyran ring (such as cannabidiol). Hence a partial list of cannabinoids includes THC, CBD, dimethyl heptylpentyl cannabidiol (DMHP-CBD), 6,12-dihydro-6-hydroxy-cannabidiol (described in U.S. Pat. No. 5,227,537, incorporated by reference); (3S,4R)-7-hydroxy- Δ^6 -tetrahydrocannabinol homologs and derivatives described in U.S. Pat. No. 4,876,276, incorporated by reference; (+)-4-[4-DMH-2,6-diacetoxy-phenyl]-2-carboxy-6,6-dimethylbicyclo[3.1.1]hept-2-en, and other 4-phenylpinene derivatives disclosed in U.S. Pat. No. 5,434,295, which is incorporated by reference; and cannabidiol (–) (CBD) analogs such as (–)CBD-monomethylether, (–)CBD dimethyl ether; (–)CBD diacetate; (–)3'-acetyl-CBD monoacetate; and $\pm$ AF11, all of which are disclosed in Consroe et al., J. Clin. Pharmacol. 21:428S-436S, 1981, which is also incorporated by reference. Many other cannabinoids are similarly disclosed in Agurell et al., Pharmacol. Rev. 38:31-43, 1986, which is also incorporated by reference.

(146) As claimed herein, the term “cannabinoid” may also be generically applied to describe all cannabinoids, short-chain fatty acid phenolic compounds, endocannabinoids, phytocannabinoids, as well as terpenes that have affinity for one or more FABP-carrier proteins, or their homologs as generally described herein. Moreover, as used herein, the term “solubilized cannabinoid” describes a “cannabinoid,” that binds to or interacts with one or more FABP-carrier proteins, or their homologs as generally described herein.

(147) Examples of cannabinoids are tetrahydrocannabinol, cannabidiol, cannabigerol, cannabichromene, cannabicyclol, cannabivarin, cannabielsoin, cannabicitran, cannabigerolic acid, cannabigerolic acid monomethylether, cannabigerol monomethylether, cannabigerovarinic acid, cannabigerovarin, cannabichromenic acid, cannabichromevarinic acid, cannabichromevarin, cannabidolic acid, cannabidiol monomethylether, cannabidiol-C4, cannabidivarinic acid, cannabidiorcol, delta-9-tetrahydrocannabinolic acid A, delta-9-tetrahydrocannabinolic acid B, delta-9-tetrahydrocannabinolic acid-C4, delta-9-tetrahydrocannabivarinic acid, delta-9-tetrahydrocannabivarin, delta-9-tetrahydrocannabiorcolic acid, delta-9-tetrahydrocannabiorcol, delta-7-cis-iso-tetrahydrocannabivarin, delta-8-tetrahydrocannabinolic acid, delta-8-tetrahydrocannabinol, cannabicyclolic acid, cannabicyclovarin, cannabielsoic acid A, cannabielsoic acid B, cannabinolic acid, cannabinol methylether, cannabinol-C4, cannabinol-C2, cannabiorcol, 10-ethoxy-9-hydroxy-delta-6a-tetrahydrocannabinol, 8,9-dihydroxy-delta-6a-tetrahydrocannabinol, cannabitriolvarin, ethoxy-cannabitriolvarin, dehydrocannabifuran, cannabifuran, cannabichromanone, cannabicitran, 10-oxo-delta-6a-tetrahydrocannabinol, delta-9-cis-tetrahydrocannabinol, 3,4,5,6-tetrahydro-7-hydroxy-alpha-alpha-2-trimethyl-9-n-propyl-2,6-methano-2H-1-benzoxocin-5-methanol-cannabiripsol, trihydroxy-delta-9-tetrahydrocannabinol, and cannabinol. Examples of cannabinoids within the context of this disclosure include tetrahydrocannabinol and cannabidiol.

(148) The term “endocannabinoid” refers to compounds including arachidonoyl ethanolamide (anandamide, AEA), 2-arachidonoyl ethanolamide (2-AG), 1-arachidonoyl ethanolamide (1-AG), and docosahexaenoyl ethanolamide (DHEA, synaptamide), oleoyl ethanolamide (OEA), eicsapentaenoyl ethanolamide, prostaglandin ethanolamide, docosahexaenoyl ethanolamide, linolenoyl ethanolamide, 5(Z),8(Z),11(Z)-eicosatrienoic acid ethanolamide (mead acid ethanolamide), heptadecanoyl ethanolamide, stearoyl ethanolamide, docosaenoyl ethanolamide, nervonoyl ethanolamide, tricosanoyl ethanolamide, lignoceroyl ethanolamide, myristoyl ethanolamide, pentadecanoyl ethanolamide, palmitoleoyl ethanolamide, docosahexaenoic acid (DHA). Particularly preferred endocannabinoids are AEA, 2-AG, 1-AG, and DHEA.

(149) Terpenoids a.k.a. isoprenoids, are a large and diverse class of naturally occurring organic chemicals similar to terpenes, derived from five-carbon isoprene units assembled and modified in a number of varying configurations. Most are multi-cyclic structures that differ from one another not only in functional groups but also in their basic carbon skeletons. Terpenoids are essential for plant metabolism, influencing general development, herbivory defense, pollination and stress response. These compounds have been extensively used as flavoring and scenting agents in cosmetics, detergents, food and pharmaceutical products. They also display multiple biological activities in humans, such as anti-inflammatory, anti-microbial, antifungal and antiviral.

(150) *Cannabis* terpenoid profiles define the aroma of each plant and share the same precursor (geranyl pyrophosphate) and the same synthesis location (glandular trichomes) as phytocannabinoids. The terpenoids most commonly found in *Cannabis* extracts include: limonene, myrcene, alpha-pinene,

linalool, beta-caryophyllene, caryophyllene oxide, nerolidol, and phytol. Terpenoids are mainly synthesized in two metabolic pathways: mevalonic acid pathway (a.k.a. HMG-CoA reductase pathway, which takes place in the cytosol) and MEP/DOXP pathway (a.k.a. The 2-C-methyl-D-erythritol 4-phosphate/1-deoxy-D-xylulose 5-phosphate pathway, non-mevalonate pathway, or mevalonic acid-independent pathway, which takes place in plastids). Geranyl pyrophosphate (GPP), which is used by *Cannabis* plants to produce cannabinoids, is formed by condensation of dimethylallyl pyrophosphate (DMAPP) and isopentenyl pyrophosphate (IPP) via the catalysis of GPP synthase. Alternatively, DMAPP and IPP are ligated by FPP synthase to produce farnesyl pyrophosphate (FPP), which can be used to produce sesquiterpenoids. Geranyl pyrophosphate (GPP) can also be converted into monoterpenoids by limonene synthase.

(151) Some examples of terpenes, and their classification, are as follows. Hemiterpenes: Examples of hemiterpenes, which do not necessarily have an odor, are 2-methyl-1,3-butadiene, hemialboside, and hymenoside. Monoterpenes: pinene, α -pinene, β -pinene, cis-pinane, trans-pinane, cis-pinanol, trans-pinanol (Erman and Kane (2008) Chem. Biodivers. 5:910-919), limonene; linalool; myrcene; eucalyptol; α -phellandrene; β -phellandrene; α -ocimene; β -ocimene, cis-ocimene, ocimene, Δ -3-carene; fenchol; sabinene, borneol, isoborneol, camphene, camphor, phellandrene, α -phellandrene, α -terpinene, geraniol, linalool, nerol, menthol, myrcene, terpinolene, α -terpinolene, β -terpinolene, γ -terpinolene, Δ -terpinolene, α -terpineol, and trans-2-pinanol. Sesquiterpenes: caryophyllene, caryophyllene oxide, humulene, α -humulene, α -bisabolene; β -bisabolene; santalol; selinene; nerolidol, bisabolol; α -cedrene, β -cedrene, β -eudesmol, eudesm-7(11)-en-4-ol, selina-3,7(11)-diene, guaial, valencene, α -guaiene, β -guaiene, Δ -guaiene, guaene, farnesene, α -farnesene, β -farnesene, elemene, α -elemene, β -elemene, γ -elemene, Δ -elemene, germacrene, germacrene A, germacrene B, germacrene C, germacrene D, and germacrene E. Diterpenes: oridonin, phytol, and isophytol. Triterpenes: ursolic acid, oleanolic acid. Terpenoids, also known as isoprenoids, are a large and diverse class of naturally occurring organic chemicals similar to terpenes, derived from five-carbon isoprene units assembled and modified in a number of ways. Most are multicyclic structures that differ from one another not only in functional groups but also in their basic carbon skeletons. Plant terpenoids are used extensively for their aromatic qualities.

(152) The term “prodrug” refers to a precursor of a biologically active pharmaceutical agent (drug). Prodrugs must undergo a chemical or a metabolic conversion to become a biologically active pharmaceutical agent. A prodrug can be converted ex vivo to the biologically active pharmaceutical agent by chemical transformative processes. In vivo, a prodrug is converted to the biologically active pharmaceutical agent by the action of a metabolic process, an enzymatic process, or a degradative process that removes the prodrug moiety to form the biologically active pharmaceutical agent.

(153) As used herein, the term “transformation” refers to the transfer of one or more nucleic acid molecule(s) into a cell. A plant is “transformed” or “genetically modified” by a nucleic acid molecule transduced into the plant when the nucleic acid molecule becomes stably replicated by the plant. As used herein, the term “transformation” or “genetically modified” encompasses all techniques by which a nucleic acid molecule can be introduced into an organism or cell. A protein has “homology” or is “homologous” to a second protein if the amino acid sequence encoded by a gene has a similar amino acid sequence to that of the second gene. Alternatively, a protein has homology to a second protein if the two proteins have “similar” amino acid sequences. (Thus, the term “homologous proteins” is defined to mean that the two proteins have similar amino acid sequences). More specifically, in certain embodiments, the term “homologous” with regard to a contiguous nucleic acid sequence, refers to contiguous nucleotide sequences that hybridize under appropriate conditions to the reference nucleic acid sequence. For example, homologous sequences may have from about 75%-100, or more generally 80% to 100% sequence identity, such as about 81%; about 82%; about 83%; about 84%; about 85%; about 86%; about 87%; about 88%; about 89%; about 90%; about 91%; about 92%; about 93%; about 94% about 95%; about 96%; about 97%; about 98%; about 98.5%; about 99%; about 99.5%; and about 100%. The property of substantial homology is closely related to specific hybridization. For example, a nucleic acid molecule is specifically hybridizable when there is a sufficient degree of complementarity to avoid non-specific binding of the nucleic acid to non-target sequences under conditions where specific binding is desired, for example, under stringent hybridization conditions. For example, in one embodiment, SEQ ID NO. 1 is 84% homologous to *Homo sapiens* FABP1 identified as SEQ ID NO. 21), and would fall within

the range of a homolog. In another embodiment, expression optimization, for example for a mammalian FABP, to be expressed in yeast may be considered homologous and having a variable sequence identity due to the variable codon positions. Additional embodiments may also include homology to include redundant nucleotide codons.

(154) The term “homolog”, used with respect to an original enzyme or gene of a first family or species, refers to distinct enzymes or genes of a second family or species which are determined by functional, structural or genomic analyses to be an enzyme or gene of the second family or species which corresponds to the original enzyme or gene of the first family or species. Most often, homologs will have functional, structural or genomic similarities. Techniques are known by which homologs of an enzyme or gene can readily be cloned using genetic probes and PCR. Identity of cloned sequences as homolog can be confirmed using functional assays and/or by genomic mapping of the genes.

(155) The term “operably linked,” when used in reference to a regulatory sequence and a coding sequence, means that the regulatory sequence affects the expression of the linked coding sequence. “Regulatory sequences,” or “control elements,” refer to nucleotide sequences that influence the timing and level/amount of transcription, RNA processing or stability, or translation of the associated coding sequence. Regulatory sequences may include promoters; translation leader sequences; introns; enhancers; stem-loop structures; repressor binding sequences; termination sequences; polyadenylation recognition sequences; etc. Particular regulatory sequences may be located upstream and/or downstream of a coding sequence operably linked thereto. Also, particular regulatory sequences operably linked to a coding sequence may be located on the associated complementary strand of a double-stranded nucleic acid molecule.

(156) As used herein, the term “promoter” refers to a region of DNA that may be upstream from the start of transcription, and that may be involved in recognition and binding of RNA polymerase and other proteins to initiate transcription. A promoter may be operably linked to a coding sequence for expression in a cell, or a promoter may be operably linked to a nucleotide sequence encoding a signal sequence which may be operably linked to a coding sequence for expression in a cell. An “inducible” promoter may be a promoter which may be under environmental control. Tissue-specific, tissue-preferred, cell type specific, and inducible promoters constitute the class of “non-constitutive” promoters. A “constitutive” promoter is a promoter which may be active under most environmental conditions or in most cell or tissue types.

(157) The term “vector” refers to some means by which DNA, RNA, a protein, or polypeptide can be introduced into a host. The polynucleotides, protein, and polypeptide which are to be introduced into a host can be therapeutic or prophylactic in nature; can encode or be an antigen; or can be regulatory in nature, etc. There are various types of vectors including virus, plasmid, bacteriophages, cosmids, and bacteria.

(158) As is known in the art, different organisms preferentially utilize different codons for generating polypeptides. Such “codon usage” preferences may be used in the design of nucleic acid molecules encoding the proteins of the invention in order to optimize expression in a particular host cell system.

(159) An “expression vector” is nucleic acid capable of replicating in a selected host cell or organism. An expression vector can replicate as an autonomous structure, or alternatively can integrate, in whole or in part, into the host cell chromosomes or the nucleic acids of an organelle, or it is used as a shuttle for delivering foreign DNA to cells, and thus replicate along with the host cell genome. Thus, an expression vector are polynucleotides capable of replicating in a selected host cell, organelle, or organism, e.g., a plasmid, virus, artificial chromosome, nucleic acid fragment, and for which certain genes on the expression vector (including genes of interest) are transcribed and translated into a polypeptide or protein within the cell, organelle or organism; or any suitable construct known in the art, which comprises an “expression cassette.” In contrast, as described in the examples herein, a “cassette” is a polynucleotide containing a section of an expression vector of this invention. The use of a cassette assists in the assembly of the expression vectors. An expression vector is a replicon, such as plasmid, phage, virus, chimeric virus, or cosmid, and which contains the desired polynucleotide sequence operably linked to the expression control sequence(s).

(160) A polynucleotide sequence is operably linked to an expression control sequence(s) (e.g., a promoter and, optionally, an enhancer) when the expression control sequence controls and regulates the

transcription and/or translation of that polynucleotide sequence.

(161) Unless otherwise indicated, a particular nucleic acid sequence also implicitly encompasses conservatively modified variants thereof (e.g., degenerate codon substitutions), the complementary (or complement) sequence, and the reverse complement sequence, as well as the sequence explicitly indicated. Specifically, degenerate codon substitutions may be achieved by generating sequences in which the third position of one or more selected (or all) codons is substituted with mixed-base and/or deoxyinosine residues (see e.g., Batzer et al., *Nucleic Acid Res.* 19:5081 (1991); Ohtsuka et al., *J. Biol. Chem.* 260:2605-2608 (1985); and Rossolini et al., *Mol. Cell. Probes* 8:91-98 (1994)). Because of the degeneracy of nucleic acid codons, one can use various different polynucleotides to encode identical polypeptides. The Table below, contains information about which nucleic acid codons encode which amino acids.

(162) Amino Acid Nucleic Acid Codons

(163) TABLE-US-00004 Amino Acid Nucleic Acid Codons Ala/A GCT, GCC, GCA, GCG Arg/R CGT, CGC, CGA, CGG, AGA, AGG Asn/N AAT, AAC Asp/D GAT, GAC Cys/C TGT, TGC Gln/Q CAA, CAG Glu/E GAA, GAG Gly/G GGT, GGC, GGA, GGG His/H CAT, CAC Ile/I ATT, ATC, ATA Leu/L TTA, TTG, CTT, CTC, CTA, CTG Lys/K AAA, AAG Met/M ATG Phe/F TTT, TTC Pro/P CCT, CCC, CCA, CCG Ser/S TCT, TCC, TCA, TCG, AGT, AGC Thr/T ACT, ACC, ACA, ACG Trp/W TGG Tyr/Y TAT, TAC Val/V GTT, GTC, GTA, GTG

(164) Moreover, because the proteins are described herein, one can chemically synthesize a polynucleotide which encodes these polypeptides described herein. Oligonucleotides and polynucleotides that are not commercially available can be chemically synthesized e.g., according to the solid phase phosphoramidite triester method first described by Beaucage and Caruthers, *Tetrahedron Letts.* 22:1859-1862 (1981), or using an automated synthesizer, as described in Van Devanter et al., *Nucleic Acids Res.* 12:6159-6168 (1984). Other methods for synthesizing oligonucleotides and polynucleotides are known in the art. Purification of oligonucleotides is by either native acrylamide gel electrophoresis or by anion-exchange HPLC as described in Pearson & Reanier, *J. Chrom.* 255:137-149 (1983).

(165) The term “plant” or “plant system” includes whole plants, plant organs, progeny of whole plants or plant organs, embryos, somatic embryos, embryo-like structures, protocorms, protocorm-like bodies (PLBs), and culture and/or suspensions of plant cells. Plant organs comprise, e.g., shoot vegetative organs/structures (e.g., leaves, stems and tubers), roots, flowers and floral organs/structures (e.g., bracts, sepals, petals, stamens, carpels, anthers and ovules), seed (including embryo, endosperm, and seed coat) and fruit (the mature ovary), plant tissue (e.g., vascular tissue, ground tissue, and the like) and cells (e.g., guard cells, egg cells, trichomes and the like). The invention may also include Cannabaceae and other *Cannabis* strains, such as *C. sativa* generally.

(166) The term “expression,” as used herein, or “expression of a coding sequence” (for example, a gene or a transgene) refer to the process by which the coded information of a nucleic acid transcriptional unit (including, e.g., genomic DNA or cDNA) is converted into an operational, non-operational, or structural part of a cell, often including the synthesis of a protein. Gene expression can be influenced by external signals; for example, exposure of a cell, tissue, or organism to an agent that increases or decreases gene expression. Expression of a gene can also be regulated anywhere in the pathway from DNA to RNA to protein. Regulation of gene expression occurs, for example, through controls acting on transcription, translation, RNA transport and processing, degradation of intermediary molecules such as mRNA, or through activation, inactivation, compartmentalization, or degradation of specific protein molecules after they have been made, or by combinations thereof. Gene expression can be measured at the RNA level or the protein level by any method known in the art, including, without limitation, Northern blot, RT-PCR, Western blot, or in vitro, in situ, or in vivo protein activity assay(s).

(167) The term “nucleic acid” or “nucleic acid molecules” include single- and double-stranded forms of DNA; single-stranded forms of RNA; and double-stranded forms of RNA (dsRNA). The term “nucleotide sequence” or “nucleic acid sequence” refers to both the sense and antisense strands of a nucleic acid as either individual single strands or in the duplex. The term “ribonucleic acid” (RNA) is inclusive of iRNA (inhibitory RNA), dsRNA (double stranded RNA), siRNA (small interfering RNA), mRNA (messenger RNA), miRNA (micro-RNA), hpRNA (hairpin RNA), tRNA (transfer RNA), whether charged or discharged with a corresponding acetylated amino acid), and cRNA (complementary RNA).

The term “deoxyribonucleic acid” (DNA) is inclusive of cDNA, genomic DNA, and DNA-RNA hybrids. The terms “nucleic acid segment” and “nucleotide sequence segment,” or more generally “segment,” will be understood by those in the art as a functional term that includes both genomic sequences, ribosomal RNA sequences, transfer RNA sequences, messenger RNA sequences, operon sequences, and smaller engineered nucleotide sequences that encoded or may be adapted to encode, peptides, polypeptides, or proteins.

(168) The term “gene” or “sequence” refers to a coding region operably joined to appropriate regulatory sequences capable of regulating the expression of the gene product (e.g., a polypeptide or a functional RNA) in some manner. A gene includes untranslated regulatory regions of DNA (e.g., promoters, enhancers, repressors, etc.) preceding (up-stream) and following (down-stream) the coding region (open reading frame, ORF) as well as, where applicable, intervening sequences (i.e., introns) between individual coding regions (i.e., exons). The term “structural gene” as used herein is intended to mean a DNA sequence that is transcribed into mRNA which is then translated into a sequence of amino acids characteristic of a specific polypeptide. It should be noted that any reference to a SEQ ID or sequence specifically encompasses that sequence, as well as all corresponding sequences that correspond to that first sequence. For example, for any amino acid sequence identified, the specific specifically includes all compatible nucleotide (DNA and RNA) sequences that give rise to that amino acid sequence or protein, and vice versa.

(169) A nucleic acid molecule may include either or both naturally occurring and modified nucleotides linked together by naturally occurring and/or non-naturally occurring nucleotide linkages. Nucleic acid molecules may be modified chemically or biochemically, or may contain non-natural or derivative nucleotide bases, as will be readily appreciated by those of skill in the art. Such modifications include, for example, labels, methylation, substitution of one or more of the naturally occurring nucleotides with an analog, internucleotide modifications (e.g., uncharged linkages: for example, methyl phosphonates, phosphotriesters, phosphoramidates, carbamates, etc.; charged linkages: for example, phosphorothioates, phosphorodithioates, etc.; pendent moieties: for example, peptides; intercalators: for example, acridine, psoralen, etc.; chelators; alkylators; and modified linkages: for example, alpha anomeric nucleic acids, etc.). The term “nucleic acid molecule” also includes any topological conformation, including single-stranded, double-stranded, partially duplexed, triplexed, hair-pinned, circular, and padlocked conformations.

(170) As used herein with respect to DNA, the term “coding sequence,” “structural nucleotide sequence,” or “structural nucleic acid molecule” refers to a nucleotide sequence that is ultimately translated into a polypeptide, via transcription and mRNA, when placed under the control of appropriate regulatory sequences. With respect to RNA, the term “coding sequence” refers to a nucleotide sequence that is translated into a peptide, polypeptide, or protein. The boundaries of a coding sequence are determined by a translation start codon at the 5'-terminus and a translation stop codon at the 3'-terminus. Coding sequences include, but are not limited to: genomic DNA; cDNA; EST; and recombinant nucleotide sequences.

(171) The term “sequence identity” or “identity,” as used herein in the context of two nucleic acid or polypeptide sequences, refers to the residues in the two sequences that are the same when aligned for maximum correspondence over a specified comparison window.

(172) The term “recombinant” when used with reference, e.g., to a cell, or nucleic acid, protein, or vector, indicates that the cell, organism, nucleic acid, protein, or vector has been modified by the introduction of a heterologous nucleic acid or protein, or the alteration of a native nucleic acid or protein, or that the cell is derived from a cell so modified. Thus, for example, recombinant cells may express genes that are not found within the native (nonrecombinant or wild-type) form of the cell or express native genes that are otherwise abnormally expressed—over-expressed, under expressed, or not expressed at all.

(173) The terms “approximately” and “about” refer to a quantity, level, value, or amount that varies by as much as 30%, or in another embodiment by as much as 20%, and in a third embodiment by as much as 10% to a reference quantity, level, value or amount. As used herein, the singular form “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

(174) As used herein, “heterologous” or “exogenous” in reference to a nucleic acid is a nucleic acid that

originates from a foreign species, or is synthetically designed, or, if from the same species, is substantially modified from its native form in composition and/or genomic locus by deliberate human intervention. A heterologous protein may originate from a foreign species or, if from the same species, is substantially modified from its original form by deliberate human intervention. By “host cell” is meant a cell which contains an introduced nucleic acid construct and supports the replication and/or expression of the construct.

(175) The invention now being generally described will be more readily understood by reference to the following examples, which are included merely for the purposes of illustration of certain aspects of the embodiments of the present invention. The examples are not intended to limit the invention, as one of skill in the art would recognize from the above teachings and the following examples that other techniques and methods can satisfy the claims and can be employed without departing from the scope of the claimed invention. Indeed, while this invention has been particularly shown and described with references to preferred embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the invention encompassed by the appended claims.

EXAMPLES

Example 1: Genetic Modification of FABP and Predictive Binding Affinity Determination Utilizing in Silico Directed Evolution

(176) In one embodiment the present invention demonstrated an increase water solubility, transport, and storage stability of cannabinoids, terpenoids, and volatiles through the use of genetically modified, or engineered fatty acid-binding proteins (FABPs). The present inventors selected FABP isoforms 1, 3, 5, and 7 as an initial set of exemplary targets as these have been experimentally shown to bind the cannabinoids THC and CBD. The present inventors generated crystal structures for the human versions of these four FABP isoforms, while corresponding homology models were built for various mammalian versions (cow, pig, sheep, horse, rabbit, and chicken) as well as for a plant-based FABP identified from a rainforest tree named *Rhodamnia argentea*. Computational docking of CBD was then combined with in silico directed evolution (via random mutagenesis) to identify FABP variants with enhanced binding affinity to CBD as an exemplary cannabinoids. FABP variants were screened using two separate scoring schemes. Variants selected for experimental testing comprised those with the best predicted binding affinities based on wither score. As generally shown in Table 1, exemplary top predicted CBD-binding variants were identified for chicken FABP7 (SEQ ID NOs. 74, and 79-80), pig FABP1 (SEQ ID NOs. 75, and 81-82), horse FABP1 (SEQ ID NOs. 76, and 83-84) and plant FABP (SEQ ID NOs. 77, and 85-86). The top scoring variants using either scoring scheme were selected, leading to 8 engineered FABP sequences provided for experimental testing.

Example 2: FABP Binding to Cannabinoids by ANS Displacement

(177) As noted above, FABPs with higher predicted binding affinity to cannabinoids were selected for overexpression, purification and binding assays. In addition, select FABPs were modified to enhance binding affinity to cannabinoids. Binding of cannabinoids to FABPs was measured by ANS displacement. Arachidonic acid was used as a positive control. The decrease in fluorescence compared to protein without ligand provides a measure of binding of the ligand, for example one or more cannabinoid such as CBD or THC to the target protein. As shown generally in FIGS. 4-6, representative spectral images for binding experiments conducted by the present inventors with the selected FABPs are demonstrated. As specifically noted in FIG. 7, the present inventors demonstrated the quantified binding results as changes in fluorescence from protein without ligand.

(178) As shown in FIG. 4A, no, or negligible binding to cannabinoids, which in this exemplary embodiment included CBD and THC, was observed with the wild type target FABP identified from chicken (SEQ ID NO. 74). Notably, the present inventor demonstrated that the genetically modified target FABP identified from chicken (SEQ ID NO. 79) demonstrated cannabinoid binding, and particularly binding to CBD (FIG. 4C). As further shown in FIG. 7, an exemplary rosetta-engineered FABP from pig (SEQ ID NO. 81) also demonstrated improved binding to both CBD and THC relative to the wild type based on relative change in fluorescence.

Example 3: Materials and Methods

(179) Cloning, transformation and protein expression in *E. coli*: and FABPs were cloned in a bacteria

expression system using a modified pET 24a(+) vector (from GenScript, FIG. 2) and transformed in BL21 (DE3) competent cells. This vector is under the control of the strong T7 promoter, and has 6×His tag at the C-terminal of the protein sequence for purification. One colony was inoculated in 10 ml of LB and grown overnight for small scale protein expression. Next day, the culture was diluted 1:100 in LB medium and grown until OD reached 0.5. Protein expression was induced with 400 μM of isopropyl-β-D-thio-galactoside (IPTG) for 3 hours at 30 C and with shaking at 250 rpm. After 3 hours of growth, the cells were harvested and washed with 50 mM Tris-HCl and cell pellets were stored at -80° C. for further protein purification.

(180) Protein purification: Cell pellets of 500 ml cell culture were thawed and resuspended in 15 ml of cell lysis containing 50 mM of Tris-HCl and protease inhibitors. Cells were lysed using Ultrasonic-Homogenizer, Biologics Inc Model 3000. After sonication lysed cells were spun down at 14,000 rpm for 10 min. For soluble proteins, supernatants were used for affinity chromatography purification using TALON resin. For insoluble proteins, pellets were dissolved in the detergent-based buffer SoluLyse (Genlantis, San Diego, CA) with multiple washing steps to extract protein from inclusion bodies according to SoluLyse manufacturers. Proteins from inclusion bodies were unfolded in 9M Urea and 5 mM DTT and refolded by dilution with 50 mM Tris-HCl and 150 mM NaCl pH 8.

(181) Ligand binding assays-ANS binding studies: Binding assays of cannabinoids to proteins were assessed by 8-anilino-1-naphthalenesulfonic acid (ANS, Thermofisher scientific, Waltham, MA) displacement. ANS is a fluorescent probe commonly used to measure conformational changes due to ligand binding. ANS binds mostly to hydrophobic sites in the protein (Yu and Strobel, 1996; Huang et al., 2016). 2 μM of protein was labelled with 20 μM of ANS. 100 μM stocks of cannabidiol (CBD), delta 9 tetrahydrocannabinol (THC) and Arachidonic acid were prepared in 10% of MeOH. Protein-ANS complex were excited at 390 nm and emission scan were recorded from 400 to 550 nm. All the experiments were done at 20 C on a FluroMax Spectrofluorometer.

(182) TABLE-US-00005 TABLE 1 Predicted CBD binding affinities for the WT and engineered FABPs. CBD binding Species and affinity using isoform Variant Rosetta (REU) Smina (kcal/mol) Chicken mdl_r5_0225_ro -31.444 -8.074 FABP7 mdl_r3_1403_sm -29.481 -10.116 WT -20.864 -7.606 Pig FABP1 mdl_r5_1652_ro -33.233 -7.516 mdl_r4_0372_sm -31.764 -10.065 WT -24.231 -7.265 Horse mdl_r5_0954_ro -33.131 -9.451 FABP1 mdl_r5_1179_sm -31.288 -10.995 WT -27.447 -8.759 Plant FABP mdl_r3_0887_ro_sm -35.135 -10.536 mdl_r4_1489_ro -31.257 -9.026 WT -23.775 -9.022

(183) TABLE-US-00006 SEQUENCE LISTINGS Amino Acid Hepatic Fatty acid-binding protein 1 *Mus musculus* SEQ ID NO. 1
 MNFSGKYQLQSQENFEPFMKAIGLPEDLIQKGKDIKGVSEIVHEGKKIKLTITYGPKVVRNEFT
 LGEECELETMTGEKVKAVVKLEGDNKMVTTFKGIKSVTELNGDTITNTMTLGDIVYKRVSKRI
 Amino Acid Hepatic Fatty acid-binding protein 1 *Bos taurus* SEQ ID NO. 2
 MNFSGKYQVQTQENYEAFMKAVGMPDDIIQKGKDIKGVSEIVQNGKHFKFIITAGSKVIQNEFT
 LGEECEMEFMTGEKIKAVVQQEGDNKLVTTFKGIKSVTEFNGDTVSTMTKGDVVFKRVSRI
 N-terminal secretion signal *S. cerevisiae* SEQ ID NO. 3
 MRFPSIFTAVLFAASSALAAPVNTTTEDETAQIPAEAVIGYSDLEGDFDVAVL PFSNSTNNGLL
 FINTTASIAAKEEGVSLEKR Amino Acid Exportable Hepatic Fatty acid-binding protein 1
Mus musculus SEQ ID NO. 4
 MRFPSIFTAVLFAASSALAAPVNTTTEDETAQIPAEAVIGYSDLEGDFDVAVL PFSNSTNNGLL
 FINTTASIAAKEEGVSLEKRMNFSGKYQLQSQENFEPFMKAIGLPEDLIQKGKDIKGVSEIVH
 EGKKIKLTITYGPKVVRNEFTLGEECELETMTGEKVKAVVKLEGDNKMVTTFKGIKSVTELNGD
 TITNTMTLGDIVYKRVSKRI Amino Acid Exportable Hepatic Fatty acid-binding protein 1
Bos taurus SEQ ID NO. 5
 MRFPSIFTAVLFAASSALAAPVNTTTEDETAQIPAEAVIGYSDLEGDFDVAVL PFSNSTNNGLL
 FINTTASIAAKEEGVSLEKRMNFSGKYQVQTQENYEAFMKAVGMPDDIIQKGKDIKGVSEIVQ
 NGKHFKFIITAGSKVIQNEFTLGEECEMEFMTGEKIKAVVQQEGDNKLVTTFKGIKSVTEFNGD
 TVTSTMTKGDVVFKRVSRI Amino Acid Catalase *Arabidopsis thaliana* SEQ ID NO. 6
 MDPYKYR PASSYNPFFTTNSGAPVWNNNSSMTVGPRGLILLEDYHLVEKLANFDRERIPERVV
 HARGASAKGFFEVT HDISNLT CADFLRAPGVQTPVIVRFSTVIHARGSPETLRDPRGFAVKFYT

REGNFDLVGNDFVNFVFFIRDFGNFHLKPNPKSHIQENWRILDFFSHHPESLNMFTFLFDD
IGIPQDYRHMDGSGVNTYMLINKAGKAHYVKFHWKPTCGVKSLLLEEDAIRLGGTNHSHATQDLY
DSIAAGNYPEWKLFIQIIDPADEDKFDFFDPLDVTKTWPEDILPLQPVGRMVLNKNIDNFFAENE
QLAFCPAIIVPGIHYSDDKLLQTRVFSYADTQRHRLGPNYLQLPVNAPKCAHHNNHHEGFMNFM
HRDEEVNYFPSRYDQVRHAEKYPTPPAVCSGKRERCIIEKENNFKEPGERYRTFTPERQERFIQ
RWIDALSDPRITHEIRSIWISYWSQADKSLGQKLASRLNVRPSI Amino Acid Catalase HP11
(KatE) *Escherichia coli* SEQ ID NO. 7
MSQHNEKNPHQHQSPLHDSSEAKPGMDSLAPEDGSHRPAAEPTPPGAQPTAPGSLKAPDTRNEK
LNSLEDVRKGSSENYALTNNQGVRIADDQNSLRAGSRGPTLLEDFILREKITHFDHERIPERIVH
ARGSAAHGYFQPYKSLSDITKADFLSDPNKITPVFVRFSTVQGGAGSADTVRDIRGFATKFYTE
EGIFDLVGNNTPIFFIQDAHKFPDFVHAVKPEPHWAIPQGGQSAHDTFWDYVSLQPETLHNVMWA
MSDRGIPRSYRTMEGFGIHTFRLINAEGKATFVRFWKPLAGKASLVWDEAQKLTGRDPDFHRR
ELWEAIEAGDFPEYELGFLIPEEDEFKFDFFDLDPTKLIPEELVPVQRVGKMVLNRNPDNFFA
ENEQAAFHPGHIVPGLDFTNDPLLQGRLFSYTDQTISRLGGPNFHEIPINRPTCPYHNFQRDGM
HRMGIDTNPANYEPNSINDNWPRETTPPGPKRGGFESYQERVEGNKVRERSPSFGEYYSHPLFW
LSQTPFEQRHIVDGFSELSKVVRPYIRERVVDQLAHIDTLAQAVAKNLGIELTDDQLNITPP
PDVNGLKKDPSLSLYAIPDGDVKGRRVAILLNDEVRSADLLAILKALKAKGVHAKLLYSRMGEV
TADDGTVLPIAATFAGAPSLTVDAVIVPCGNIADIADNGDANYYLMEAYKHLKPIALAGDARKF
KATIKIADQGEEGIVEADSADGSFMDLTLMAAHRVWSRIPKIDKIPA Amino Acid Catalase 1
Arabidopsis thaliana SEQ ID NO. 8
MDPYRVRPSSAHDSPFFTTNSGAPVWNNNSSLTVGTRGPILLEDYHLLKLANFDRERIPERVV
HARGASAKGFFEVTHTDITQLTSADFLRGPGVQTPVIVRFSTVIHERGSPETLRDPRGFAVKFYT
REGNFDLVGNDFVNFVFFVRDGMKFPDMVHALKPNPKSHIQENWRILDFFSHHPESLHMFSFLFDD
LGIPQDYRHMEGAGVNTYMLINKAGKAHYVKFHWKPTCGIKCLSDEEAIKRVGGANHSHATKDLY
DSIAAGNYPQWNLFVQVMDPAHEDKFDFFDPLDVTKIWPEDILPLQPVGRVLNKNIDNFFNENE
QIAFCPALVVPGIHYSDDKLLQTRIFSYADSQRHRLGPNYLQLPVNAPKCAHHNNHHDGFMNFM
HRDEEVNYFPSRLDPVRHAEKYPTTPIVCSGNREKCFIGKENNFQKQPGERYRSWDSRQERFVK
RFVEALSEPRVTHEIRSIWISYWSQADKSLGQKLATRLNVRPNF Amino Acid Catalase 2
Arabidopsis thaliana SEQ ID NO. 9
MDPYKYRPASSYNPFFTTNSGAPVWNNNSSMTVGPRGPILLEDYHLVEKLANFDRERIPERVV
HARGASAKGFFEVTHTDISNLTCADFLRAPGVQTPVIVRFSTVIHERGSPETLRDPRGFAVKFYT
REGNFDLVGNDFVNFVFFIRDGMKFPDMVHALKPNPKSHIQENWRILDFFSHHPESLNMFTFLFDD
IGIPQDYRHMDGSGVNTYMLINKAGKAHYVKFHWKPTCGVKSLLLEEDAIRVGGTNHSHATQDLY
DSIAAGNYPEWKLFIQIIDPADEDKFDFFDPLDVTKTWPEDILPLQPVGRMVLNKNIDNFFAENE
QLAFCPAIIVPGIHYSDDKLLQTRVFSYADTQRHRLGPNYLQLPVNAPKCAHHNNHHEGFMNFM
HRDEEVNYFPSRYDQVRHAEKYPTPPAVCSGKRERCIIEKENNFKEPGERYRTFTPERQERFIQ
RWIDALSDPRITHEIRSIWISYWSQADKSLGQKLASRLNVRPSI Amino Acid Catalase 3
Arabidopsis thaliana SEQ ID NO. 10
MDPYKYRPSSAYNAPFYTTNGGAPVSNNISSLTIGERGPVLLEDYHLIEKLVANFTTRERIPERVV
HARGISAKGFFEVTHTDISNLTCADFLRAPGVQTPVIVRFSTVVHERASPETMRDIRGFAVKFYT
REGNFDLVGNNTPVFFIRDGIQFPDVVHALKPNPKTNIQEYWRILDYMSHLPESLLTWCWMFDD
VGIPQDYRHMEGFGVHTYTLIAKSGKVLVFKFHWKPTCGIKNLTDEEAKVVGGANHSHATKDLH
DAIASGNYPEWKLFIQTMDPADEDKFDFFDPLDVTKIWPEDILPLQPVGRVLNRTIDNFFNETE
QLAFNPGLVVPGIYYSDDKLLQCRIFAYGDTQRHRLGPNYLQLPVNAPKCAHHNNHHEGFMNFM
HRDEEINYPSKFDPVRCAEKVPTPTNSYTGIRTKCVIKKENNFQKQAGDRYRSWAPDRQDRFVK
RWVEILSEPRLTHERGIWISYWSQADRSLGQKLASRLNVRPSI Amino Acid THCA Synthase
Trichome targeting domain *Cannabis* SEQ ID NO. 11 MNCSAFSFWFVCKIIFFFLSFHIQISIA
Amino Acid CBDA Synthase Trichome targeting domain *Cannabis* SEQ ID NO. 12
MKCSTFSFWFVCKIIFFFFSFNIQTSIA Amino Acid Cytosolic targeted THCA Synthase
(ctTHCAs) *Cannabis* SEQ ID NO. 13
NPRENFLKCFSKHIPNNVANPKLVYTQHDQLYMSILNSTIQNLRFISDTPPKPLVIVTPSNNSH
IQATILCSKKVGLQIRTRSGGHDAEGMSYISQVPFVVVDLRNMHSIKIDVHSQTAWVEAGATLG
EVYYWINEKNENLSFPGGYCPTVGVGGHFSGGGYGALMRNYGLAADNIIDAHLVNVDGKVLDRK

SMGEDLFWALRGGGGENFGIIAAWKSLVDVPSKIFSVKKNMEIHLVKLFNKNWQNIAYKYD
KDLVLMTHFITKNITDNHKGKNTTVHGYFSSIFHGGVDSLVDLMNKSFPPELGIIKKTDCKEFSWI
DTTIFYSGVVNFNTANFKKEILLDRSAGKKTAFSIKLDYVKKPIPETAMVKILEKLYEEDVGAG
MYVLYPYGGIMEEISESAIPFPHRAGIMYELWYTASWEKQEDNEKHINWVRSVYNFTTPYVSQN
PRLAYLNYRDLDLGKTNHASPNNYTQARIWGEKYFGKNFNRLVKVKTCLVDPNNFFRNEQSIPPL
PPHHH DNA Cytosolic CBDA synthase (cytCBDAs) *Cannabis sativa* SEQ ID NO. 14
ATGAATCCTCGAGAAAACCTTCCTTAAATGCTTCTCGCAATATATTCCCAATAATGCAACAAATC
TAAAACTCGTATACACTCAAAACAACCCATTGTATATGTCTGTCCTAAATTGACAAATACACAA
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GTCATGATTCTGAGGGCATGTCCTACATATCTCAAGTCCCATTGTTATAGTAGACTTGAGAAA
CATGCGTTCAATCAAATAGATGTTTCATAGCCAAACTGCATGGGTGAAGCCGGAGCTACCCTT
GGAGAAGTTTATTATTGGGTAAATGAGAAAAATGAGAATCTTAGTTTGGCGGCTGGGTATTGCC
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CCTCGCGGCTGATAATATCATTGATGCACACTTAGTCAACGTTTCATGGAAAAGTGCTAGATCGA
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GATACTATCATCTTCTATAGTGGTGTGTAAATTACGACACTGATAATTTTAAACAAGGAAATTT
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CGAAAAGCATCTAAACTGGATTAGAAATATTTATAACTTCATGACTCCTTATGTGTCCAAAAAT
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ATAATTACACACAAGCACGTATTTGGGGTGAGAAGTATTTTGGTAAAAATTTTGACAGGCTAGT
AAAAGTGAAAACCCTGGTTGATCCCAATAACTTTTTTAGAAACGAACAAAGCATCCACCTCTA
CCACGGCATCGTCATTAA Amino Acid Cytosolic CBDA synthase (cytCBDAs) *Cannabis*

sativa SEQ ID NO. 15
MNPRENFLKCF SQYIPNNATNLKLVYTQNNPLYMSVLNSTIHNLRFTSDTTPKPLVIVTPSHVS
HIQGTILCSKKVGLQIRTRSGGHDSEGMSYISQVPFVIVDLRNMRSIKIDVHSQTAWVEAGATL
GEVYYWVNEKNENLSLAAGYCPTVCAGGHFGGGGYGPLMRNYGLAADNIIDAHLVNVHGKVLDR
KSMGEDLFWALRGGGAESFGIIVAWKIRLVAVPKSTMFSVKKIMEIHELKLVNKNWQNIAYKYD
KDLLLMTHFITRNTDNQGKNKTAIHTYFSSVFLGGVDSLVDLMNKSFPPELGIIKKTDCRQLSWI
DTIIFYSGVVNYDNTDNFNKEILLDRSAGQNGAFKIKLDYVKKPIPESVFVQILEKLYEEDIGAG
MYALYPYGGIMDEISESAIPFPHRAGILYELWYICSWEKQEDNEKHLNWIRNIYNFMTPYVSKN
PRLAYLNYRDLDIGINDPKPNPNYTQARIWGEKYFGKNFDRLVKVKTCLVDPNNFFRNEQSIPPL
PRHRH DNA MYB12-like *Cannabis* SEQ ID NO. 16
ATGAAGAAGAACAAATCAACTAGTAATAATAAGAACAACAACAGTAATAATATCATCAAAAACG
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TCATAATGAGAAAGTTACTGTCAGTACTGATCATATTATTAATCTTGATGATAAGCAGAAACGA
CAATTATGTCGTTGTCGTTTAGAAAAAGAAGAAGAAGAAGAAGGAAGTGGTGGTTGTGGTGAGA
CAGTAGTAATGATGCTAGGGTCAGTATCTCCTGCTGCTGCTACTGCTGCTGCAGCTGGGGGCTC
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AAGGTGGGGCAGCTGATCAGATTGACAAGACTACTTGTAATAATACTACTATTACTACTAATGA
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 ATCCAGAAACTATTAAGAACGTTTTTGGTTAAGGATTGTTTGTCTGTTTTTACTAACAGAAGAGA
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 GTTGAACATTTGTATGTTTCCAAACGATTCTATTGAATTTTTTAAGAAGTTTGTTGATAGAATG
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GCTTTGATGAACATGAAGTTGGCTGTTACTAAGGTTTTTGCAAACTTTTCTTTTCAACCATGTC
AAGAACTCAAATTCATTGAAGTTGTCTAGACAAGGTATTTTGCAACCAGAAAAGCCAATTGT
TTTGAAGGTTGTTCCAAGAGATGCTGTTATTACTGGTGCTTAA Amino Acid Cytochrome

P450 (CYP3A4) *Mus musculus* SEQ ID NO. 23

MNLFSAISLDLTVLLAIILVLLYRYGTRTHGLFKKQGIPGPKPLPFLGTVLNYTGIWKFDMEC
YEKYGKTWGLFDGQTPLLITDPETIKNVLVKDCLSVFTNRREFGPGVIMSKAISISKDEEWKR
YRALLSPTFTSGRLKEMFPVIEQYGDILVKYLRQEAKEKGMPPVAMKDVLGAYSMDVITSTSFGVN
VDSLNNPEDPFVEEAKKFLRVDFDPLLFSVVLFPLLTPVYEMLNICMFPNDSIEFFKKFVDRM
QESRLDSNQKHRVDFLQLMMNSHNNSKDKDSHKAFSNMEITVQSIIFISAGYETTSSTLSFTLY
CLATHPDIQKKLQAEIDKALPNKATPTCDTVMEMEYLDMLVNETLRLYPIVTRLERVCKKDVEL
NGVYIPKGSMMVIPSIALHHDPPQHWPDPEEFQPERFSKENKGSIDPYVYLPFGIGPRNCIGMRF
ALMNMKLAVTKVLQNFSPQPCQETQIPLKLSRQGIQPEKPIVLKVVPRDAVITGA DNA P450

oxidoreductase gene (CYP oxidoreductase) *Mus musculus* SEQ ID NO. 24

ATGGGTGATTCTCATGAAGATACTTCTGCTACTGTTCCAGAAGCTGTTGCTGAAGAAGTTTCTT
TGTTTTCTACTACTGATATTGTTTTGTTTTCTTTGATTGTTGGTGTTTTGACTTACTGGTTTAT
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AAGGAATCTTCTTTTGTGAAAAGATGAAGAAGACTGGTAGAAACATTATTGTTTTTTACGGTT
CTCAAACACTGGTACTGCTGAAGAATTTGCTAACAGATTGTCTAAGGATGCTCATAGATACGGTAT
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GATAAGTCTTTGGTTGTTTTTTGTATGGCTACTTACGGTGAAGGTGATCCAACACTGATAACGCTC
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AACAAGCTCATAAGGTTTACGTTCAACATTTGTTGAAGAGAGATAAGGAACATTTGTGGAAGTT
GATTCATGAAGGTGGTGCTCATATTTACGTTTGTGGTGATGCTAGAAACATGGCTAAGGATGTT
CAAAACACTTTTTACGATATTGTTGCTGAATTTGGTCCAATGGAACATACTCAAGCTGTTGATT
ACGTTAAGAAGTTGATGACTAAGGGTAGATACTCTTTGGATGTTTGGTCTTAA Amino Acid

P450 oxidoreductase (CYP oxidoreductase) *Mus musculus* SEQ ID NO. 25

MGDSHEDTSATVPEAAEEVSLFSTTDIVLFLSLIVGVLTYWFIKFKKKEEIPFSKIQTAPPV
KESSFVEKMKKTGRNIIVFYGSQTGTAEFFANRLSKDAHRYGMRGMSADPEEYDLADLSSLPEI

DKSLVVFCDTGYGEGDPTDNDQAQDFYDQWLDQETDGTGKFAVFGLGKNTYEHFNAMGKYVDQRL
EQLGAQRIFELGLGDDDDGNLEEDFITWREQFWPAVCEFFGVEATGEESSIRQYELVVHEDMDTA
KVYTGEMGRLKSYENQKPPFDAKNPFLAAVTTNRKLNQGTERHLMHLELDISDSKIRYESGDHV
AVYPANDSTLVNQIGEILGADLDVIMSLNNLDEESNKKHPFPCPTTYRTALTYYLDITNPRTN
VLYELAQYASEPSEQEHLHKMASSSGEGKELYLSWVVEARRHILAILQDYP SLRPPIDHLCCELL
PRLQARYYSIASSSKVHPNSVHICAVAVEYEAKSGRVNKG VATSWLRTKEPAGENGRRALVPMF
VRKSQFRLPFKPTTPVIMVGP GTGVAPFMGFIQERAWLREQGKEVGETLLYYGCRRSDEDYLYR
EELARFHKDGALTQLNVAFSREQAHKVYVQHLLKRDKEHLWKLIHEGGAHIYVCGDARNMAKDV
QNTFYDIVAEFGPMEHTQAVDYVKKLMTKGRYSLDVWS DNA Cytochrome P450 (CYP3A4)
Human SEQ ID NO. 26

ATGGCTTTGATTCCTGATTTGGCTATGGAACTAGATTGTTGTTGGCTGTTTCATTGGT TTTTGT
TGTATTTGTATGGA ACTCATTACATGGATTGTTTAAAAAATTGGGAATTCCTGGACCTACTCC
TTTGCCTTTTTTTGGGAAATATTTTGT CATATCATAAAGGATTTTGCATGTTTGATATGGAATGC
CATAAAAAATATGGAAAAGTTTGGGGATTTTATGATGGACAACAACCTGTTTTGGCTATTACTG
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AAACTCAAATTCCTTTGAAATTGTCATTGGGAGGATTGTTGCAACCTGAAAAACCTGTTGTTTT
GAAAGTTGAATCAAGAGATGGA ACTGTTTCAGGAGCT Amino Acid Cytochrome P450
(CYP3A4) Human SEQ ID NO. 27

MALIPDLAMETRLLLAVSLVLLYLYGTHSHGLFKKLGI PGPTPLPFLGNILSYHKGFCMFDMEC
HKKYGKVWGFYDGGQQPVLAITDPD MIKTVLVKECYSVFTNRRPFGPVGFMKSAISIAEDEEWKR
LRSLLSPTFTSGKLKEMVPIIAQYGDVLVRNLRREAETGKPVTLKDVF GAYSMDVITSTSFGVN
IDSLNNPQDPFVENTKKLLRFDFLDPFFLSITVFPFLIPILEV LNICVFPREVTNFLRKSVKRM
KESRLEDTQKHRVDFLQLMIDSQNSKETESHKALSDLELVAQSIIFIFAGCETTSSVLSFIMYE
LATHPDVQQKLQEEIDAVLPNKAPPTYDTVLQMEYLD MVVNETLRLFP IAMRLERVCKKDVEIN
GMFIPKGVVVMIPSYALHRDPKYWTEPEKFLPERFSKKNKD NIDPYIYTPFGSGPRNCIGMRFA
LMNMKLALIRVLQNF SFKPCKETQIPLKLSLGGLLQPEKPVVLKVESRDGTVSGA DNA P450
oxidoreductase gene (oxred) Human SEQ ID NO. 28

ATGATTAATATGGGAGATTCACATGTTGATACTTCATCAACTGTTTCAGAAAGCTGTTGCTGAAG
AAGTTTCATTGTTTTCAATGACTGATATGATTTTGT TTTTCATTGATTGTTGGATTGTTGACTTA
TTGGTTTTTTGTTTAGGAAAAAAGAAAGAAATTCTGAATTTACTAAAATTCAAAC TTTGACT
TCATCAGTTAGAGAATCATCATTGTTGAAAAAATGAAAAAACTGGAAGAAATATTATTGTTT
TTTATGGATCACAACTGGA ACTGCTGAAGAATTTGCTAATAGATTGTCAAAGATGCTCATAG
ATATGGAATGAGAGGAATGTCAGCTGATCCTGAAGAATATGATTTGGCTGATTGTGCATCATTG
CCTGAAATTGATAATGCTTTGGTTGTTTTTTGCATGGCTACTTATGGAGAAGGAGATCCTACTG
ATAATGCTCAAGATTTTTATGATTGGTTGCAAGAACTGATGTTGATTGTGTCAGGAGTTAAATT
TGCTGTTTTTGGATTGGGAAATAAACTTATGAACATTTTAATGCTATGGGAAAATATGTTGAT

AAAAGATTGGAACTAATTGGGAGCTCAAAGAATTGGGATTGGGAGATGATGATGGAA
ATTTGGAAGAAGATTTTATTACTTTGGAGAGAACAAATTTTGGTTGGCTGTTTGCGAACATTTTGG
AGTTGAAGCTACTGGAGAAGAATCATCAATTAGACAATATGAATTGGTTGTTTCATACTGATATT
GATGCTGCTAAAGTTTATATGGGAGAAATGGGAAGATTGAAATCATATGAAAATCAAAAACCTC
CTTTTGATGCTAAAAATCCTTTTTTTGGCTGCTGTTACTACTAATAGAAAATTGAATCAAGGAAC
TGAAAGACATTTGATGCATTTGGAATTGGATATTTTCAGATTCAAAAATTAGATATGAATCAGGA
GATCATGTTGCTGTTTATCCTGCTAATGATTCAGCTTTGGTTAATCAATTGGGAAAAATTTTGG
GAGCTGATTTGGATGTTGTTATGTCATTGAATAATTTGGATGAAGAATCAAATAAAAAACATCC
TTTTCTTGCCCTACTTCATATAGAACTGCTTTGACTTATTATTTGGATATTACTAATCCTCCT
AGAACTAATGTTTTGTATGAATTGGCTCAATATGCTTCAGAACCTTCAGAACAAAGAATTGTTGA
GAAAAATGGCTTCATCATCAGGAGAAGGAAAAGAATTGTATTTGTCATGGGTTGTTGAAGCTAG
AAGACATATTTTGGCTATTTTGCAGATTGCCCTTCATTGAGACCTCCTATTGATCATTTGTGC
GAATTGTTGCCTAGATTGCAAGCTAGATATTATTCAATTGCTTCATCATCAAAAGTTCATCCTA
ATTCAGTTCATATTTGCGCTGTTGTTGTTGAATATGAACTAAAGCTGGAAGAATTAATAAAGG
AGTTGCTACTAATTGGTTGAGAGCTAAAGAACCTGTTGGAGAAAATGGAGGAAGAGCTTTGGTT
CCTATGTTTGTAGAAAATCACAATTTAGATTGCCTTTTAAAGCTACTACTCCTGTTATTATGG
TTGGACCTGGAAGTGGAGTTGCTCCTTTTTATTGGATTTATTCAAGAAAGAGCTTGGTTGAGACA
ACAAGGAAAAGAAGTTGGAGAACTTTGTTGTATTATGGATGCAGAAGATCAGATGAAGATTAT
TTGTATAGAGAAGAATTGGCTCAATTCATAGAGATGGAGCTTTGACTCAATTGAATGTTGCTT
TTTCAAGAGAACAAATCACATAAAGTTTATGTTCAACATTTGTTGAAACAAGATAGAGAACATTT
GTGGAATTGATTGAAGGAGGAGCTCATATTTATGTTTGCAGGAGATGCTAGAAATATGGCTAGA
GATGTTCAAATACTTTTTATGATATTGTTGCTGAATTGGGAGCTATGGAACATGCTCAAGCTG
TTGATTATATTAATAAATTGATGACTAAAGGAAGATATTCATTGGATGTTTGGTCA Amino
Acid P450 oxidoreductase Human SEQ ID NO. 29

MINMGDSHVDTSSTVSEAVAEVSLFSMTDMILFSLIVGLLTYWFLFRKKKEEVPEFTKIQTLT
SSVRESSFVEKMKKTGRNIIIFYGSQTGTAEFANRLSKDAHRYGMRGMSADPEEYDLADLSSL
PEIDNALVVFCMATYGECDPTDNAQDFYDWLQETDVDLSGVKFAVFGLGNKTYEHFNAMGKYVD
KRLEQLGAQRIFELGLGDDDDGNLEEDFITWREQFWLAVCEHFGVEATGEESSIRQYELVVHTDI
DAAKVYMGEMGRLKSYENQKPPFDAKNPFLAAVTTNRKLNQGTERHLMHLELDISDSKIRYESG
DHVAVYPANDSALVNQLGKILGADLDVVMNLNDEESNKKHPFPCPTSYRTALTYYLDTNPP
RTNVLYELAQYASEPSEQELLRKMASSSGEGKELYLSWVVEARRHILAILQDCPSLRPPIDHLC
ELLPRLQARYYSIASSSKVHPNSVHICAVVVEYETKAGRINKGVATNWLRAKEPVGENGGRALV
PMFVRKSQFRLPFKATTPVIMVGPVGTVAPFIGFIQERAWLRQQGKEVGETLLYYGCRRSDDEDY
LYREELAQFHRDGALTQLNVAFSREQSHKVYVQHLLKQDREHLWKLIEGGAHIYVCGDARNMAR
DVQNTFYDIVAELGAMEHAQAVDYIKKLMTKGRYSLDVWS DNA cannabidiolic acid

(CBDA) synthase *Cannabis sativa* SEQ ID NO. 43
ATGAATCCTCGAGAAAACCTTCCTTAAATGCTTCTCGCAATATATTCCCAATAATGCAACAAATC
TAAAACTCGTATACACTCAAACAACCCATTGTATATGTCTGTCCTAAATTCGACAATACACAA
TCTTAGATTCACCTCTGACACAACCCCAAACCACTTGTATCGTCACTCCTTCACATGTCTCT
CATATCCAAGGCACTATTCTATGCTCCAAGAAAGTTGGCTTGCAGATTCGAACTCGAAGTGGTG
GTCATGATTCTGAGGGCATGTCCTACATATCTCAAGTCCCATTTGTTATAGTAGACTTGAGAAA
CATGCGTTCAATCAAATAAGATGTTTCATAGCCAACTGCATGGGTTGAAGCCGGAGCTACCCTT
GGAGAAGTTTATTATTGGGTTAATGAGAAAAATGAGAATCTTAGTTTGGCGGCTGGGTATTGCC
CTACTGTTTGCAGGTTGGACACTTTGGTGGAGGAGGCTATGGACCATTGATGAGAACTATGG
CCTCGCGGCTGATAATATCATTGATGCACACTTAGTCAACGTTTCATGGAAAAGTGCTAGATCGA
AAATCTATGGGGGAAGATCTCTTTTGGGCTTTACGTGGTGGTGGAGCAGAAAGCTTCGGAATCA
TTGTAGCATGGAAAATTAGACTGGTTGCTGTCCCAAAGTCTACTATGTTTAGTGTTAAAAAGAT
CATGGAGATACATGAGCTTGTCAAGTTAGTTAACAAATGGCAAAATATTGCTTACAAGTATGAC
AAAGATTTATTACTCATGACTCACTTCATAACTAGGAACATTACAGATAATCAAGGGAAGAATA
AGACAGCAATACACACTTACTTCTCTTCAGTTTTCTTTGGTGGAGTGGATAGTCTAGTCGACTT
GATGAACAAGAGTTTTCTGAGTTGGGTATTAAAAAACGGATTGCAGACAATTGAGCTGGATT
GATACTATCATCTTCTATAGTGGTGTGTAAATTACGACACTGATAATTTTAAACAAGGAAATTT
TGCTTGATAGATCCGCTGGGCAGAACGGTGCTTTCAAGATTAAGTTAGACTACGTTAAGAAACC

AATTCCAGAACTGTGTTCCAAATTTTGGAAAAATTTATATGAAGAAGATATAGGAGCTGGG
ATGTATGCGTTGTACCCTTACGGTGGTATAATGGATGAGATTTCAGAATCAGCAATTCCATTCC
CTCATCGAGCTGGAATCTTGTATGAGTTATGGTACATATGTAGTTGGGAGAAGCAAGAAGATAA
CGAAAAGCATCTAAACTGGATTAGAAATATTTATAACTTCATGACTCCTTATGTGTCCAAAAAT
TCAAGATTGGCATATCTCAATTATAGAGACCTTGATATAGGAATAAATGATCCCAAGAATCCAA
ATAATTACACACAAGCACGTATTTGGGGTGAGAAGTATTTTGGTAAAAATTTTGACAGGCTAGT
AAAAGTGAAAACCCTGGTTGATCCCAATAACTTTTTTAGAAACGAACAAAGCATCCCACCTCAA
CCACGGCATCGTCATTAA Amino Acid Cannabidiolic acid (CBDA) synthase *Cannabis*
sativa SEQ ID NO. 31

MNPRENFLKCF SQYIPNNATNLKLVYTQNNPLYMSVLNSTIHNL RFTSDTTPKPLVIVTPSHVS
HIQGTILCSKKVGLQIRTRSGGHDSEGMSYISQVPFVIVDLRNMRSIKIDVHSQTAWVEAGATL
GEVYYWVNEKNENLSLAAGYCPTVCAGGHFGGGGYGPLMRNYGLAADNIIDAHLVNVHGKVLDR
KSMGEDLFWALRGGGAESFGIIVAWKIRLVAVPKSTMFSVKKIMEIHELVLKLVNKWQNIAYKYD
KDLLLMTHFITR NITDNQGKNKTAIHTYFSSVFLGGVDSLVDLMNKSFP ELGIKKTDCRQLSWI
DTIIFYSGVVNYD TDNFNKEILLDRSAGQNGAFKIKLDYVKKPIESV FVQILEKLYEEDIGAG
MYALYPYGGIMDEISESAIPFPHRAGILYELWYICSWEKQEDNEKHLNWIRNIYNFMTPYVSKN
SRLAYLNYRDL DIGINDPKNPNNYTQARIWGEKYFGKNFDRLVKVKTLPDPNNFFRNEQSIPPQ
PRHRH DNA UDP glycosyltransferase 76G1 *Stevia rebaudiana* SEQ ID NO. 32

ATGGAAAATAAACTGAACTACTGTTAGAAGAAGAAGAATTATTTTGTTCCTGTTCCCTT
TTCAAGGACATATTAATCCTATTTTGCAATTGGCTAATGTTTTGTATTCAA AAGGATTTTCAAT
TACTATTTTTTCATACTAATTTTAATAAACCTAA AACTTCAAATTATCCTCATTTTACTTTTAGA
TTTATTTTGGATAATGATCCTCAAGATGAAAGAATTTCAAATTTGCCTACTCATGGACCTTTGG
CTGGAATGAGAATTCCTATTATTAATGAACATGGAGCTGATGAATTGAGAAGAGAATTGGAATT
GTTGATGTTGGCTTCAGAAGAAGATGAAGAAGTTTCATGCTTGATTACTGATGCTTTGTGGTAT
TTTGCTCAATCAGTTGCTGATTCATTGAATTTGAGAAGATTGGTTTTTGATGACTTCATCATTGT
TTAATTTTCATGCTCATGTTTCATTGCCTCAATTTGATGAATTGGGATATTTGGATCCTGATGA
TAA AACTAGATTGGAAGAACAAGCTTCAGGATTTCTATGTTGAAAGTTAAAGATATTAAATCA
GCTTATTCAAATTGGCAAATTTTGAAAGAAATTTTGGGAAAAATGATTAAACAACTAGAGCTT
CATCAGGAGTTATTTGGAATTCATTTAAGAATTGGAAGAATCAGAATTGGAACTGTTATTAG
AGAAATTCCTGCTCCTTCATTTTTTGATTCCTTTGCCTAAACATTTGACTGCTTCATCATCATCA
TTGTTGGATCATGATAGAACTGTTTTTCAATGGTTGGATCAACAACCTCCTTCATCAGTTTTGT
ATGTTTCATTTGGATCAACTTCAGAAGTTGATGAAAAAGATTTTTTGGAAATTGCTAGAGGATT
GGTTGATTCAA AACAATCATTTTTTGTTGGGTTGTTAGACCTGGATTTGTTAAAGGATCAACTTGG
GTTGAACCTTTGCCTGATGGATTTTTTGGGAGAAAGAGGAAGAATTGTTAAATGGGTTCCTCAAC
AAGAAGTTTTGGCTCATGGAGCTATTGGAGCTTTTTTGGACTCATTCAGGATGGAATTCAACTTT
GGAATCAGTTTGC GAAGGAGTTCCTATGATTTTTTCAGATTTTGGATTGGATCAACCTTTGAAT
GCTAGATATATGTCAGATGTTTTGAAAGTTGGAGTTTATTTGGAAAATGGATGGGAAAGAGGAG
AAATTGCTAATGCTATTAGAAGAGTTATGGTTGATGAAGAAGGAGAATATATTAGACAAAATGC
TAGAGTTTTGAAACAAAAAGCTGATGTTTCATTGATGAAAGGAGGATCATCATATGAATCATTG
GAATCATTGGTTTCATATATTTTCATCATTG Amino Acid UPD glycosyltransferase 76G1

Stevia rebaudiana SEQ ID NO. 33
MENKTETTVRRRRRIILFPVPFQGHINPILQLANVLYSKGFSITIFHTNFNKP KTSNYPHFTFR
FILDNDPQDERISNLPTHGPLAGMRIPIINEHGADEL RRELELLMLASEEDEEV SCLITDALWY
FAQSVADSLNLRRLV LMTSSLFNFHAHVSLPQFDELGYLDPDDKTRLEE QASGF PMLKV KDIKS
AYS NWQILKEILGKMIKQTRASSGVIWNSFKELEESELET VIREIPAPSFLIPLPKHLTASSSS
LLDHDRTVFQWLDQ QPPSSVLYVSFGSTSEVDEKDFLEIARGLVDSKQSFLWVVRPGFVKGSTW
VEPLPDGFLGERGRIVKWVPQ QEVLAHGAIGAFWTHSGWNSTLESVCEGVPMIFSDFGLDQPLN
ARYMSDVLKVG VYLENGWERGEIANAIRRMVDEEGEYIRQNARVLKQKADVSLMKGGSSYESL
ESLVSYISSL Amino Acid Glycosyltransferase (NtGT5a) *Nicotiana tabacum* SEQ ID NO.
34 MGSIGAELTKPHAVCIPYPAQGHINPMLKLAKILHHKGFHITFVNTEFNHRLLKSRGPDSLKG
LSSFRFETIPDGLPPCEADATQDIPSLCESTTNTCLAPFRDLLAKLNDTNTSNVPPVSCIVSDG
VMSFTLAAAQELGVPEVLFWTT SACGFLGYMHYCKVIEKGYAPLKDASDLTNGYLETTLD FIPG
MKDVRLRDLPSFLRTTNPDEFMIKFVLQETERARKASAILNTFETLEAEVLES LRNLLPPVYP

IGPLHFLVKHVDNENLKGRLSSLWKEEPECIQWLDTKEPNSVYVNFSGSITVMTPNQLIEFAWG
LANSQQTFLWIIRPDIVSGDASILPPEFVEETKNRGM LASWCSQEEVL SHPAIVGFLTHSGWNS
TLESISSGVPMICWPFFAEQQTNCWFSVTKWDVGM EIDSDVKRDEVESLVRELMVGGKGKKMKK
KAMEWKELAEASAKEHSGSSYVNIEKLVNDILLSSKH DNA Glycosyltransferase (NtGT5a)

Nicotiana tabacum SEQ ID NO. 35

ATGGGTTCCATTGGTGCTGAATTAACAAAGCCACATGCAGTTTGCATACCATATCCCGCCCAAG
GCCATATTAACCCCATGTTAAAGCTAGCCAAAATCCTTCATCACAAAGGCTTTCACATCACTTT
TGTC AATACTGAATTTAACCACCGACGTCCTTAAATCTCGTGGCCCTGATTCTCTCAAGGGT
CTTTCTTCTTTCCGTTTTTGAGACCATTCTGATGGACTTCCGCCATGTGAGGCAGATGCCACAC
AAGATATACCTTCTTTGTGTGAATCTACAACCAATACTTGCTTGGCTCCTTTTAGGGATCTTCT
TGCGAAACTCAATGATACTAACACATCTAACGTGCCACCCGTTTCGTGCATCGTCTCGGATGGT
GTCATGAGCTTCACCTTAGCCGCTGCACAAGAATTGGGAGTCCCTGAAGTTCTGTTTTGGACCA
CTAGTGCTTGTGGTTTCTTAGGTTACATGCATTACTGCAAGGTTATTGAAAAAGGATATGCTCC
ACTTAAAGATGCGAGTGACTTGACAAATGGATACCTAGAGACAACATTGGATTTTATACCAGGC
ATGAAAGACGTACGTTTAAGGGATCTTCCAAGTTTCTTGAGAACTACAAATCCAGATGAATTCA
TGATCAAATTTGTCCTCCAAGAAACAGAGAGAGCAAGAAAGGCTTCTGCAATTATCCTCAACAC
ATTTGAAACACTAGAGGCTGAAGTTCTTGATCGCTCCGAAATCTTCTTCTCCTCCAGTCTACCCC
ATAGGGCCCTTGCATTTTCTAGTGAAACATGTTGATGATGAGAATTTGAAGGGACTTAGATCCA
GCCTTTGGAAAGAGGAACCAGAGTGTATACAATGGCTTGATACCAAAGAACCAAATTCTGTTGT
TTATGTTAACTTTGGAAGCATTACTGTTATGACTCCTAATCAGCTTATTGAGTTTGCTTGGGGA
CTTGCAAACAGCCAGCAAACATTCTTATGGATCATAAGACCTGATATTGTTTCAGGTGATGCAT
CGATTCTTCCACCCGAATTCGTGGAAGAAACGAAGAACAGAGGTATGCTTGCTAGTTGGTGTTT
ACAAGAAGAAGTACTTAGTCACCCTGCAATAGTAGGATTCTTGACTCACAGTGGATGGAATTCG
ACACTCGAAAGTATAAGCAGTGGGGTGCCTATGATTTGCTGGCCATTTTTTCGCTGAACAGCAA
CAAATTGTTGGTTTTTCCGTCACTAAATGGGATGTTGGAATGGAGATTGACAGTGATGTGAAGAG
AGATGAAGTGGAAGGCCTTGTAAGGGAATTGATGGTTGGGGGAAAAGGCCAAAAAGATGAAGAAA
AAGGCAATGGAATGGAAGGAATTGGCTGAAGCATCTGCTAAAGAACATTCAGGGTCATCTTATG
TGAACATTGAAAAGTTGGTCAATGATATTCTTCTTTCATCCAAACATTAA Amino Acid

Glycosyltransferase (NtGT5b) *Nicotiana tabacum* SEQ ID NO. 36

MGSIGAEFTKPHAVCIPYPAQGHINPMLKLAKILHHKGFHITFVNTEFNHRRLKSRGPD SLKG
LSSFRFETIPDGLPPCDADATQDIPSLCESTTNTCLGPFRDLLAKLNDTNTSNVPPVSCIISDG
VMSFTLAA AQELGVPEVLFWTT SACGFLGYMHYYKVIEKGYAPLKDASDLTNGYLETTLD FIPC
MKDVRLRLDLP SFLRTTNPDEFMIKFVLQETERARKASAILNTYETLEAEVLES LRNLLPPVYP
IGPLHFLVKHVDNENLKGRLSSLWKEEPECIQWLDTKEPNSVYVNFSGSITVMTPNQLIEFAWG
LANSQQSFLWIIRPDIVSGDASILPPEFVEETKKRGM LASWCSQEEVL SHPAIGGFLTHSGWNS
TLESISSGVPMICWPFFAEQQTNCWFSVTKWDVGM EIDCDVKRDEVESLVRELMVGGKGKKMKK
KAMEWKELAEASAKEHSGSSYVNIEKVVDILLSSKH DNA Glycosyltransferase (NtGT5b)

Nicotiana tabacum SEQ ID NO. 37

ATGGGTTCCATTGGTGCTGAATTTACAAAGCCACATGCAGTTTGCATACCATATCCCGCCCAAG
GCCATATTAACCCCATGTTAAAGCTAGCCAAAATCCTTCATCACAAAGGCTTTCACATCACTTT
TGTC AATACTGAATTTAACCACAGACGTCCTTAAATCTCGTGGCCCTGATTCTCTCAAGGGT
CTTTCTTCTTTCCGTTTTTGAGACAATTCTGATGGACTTCCGCCATGTGATGCAGATGCCACAC
AAGATATACCTTCTTTGTGTGAATCTACAACCAATACTTGCTTGGGTCCTTTTAGGGATCTTCT
TGCGAAACTCAATGATACTAACACATCTAACGTGCCACCCGTTTCGTGCATCATCTCAGATGGT
GTCATGAGCTTCACCTTAGCCGCTGCACAAGAATTGGGAGTCCCTGAAGTTCTGTTTTGGACCA
CTAGTGCTTGTGGTTTCTTAGGTTACATGCATTATTACAAGGTTATTGAAAAAGGATACGCTCC
ACTTAAAGATGCGAGTGACTTGACAAATGGATACCTAGAGACAACATTGGATTTTATACCATGC
ATGAAAGACGTACGTTTAAGGGATCTTCCAAGTTTCTTGAGAACTACAAATCCAGATGAATTCA
TGATCAAATTTGTCCTCCAAGAAACAGAGAGAGCAAGAAAGGCTTCTGCAATTATCCTCAACAC
ATATGAAACACTAGAGGCTGAAGTTCTTGAATCGCTCCGAAATCTTCTTCTCCTCCAGTCTACCCC
ATTGGGCCCTTGCATTTTCTAGTGAAACATGTTGATGATGAGAATTTGAAGGGACTTAGATCCA
GCCTTTGGAAAGAGGAACCAGAGTGTATACAATGGCTTGATACCAAAGAACCAAATTCTGTTGT
TTATGTTAACTTTGGAAGCATTACTGTTATGACTCCTAATCAACTTATTGAATTTGCTTGGGGA

CTTGCAACCAACCAATCATCTTATGGATCATAAAGACCTGATATTGTTTACAGGTGATGCAT
CGATTCTTCCCCCGAATTCGTGGAAGAAACGAAGAAGAGAGGTATGCTTGCTAGTTGGTGTTC
ACAAGAAGAAGTACTTAGTCACCCTGCAATAGGAGGATTCTTGACTCACAGTGGATGGAATTCG
ACACTCGAAAGTATAAGCAGTGGGGTGCCTATGATTTGCTGGCCATTTTTTCGCTGAACAGCAA
CAAATTGTTGGTTTTTCCGTCACTAAATGGGATGTTGGAATGGAGATTGACTGTGATGTGAAGAG
GGATGAAGTGGAAAGCCTTGTAAGGGAATTGATGGTTGGGGGAAAAGGCCAAAAAGATGAAGAAA
AAGGCAATGGAATGGAAGGAATTGGCTGAAGCATCTGCTAAAGAACATTACAGGGTCATCTTATG
TGAACATTGAGAAGGTGGTCAATGATATTCTTCTTTTCGTCCAAACATTAA Amino Acid UDP-
glycosyltransferase 73C3 (NtGT4) *Nicotiana tabacum* SEQ ID NO. 38

MATQVHKLHFIPLMAPGHMIPMIDIAKLLANRGVITTIITTPVNANRFSSTITRAIKSGLRI
QILTLKFPSVEVGLPEGCENIDMLPSLDLASKFFAAISMLKQQVENLLEGINPSPSCVISDMGF
PWTTQIAQNFNIPRIVFHGTCCFSLCSYKILSSNILENITSDSEYFVVPDLPDRVELTKAQVS
GSTKNNTSVSSSVLKEVTEQIRLAESSYGVIVNSFEELEQVYEKEYRKARGKKVWCVGPVSLC
NKEIEDLVTRGNKTAIDNQDCLKWLDNFETESVYASLGSLSRLTLLQMVELGLGLEESNRPFV
WVLGGGDKLNDLEKWILENGFEQRIKERGVLIRGWAPQVLILSHPAIGGVLTTHCGWNSTLEGIS
AGLPMVTWPLFAEQFCNEKLVVQVLKIGVSLGVKVPVKWGDENVGVLVKKDDVKKALDKLMDE
GEEGQVRRTKAKELGELAKKAFGEGGSSYVNLTSLIEDIIEQQNHKEK DNA UDP-
glycosyltransferase 73C3 (NtGT4) *Nicotiana tabacum* SEQ ID NO. 39

ATGGCAACTCAAGTGCACAACTTCATTTTCACTATTCCCTTTAATGGCTCCAGGCCACATGA
TTCCTATGATAGACATAGCTAAACTTCTAGCAAATCGCGGTGTCATTACCACTATCATCACCAC
TCCAGTAAACGCCAATCGTTTCAGTTCAACAATTACTCGTGCCATAAAATCCGGTCTAAGAATC
CAAATTCTTACACTCAAATTTCCAAGTGTAAGAAGTAGGATTACCAGAAGGTTGCGAAAATATTG
ACATGCTTCCTTCTCTTGACTTGGCTTCAAAGTTTTTTGCTGCAATTAGTATGCTGAAACAACA
AGTTGAAAATCTCTTAGAAGGAATAAATCCAAGTCCAAGTTGTGTTATTTTCAAGATATGGGATTT
CCTTGGACTACTCAAATTGCACAAAATTTTAATATCCCAAGAATTGTTTTTTCATGGTACTTGT
GTTTCTCACTTTTATGTTCTCTATAAAATACTTTCTCTCCAACATTCTTGAAAATATAACCTCAGA
TTCAGAGTATTTTGTGTTCTCTGATTTACCCGATAGAGTTGAACTAACGAAAGCTCAGGTTTCA
GGATCGACGAAAAATACTACTTCTGTAGTTCTTCTGTATTGAAAGAAGTTACTGAGCAAATCA
GATTAGCCGAGGAATCATCATATGGTGTAATTGTTAATAGTTTTGAGGAGTTGGAGCAAGTGTA
TGAGAAAGAATATAGGAAAGCTAGAGGGGAAAAAAGTTTTGGTGTGTTGGTCTCTGTTTCTTTGTGT
AATAAGGAAATTGAAGATTTGGTTACAAGGGGTAATAAACTGCAATTGATAATCAAGATTGCT
TGAAATGGTTAGATAATTTTGAAACAGAATCTGTGGTTTATGCAAGTCTTGGAAGTTTATCTCG
TTTGACATTATTGCAAATGGTGGAACTTGGTCTTGGTTTAGAAGAGTCAAATAGGCCTTTTGTA
TGGGTATTAGGAGGAGGTGATAAATTAATGATTTAGAGAAATGGATTCTTGAGAATGGATTG
AGCAAAGAATTAAAGAAAGAGGAGTTTTGATTAGAGGATGGGCTCCTCAAGTGCTTATACTTTC
ACACCCTGCAATTGGTGGAGTATTGACTCATTGCGGATGGAATTCTACATTGGAAGGTATTTCA
GCAGGATTACCAATGGTAACATGGCCACTATTTGCTGAGCAATTTTGCAATGAGAAGTTAGTAG
TCCAAGTGCTAAAAAATTGGAGTGAGCCTAGGTGTGAAGGTGCCTGTCAAATGGGGAGATGAGGA
AAATGTTGGAGTTTTTGGTAAAAAAGGATGATGTTAAGAAAGCATTAGACAACTAATGGATGAA
GGAGAAGAAGGACAAGTAAGAAGAACAAAAGCAAAAGAGTTAGGAGAATTGGCTAAAAAGGCAT
TTGGAGAAGGTGGTTCTTCTTATGTTAACTTAACATCTCTGATTGAAGACATCATTGAGCAACA
AAATCACAAGGAAAAATAG Amino Acid Glycosyltransferase (NtGT1b) *Nicotiana tabacum*
SEQ ID NO. 40

MKTAELVFIPAPGMGHLVPTVEVAKQLVDRHEQLSITVLIMTIPLTNIPSYTKSLSSDYSSRI
TLLPLSQPETSVTMSSFNAINFFEYISSYKGRVKDAVSETSFSSNSVKLAGFVIDMFCTAMID
VANEFGIPSYVFYTSSAAMLGLQLHFQSLSIECSPKVHNYVEPESEVLITYMNPVPVKCLPGI
ILVNDESSTMFVNHARRFRETKGIMVNTFTELESHALKALSDDEKIPPIYPVGPILNLENGNED
HNQEYDAIMKWLDKPNSSVVFLCFGSKGSFEEDQVKEIANALESSGYHFLWSLRRPPPKDKLQ
FPSEFENPEEVLPEGFFQRTKGRGKVIGWAPQLAILSHPSVGGFVSHCGWNSTLESVRSGVPIA
TWPLYAEQQSNAFQLVKDLGMAVEIKMDYREDFNTRNPPLVKAEEIEDGIRKLMDSENKIRAKV
TEMKDKSRAALLEGSSYVALGHFVETVMKN DNA Glycosyltransferase (NtGT1b) *Nicotiana tabacum* SEQ ID NO. 41
ATGAAGACAGCAGAGTTAGTATTCATTCCTGCTCCTGGGATGGGTCACCTTGTACCAACTGTGG

AGGTGGCAAAGCAACTAGTCGACAGACGACAGCAGCTTTCGATCATCAGAGTTCTAATCATGACAAAT
TCCTTTGGAAACAAATATTCCATCATATACTAAATCACTGTCCTCAGACTACAGTTCTCGTATA
ACGCTGCTTCCACTCTCTCAACCTGAGACCTCTGTTACTATGAGCAGTTTTAATGCCATCAATT
TTTTTGAGTACATCTCCAGCTACAAGGGTCGTGTCAAAGATGCTGTTAGTGAAACCTCCTTTAG
TTCGTCAAATTCTGTGAAACTTGCAGGATTTGTAATAGACATGTTCTGCACTGCGATGATTGAT
GTAGCGAACGAGTTTGAATCCCAAGTTATGTGTTCTACACTTCTAGTGCAGCTATGCTTGGAC
TACAACTGCATTTTCAAAGTCTTAGCATTGAATGCAGTCCGAAAGTTCATAACTACGTTGAACC
TGAATCAGAAGTTCTGATCTCAACTTACATGAATCCGGTTCCAGTCAAATGTTTGCCCGGAATT
ATACTAGTAAATGATGAAAGTAGCACCATGTTTGTCAATCATGCACGAAGATTCAGGGAGACGA
AAGGAATTATGGTGAACACGTTCACTGAGCTTGAATCACACGCTTTGAAAGCCCTTTCCGATGA
TGAAAAAATCCCACCAATCTACCCAGTTGGACCTATACTTAACCTTGAAAATGGGAATGAAGAT
CACAATCAAGAATATGATGCGATTATGAAGTGGCTTGACGAGAAGCCTAATTCATCAGTGGTGT
TCTTATGCTTTGGAAGCAAGGGGTCTTTCGAAGAAGATCAGGTGAAGGAAATAGCAAATGCTCT
AGAGAGCAGTGGCTACCACTTCTTGTGGTCGCTAAGGCGACCGCCACCAAAAAGACAAGCTACAA
TTCCCAAGCGAATTCGAGAATCCAGAGGAAGTCTTACCAGAGGGATTCTTTCAAAGGACTAAAG
GAAGAGGAAAGGTGATAGGATGGGCACCCCAGTTGGCTATTTTGTCTCATCCTTCAGTAGGAGG
ATTCGTGTCGCATTGTGGGTGGAATTCAACTCTGGAGAGCGTTCGAAGTGGAGTGCCGATAGCA
ACATGGCCATTGTATGCAGAGCAACAGAGCAATGCATTTCAACTGGTGAAGGATTTGGGTATGG
CAGTAGAGATTAAGATGGATTACAGGGAAGATTTTAATACGAGAAATCCACCACTGGTTAAAGC
TGAGGAGATAGAAGATGGAATTAGGAAGCTGATGGATTCAGAGAATAAAATCAGGGCTAAGGTG
ACGGAGATGAAGGACAAAAGTAGAGCAGCACTGCTGGAGGGCGGATCATCATATGTAGCTCTTG
GGCATTTTTGTGAGACTGTCATGAAAACTAG Amino Acid Glycosyltransferase (NtGT1a)

Nicotiana tabacum SEQ ID NO. 42

MKTTELVFIPAPGMGHLVPTVEVAKQLVDRDEQLSITVLIMTLPLETNIPSYTKSLSSDYSSRI
TLLQLSQPETSVSMSFFNAINFFEYISSYKDRVKDAVNETFSSSSSVKLKGFVIDMFCTAMIDV
ANEFGIPSYVFYTSNAAMLGLQLHFQSLSIEYSPKVHNYLDPSEVAISTYINPIPVKCLPGII
LDNDKSGTMFVNHARRFRETKGIMVNTFAELESHALKALSDDEKIPPIYPVGPILNLGDGNEDH
NQEYDMIMKWLD EQPHSSVFLCFGSKGSFEEDQVKEIANALERSGNRFLWSLRPPPKDTLQF
PSEFENPEEVLPVGFFQRTKGRGKVIGWAPQLAILSHPAVGGFVSHCGWNSTLESVRSGVPIAT
WPLYAEQQSNAFQLVKDLGMAVEIKMDYREDFNKTNPPLVKAEIEIDGIRKLMDSENKIRAKVM
EMKDKSRAALLEGSSYVALGHFVETVMKN DNA Glycosyltransferase (NtGT1a) *Nicotiana*

tabacum SEQ ID NO. 43

ATGAAGACAACAGAGTTAGTATTTCATTCCTGCTCCTGGCATGGGTCACCTTGTACCCACTGTGG
AGGTGGCAAAGCAACTAGTCGACAGAGACGAACAGCTTTCAATCACAGTTCTCATCATGACGCT
TCCTTTGGAAACAAATATTCCATCATATACTAAATCACTGTCCTCAGACTACAGTTCTCGTATA
ACGCTGCTTCAACTTTCTCAACCTGAGACCTCTGTTAGTATGAGCAGTTTTAATGCCATCAATT
TTTTTGAGTACATCTCCAGCTACAAGGATCGTGTCAAAGATGCTGTTAATGAAACCTTTAGTTC
GTCAAGTTCTGTGAAACTCAAAGGATTTGTAATAGACATGTTCTGCACTGCGATGATTGATGTG
GCGAACGAGTTTGAATCCCAAGTTATGTCTTCTACACTTCTAATGCAGCTATGCTTGGACTCC
AACTCCATTTTCAAAGTCTTAGTATTGAATACAGTCCGAAAGTTCATAATTACCTAGACCCTGA
ATCAGAAGTAGCGATCTCAACTTACATTAATCCGATTCCAGTCAAATGTTTGCCCGGGATTATA
CTAGACAATGATAAAAGTGGCACCATGTTTCGTCAATCATGCACGAAGATTCAGGGAGACGAAAG
GAATTATGGTGAACACATTCGCTGAGCTTGAATCACACGCTTTGAAAGCCCTTTCCGATGATGA
GAAAATCCCACCAATCTACCCAGTTGGGCCTATACTTAACCTTGAGAGATGGGAATGAAGATCAC
AATCAAGAATATGATATGATTATGAAGTGGCTCGACGAGCAGCCTCATTTCATCAGTGGTGTTC
TATGCTTTGGAAGCAAGGGATCTTTCGAAGAAGATCAAGTGAAGGAAATAGCAAATGCTCTAGA
GAGAAGTGGTAACCGGTTCTTGTGGTCGCTAAGACGACCGCCACCAAAAAGACACGCTACAATTC
CCAAGCGAATTCGAGAATCCAGAGGAAGTCTTGCCGGTGGGATTCTTTCAAAGGACTAAAGGAA
GAGGAAAGGTGATAGGATGGGCACCCCAGTTGGCTATTTTGTCTCATCCTGCAGTAGGAGGATT
CGTGTCTGCATTGTGGGTGGAATTCAACTTTGGAGAGTGTTTCGTAGTGGAGTACCGATAGCAACA
TGGCCATTGTAATGCAGAGCAACAGAGCAATGCATTTCACTGGTGAAGGATTTGGGGATGGCAG
TGGAGATTAAGATGGATTACAGGGAAGATTTTAATAAGACAAATCCACCACTGGTTAAAGCTGA
GGAGATAGAAGATGGAATTAGGAAGCTGATGGATTCAGAGAATAAAATCAGGGCTAAGGTGATG

GAGTAGAAGGACCAAGTAGAGCAGCGTATTAGAAAGCGGATCATCATATGTAGCTCTCGGGC
ATTTTGTGAGACTGTCATGAAAACTAA Amino Acid Glycosyltransferase (NtGT3)

Nicotiana tabacum SEQ ID NO. 44

MKETKKIELVFIPSPGIGHLVSTVEMAKLLIAREEQLSITVLIIQWPNDKKLDSYIQSVANFSS
RLKFIRLPQDDSIMQLLKSNIFFTFIASHKPAVRDAVADILKSESNNTLAGIVIDLFCTSMIDV
ANEFELPTYVIFYTSGAATLGLHYHIQNLRDEFNKDITKYKDEPEEKLSIATYLNPFPAKCLPSV
ALDKEGGSTMFLDLAKRFRETKGIMINTFLELESYALNSLSRDKNLPPIYPVGPVLNLNNVEGD
NLGSSDQNTMKWLDDQPASSVVFLCFGSGGSFEKHQVKEIAYALESSGCRFLWSLRPPTEDAR
FPSNYENLEEILPEGFLERTKGIGKVIGWAPQLAILSHKSTGGFVSHCGWNSTLESTYFGVPIA
TWPMYAEQQANAFQLVKDLRMGVEIKMDYRKDMKVMGKEVIVKAEIEKAIREIMDSESEIRVK
VKEMKEKSRAAQMEGGSSYTSIGGFIQIIMENSQ DNA Glycosyltransferase (NtGT3) *Nicotiana*

tabacum SEQ ID NO. 45

ATGAAAGAAACCAAGAAAATAGAGTTAGTCTTCATTCCTTCACCAGGAATTGGCCATTTAGTAT
CCACAGTTGAAATGGCAAAGCTTCTTATAGCTAGAGAAGAGCAGCTATCTATCACAGTCCTCAT
CATCCAATGGCCTAACGACAAGAAGCTCGATTCTTATATCCAATCAGTCGCCAATTTAGCTCG
CGTTTGAAATTCATTCGACTCCCTCAGGATGATTCCATTATGCAGCTACTCAAAGCAACATTT
TCACCACGTTTATTGCCAGTCATAAGCCTGCAGTTAGAGATGCTGTTGCTGATATTCTCAAGTC
AGAATCAAATAATACGCTAGCAGGTATTGTTATCGACTTGTTCTGCACCTCAATGATAGACGTG
GCCAATGAGTTCGAGCTACCAACCTATGTTTTCTACACGTCTGGTGCAGCAACCCTTGGTCTTC
ATTATCATATACAGAATCTCAGGGATGAATTTAACAAAGATATTACCAAGTACAAAGACGAACC
TGAAGAAAACTCTCTATAGCAACATATCTCAATCCATTTCCAGCAAAAATGTTTGCCGTCTGTA
GCCTTAGACAAAGAAGGTGGTTCAACAATGTTTCTTGATCTCGCAAAAAGGTTTCGAGAAACCA
AAGGTATTATGATAAACACATTTCTAGAGCTCGAATCCTATGCATTAAACTCGCTCTCACGAGA
CAAGAATCTTCCACCTATATACCCTGTCGGACCAGTATTGAACCTTAACAATGTTGAAGGTGAC
AACTTAGGTTTCATCTGACCAGAATACTATGAAATGGTTAGATGATCAGCCCGCTTCATCTGTAG
TGTTCCTTTGTGTTTGGTAGTGGTGGAAGCTTTGAAAAACATCAAGTTAAGGAAATAGCCTATGC
TCTGGAGAGCAGTGGGTGTCGGTTTTTGTGGTCGTTAAGGCGACCACCAACCGAAGATGCAAGA
TTTCCAAGCAACTATGAAAATCTTGAAGAAATTTTGCCAGAAGGATTCTTGGAAGAACAAG
GGATTGGAAGAGTGATAGGATGGGCACCTCAGTTGGCGATTTTGTACATAAATCGACGGGGGG
ATTTGTGTCGCACTGTGGATGGAATTCGACTTTGGAAAGTACATATTTTGGAGTGCCAATAGCA
ACCTGGCCAATGTACGCGGAGCAACAAGCGAATGCATTTCAATTGGTTAAGGATTTGAGAATGG
GAGTTGAGATTAAAGATGGATTATAGGAAGGATATGAAAGTGATGGGCAAAGAAGTTATAGTGAA
AGCTGAGGAGATTGAGAAAGCAATAAGAGAAATTATGGATTCCGAGAGTGAAATTCGGGTGAAG
GTGAAAGAGATGAAGGAGAAGAGCAGAGCAGCACAAATGGAAGGTGGCTCTTCTTACACTTCTA
TTGGAGGTTTCATCCAAATTATCATGGAGAATTCTCAATAA Amino Acid Glycosyltransferase

(NtGT2) *Nicotiana tabacum* SEQ ID NO. 46

MVQPHVLLVTFPAQGHINPCLQFAKRLIRMGIEVTFATSVFAHRRMAKTTTSTLSKGLNFAAFS
DGYDDGFKADEHDSQHMYMSEIKSRGSKTLKDIILKSSDEGRPVTSLVYSLLL PWA AKVAREFHI
PCALLWIQPATVLDIYYYYFNGYEDAIKGSTNDPNWCIQLPRLPLLKSQDLPSFLLSSSNEEKY
SFALPTFKEQLD TL DVEENPKVLVNTFDAL EPKEL KAIEKYNLIGIGPLIPSTFLDGKDPLDSS
FGGDLFQKSNDYIEWLNSKANSSVYISFGSLLNLSKNQKEEIAKGLIEIKPFLWVIRDQENG
KGDEKEEKLSCMMELEKQGKIVPWCSQLEVLTHPSIGCFVSHCGWNSTLESLSGVS VVA FPHW
TDQGTNAKLIEDVWKTGVRLKKNEDGVVESEEIKRCIEMVMDGGEKGEEMRRNAQKWKELAREA
VKEGGSSEMNLKAFVQEVGKGC DNA Glycosyltransferase (NtGT2) *Nicotiana tabacum* SEQ

ID NO. 47

ATGGTGCAACCCCATGTCCTCTTGGTGACTTTTCCAGCACAAGGCCATATTAATCCATGTCTCC
AATTTGCCAAGAGGCTAATTAGAATGGGCATTGAGGTAAC TTTTGCCACGAGCGTTTTCGCCCA
TCGTCTGATGGCAAAA ACTACGACTTCCACTCTATCCAAGGGCTTAAATTTTGCGGCATTCTCT
GATGGGTACGACGATGGTTTCAAGGCCGATGAGCATGATTCTCAACATTACATGTCTGGAGATAA
AAAGTCGCGGTTCTAAAACCCTAAAAGATATCATTTTGAAGAGCTCAGACGAGGGACGTCCTGT
GACATCCCTCGTCTATTCTCTTTTGCTTCCATGGGCTGCAAAGGTAGCGCGTGAATTTACATA
CCGTGCGCGTTACTATGGATTCAACCAGCAACTGTGCTAGACATATATTATTACTTCAATG
GCTATGAGGATGCCATAAAAGGTAGCACCAATGATCCAAATTGGTGTATTCAATTGCCTAGGCT

TCCACTACTAAAAAGCCAAGATCTTCCTTCTTTTTTACTTTTCTTCTAGTAATGAAGAAAAATAT
AGCTTTGCTCTACCAACATTTAAAGAGCAACTTGACACATTAGATGTTGAAGAAAATCCTAAAG
TACTTGTGAACACATTTGATGCATTAGAGCCAAAGGAAGCTCAAAGCTATTGAAAAGTACAATTT
AATTGGGATTGGACCATTGATTCCTTCAACATTTTTTGGACGGAAAAGACCCCTTTGGATTCTTCC
TTTGGTGGTGATCTTTTTCAAAGTCTAATGACTATATTGAATGGTTGAACTCAAAGGCTAACT
CATCTGTGGTTTATATCTCATTGGGAGTCTCTTGAATTTGTCAAAAAATCAAAGGAGGAGAT
TGCAAAGGGTTGATAGAGATTAAAAAGCCATTCTTGTGGGTAATAAGAGATCAAGAAAATGGT
AAGGGAGATGAAAAAGAAGAGAAATTAAGTTGTATGATGGAGTTGGAAAAGCAAGGGAAAATAG
TACCATGGTGTTCACAACCTTGAAGTCTTAACACATCCATCTATAGGATGTTTCGTGTCACATTG
TGGATGGAATTCGACTCTGGAAGTTTATCGTCAGGCGTGTTCAGTAGTGGCATTTCCTCATTGG
ACGGATCAAGGGACAAATGCTAACTAATTGAAGATGTTTGGGAAGACAGGTGTAAGGTTGAAAA
AGAATGAAGATGGTGTGGTTGAGAGTGAAGAGATAAAAAGGTGCATAGAAATGGTAATGGATGG
TGAGAGAGAAAGGAGAAGAAATGAGAAGAAATGCTCAAAAATGGAAAGAATTGGCAAGGGGAAGCT
GTAAAAGAAGGCGGATCTTCGGAATGAATCTAAAAGCTTTTGTTCAAGAAGTTGGCAAAGGTT
GCTGA Amino Acid THCA Synthase *Cannabis* SEQ ID NO. 48

ATGAAAACAACAGAAGCTTGTCTTCATACCCGCCCGGATATGGGTCACCTTGTACCCACAGTCG
AAGTCGCCAAACAAGTAGTTGATAGAGACGAACAGTTGTCTATTACCGTCTTGATAATGACGTT
ACCCCTGGAGACTAATATCCCAAGTTACACCAAGAGTTTGTCTCTGACTATTTCATCCCGTATC
ACGTTGTTACAATAAGTCAACCTGAGACGAGTGTCTCAATGAGTAGTTTTTAACGCCATAAACT
TCTTCGAATACATTAGTTCCTATAAGGATCGTGTTAAAGATGCCGTAAACGAGACATTCTCCTC
TTCATCCTCCGTCAAACCTTAAAGGATTTGTAATCGACATGTTTTGCACGGCAATGATAGACGTG
GCCAACGAGTTCGGTATTCCATCTTATGTATTCTACACGTCCAACGCTGCCATGCTAGGCCTAC
AACTTCACTTCCAATCCTTGTCCATCGAATATTCACCTAAGGTTTATAATTATTTAGACCTGA
ATCTGAGGTAGCTATATCAACGTACATTAACCCAATACCAGTAAATGCTTACCCGGTATAATT
CTTGACAATGATAAGAGTGGCACTATGTTTCGTAAACCATGCCAGGAGATTCCGTGAAACAAAGG
GTATAATGGTAAATACTTTTGCAGAATTAGAAAGTCACGCCCTAAAGGCACTTAGTGACGATGA
GAAAATTCCTCCAATCTATCCCGTCGGACCCATTCTAAACTTGGGTGATGGTAATGAGGATCAT
AACCAAGAGTACGACATGATAATGAAATGGCTGGATGAACAACCACACAGTTCAGTGGTTTTCC
TGTGCTTCGGTTCCAAAGGTTCAATTTGAAGAAGACCAGGTTAAAGAGATAGCAAATGCTTTAGA
GAGATCAGGCAATAGGTTCCCTGTGGAGTTTAAAGACGTCCCCCTCCCAAGGATACTCTTCAATTC
CCTTCCGAATTTGAAAACCCCGAGGAAGTGCTACCTGTAGGATTTTTTTCAAAGAACCAAAGGCA
GAGGAAAAGTCATCGGATGGGCACCACAGCTTGCAATTCTATCTCACCCCTGCCGTGGTGATT
CGTTTCCCCTGCGGCTGGAATAGTACTTTGGAATCAGTTAGATCAGGTGTACCCATAGCAACA
TGGCCTCTTTATGCAGAGCAGCAGTCCAATGCATTTCAATTGGTCAAGGATCTAGGTATGGCCG
TCGAAATTAAATGGATTACCGTGAGGACTTTAACAAGACTAATCCTCCATTGGTAAAGGCAGA
GGAAATAGAAGACGGCATTAGGAAGTTGATGGACTCCGAGAATAAGATTAGGGCAAAGGTGATG
GAAATGAAAGATAAGTCCAGAGCTGCATTACTGGAAGGAGGATCCTCCTATGTTGCACTGGGTC
ACTTCGTGGAGACCGTAATGAAGAACTAA Amino Acid Glycosyltransferase (NtGT1b -
generated from codon optimized sequence for yeast expression) *Nicotiana tabacum* SEQ
ID NO. 50

LDNDKMSGTMFVNHRFTKIVNTFAELESHALSKALSDDEKIPPIYPVGPILNLGDGNEHDH
NQEYDMIMKWLD EQPHSSVVF LCFGSKGSFEEDQVKEIANALERSGNRFLWSLRPPPKDTLQF
PSEFENPEEVLPVGGFFQRTKGRGKVIGWAPQLAILSHPAVGGFVSHCGWNSTLESVRSGVPIAT
WPLYAEQQSNAFQLVKDLGMAVEIKMDYREDFNKTNPPLVKAEIEIDGIRKLMDSENKIRAKVM
EMKDKSRAALLEGSSYVALGHFVETVMKN DNA Glycosyltransferase (NtGT2 - codon
optimized for yeast expression) *Nicotiana tabacum* SEQ ID NO. 51
ATGGTTCAACCACACGTCTTACTGGTTACTTTTCCAGCACAAAGGCCATATCAACCCTTGCCTAC
AATTCGCCAAAAGACTAATAAGGATGGGCATCGAAGTAAC TTTTGCCACGAGTGTATTCGCACA
TAGGCGTATGGCTAAAACTACGACATCAACTTTGTCCAAAGGACTAAACTTCGCCGCCTTCAGT
GATGGCTATGACGATGGATTCAAAGCCGACGAACATGACAGTCAACACTACATGAGTGAAATAA
AGTCCCGTGGATCTAAACACTTAAGGATATTATACTTAAATCCTCCGATGAGGGAAGACCCGT
TACCTCTTTAGTTTATTCACTGTTACTGCCCTGGGCTGCAAAAGTCGCCAGAGAGTTTCATATT
CCTTGCGCTTTATTGTGGATCCAACCAGCTACGGTATTAGACATCTACTATTACTACTTCAATG
GATACGAGGATGCAATAAAGGGATCAACAAACGACCCCAACTGGTGTATTCAACTGCCTAGACT
TCCTCTATTAAAAAGTCAGGACTTACCTAGTTTTTTTACTGTCATCCAGTAACGAAGAAAAATAT
TCATTCGCTTTACCCACCTTCAAAGAGCAGCTTGACACTTTGGATGTTGAAGAGAACCCCAAGG
TTTTGGTCAATACTTTTGACGCTTTGGAGCCAAAAGAGCTAAAGGCTATTGAAAAATATAACCT
TATCGGCATAGGACCTTTAATCCCCTCTACTTTCTTAGATGGCAAAGACCCTCTAGATTCAAGT
TTCGGAGGTGATTTGTTTCAAAGAGTAACGATTATATCGAGTGGCTAAATAGTAAAGCCAAC T
CCAGTGTGGTCTACATTTCTTTCCGAAGTCTTCTGAATTTATCAAAAAACCAAAGGAAGAGAT
CGCAAAAGGACTGATAGAGATAAAAAAACCTTTCTTATGGGTGATCAGAGACCAGGAAAACGGT
AAAGGCGATGAGAAGGAGGAAAAACTGTCCTGTATGATGGAGCTAGAGAAACAAGGAAAAATCG
TTCCCTGGTGTTCACAGTTAGAAGTGTTAACCCATCCATCCATAGGTTGCTTCGTATCACATTG
TGGTTGGAATAGTACACTTGAAAGTCTTTCATCAGGCGTCTCTGTCTCGTCGCATTCCCCCACTGG
ACGGACCAGGGCACAAACGCCAAACTGATCGAAGATGTATGGAAGACGGGGCGTCAGGCTAAAAA
AAAATGAGGATGGCGTGGTAGAGAGTGAAGAGATAAAGCGTTGCATAGAAATGGTCATGGATGG
CGGTGAAAAGGGAGAGGAAATGAGGCGTAACGCACAAAAGTGGAAGGAACTAGCCCGTGAAGCA
GTGAAAGAAGGAGGTTCTAGTGAGATGAATTTAAAAGCTTTCGTGCAGGAAGTTGGAAAAGGCT
GCTGA Amino Acid Glycosyltransferase (NtGT2 - generated from codon optimized
sequence for yeast expression) *Nicotiana tabacum* SEQ ID NO. 52
MVQPHVLLVTFPAQGHINPCLQFAKRLIRMGIEVTFATSVFAHRRMAKTTTSTLSKGLNFAAFS
DGYDDGFKADEHDSQH YMSEIKSRGSKTLKDIILKSSDEGRPV TSLVYSLLLPWAAKVAREFHI
PCALLWIQPATVLDIYYYYFNGYEDAIKGSTNDPNWCIQLPRLPLLKSQDLPSFLLSSSNEEKY
SFALPTFKEQLD TLDVEENPKVLVNTFDAL EPKELKAIEKYNLIGIGPLIPSTFLDGKDPLDSS
FGGDLFQKSNDYIEWLNSKANSSV VYISFGSLLNLSKNQKEEIAKGLIEIKPFLWVIRDQENG
KGDEKEEKLSCMMELEKQGKIVPWCSQLEVLTHPSIGCFVSHCGWNSTLESLS SGVSVVAFPHW
TDQGTNAKLIEDVWKTGVRLKKNEDGVVESEEIKRCIEMVMDGGGEKGEEMRRNAQKWKELAREA
VKEGGSSEMNLKAFVQEVGKGC DNA Glycosyltransferase (NtGT3 - codon optimized
for yeast expression) *Nicotiana tabacum* SEQ ID NO. 53
ATGAAAGAGACTAAAAAAATTGAGTTAGTTTTTATCCCCAGTCCTGGTATAGGACACTTAGTCT
CAACTGTGGAGATGGCCAAACTGTTGATAGCCCGTGAAGAGCAACTTTCTATTACTGTCCTGAT
TATACAATGGCCTAATGATAAAAAGCTAGACAGTTATATCCAGTCCGTCGCAAAC TTTAGTTCT
AGACTGAAGTTTATACGTCTGCCCCAAGATGACTCAATCATGCAACTTTTGAAATCAAACATTT
TCACGACATTCATCGCCTCTCACAAGCCAGCTGTAAGAGACGCCGTTGCTGACATACTAAAGAG
TGAAAGTAATAACACATTGGCAGGCATTGTAATCGATCTTTTCTGCACATCCATGATCGATGTA
GCCAATGAGTTTGAGCTGCCTACTTATGTGTTTTACACTAGTGGCGCAGCCACGTTGGGTCTGC
ACTACCATATTCAAATCTGCGTGATGAGTTTAATAAAGACATTACCAAATATAAGGATGAGCC
AGAAGAAAAATTAAGTATAGCCACGTACCTTAACCCATTCCCTGCTAAGTGTCTACCCTCCGTG
GCATTGGATAAGGAAGGAGGATCAACGATGTTCCCTAGACTTAGCTAAGAGGTT CAGGGAGACCA
AAGGCATAATGATTAACACTTTTCTTGAGCTGGAATCATACTGCTCTAAACTCATTGTCTAGAGA
TAAAAACTTGCCCCCTATATACCCTGTAGGCCCTGTTTTGAACTTGAACAACGTTGAGGGTGAT
AACTTGGGCTCTAGTGATCAAATACCATGAAATGGCTGGACGACCAGCCAGCTTCTTCCGTTG
TGTTCCCTATGTTTTGGCTCAGGAGGAAGTTTCGAAAAACACCAAGTCAAAGAAATAGCTTATGC

CTTAGAATCTTCGCTGCTGAGTTCTTGCTGGAGTTTGGCTAGACCTCCACGGGAAGATGCTAGG
TTCCCTTCTAATTACGAAAACCTTAGAGGAAATTTTACCAGAGGGATTCTGGAAGAACGAAAG
GCATTGGTAAGGTCATTGGATGGGCCCCACAGTTAGCAATCTTGTCTCACAAGTCCACAGGAGG
ATTCGTGTCTCATTGCGGATGGAACCTCTACCCTTGAAAGTACCTATTTTCGGCGTTCTCTATTGCT
ACTTGGCCAATGTATGCTGAACAACAGGCCAACGCTTTTCAACTTGTTAAAGATTTGAGGATGG
GTGTTGAGATCAAAATGGATTATAGGAAGGATATGAAGGTAATGGGCAAGGAGGTTATCGTTAA
GGCAGAAGAAATTGAAAAGGCCATAAGGGAAATCATGGACTCAGAATCAGAAATCAGGGTCAAG
GTCAAAGAGATGAAGGAGAAAAGTCGTGCAGCCCAAATGGAAGGAGGATCATCATATACCTCTA
TCGGCGGCTTCATTCAAATAATCATGGAGAACTCACAGTAA Amino Acid

Glycosyltransferase (NtGT3 - generated from codon optimized sequence for yeast expression) *Nicotiana tabacum* SEQ ID NO. 54

MKETKKIELVFIPSPGIGHLVSTVEMAKLLIAREEQLSITVLIHQWPNDKKLDSYIQSVANFSS
RLKFIRLPQDDSIMQLLKSNIFFTTFIASHKPAVRDAVADILKSESNNLAGIVIDLFCTSMIDV
ANEFELPTYVIFYTSGAATLGLHYHIQNLRDEFNKDITKYKDEPEEKLSIATYLNPFPAKCLPSV
ALDKEGGSTMFLDLAKRFRETKGIMINTFLELESYALNSLSRDKNLPPIYPVGPVLNLNNVEGD
NLGSSDQNTMKWLDDQPASSVVFLCFGSGGSFEKHQVKEIAYALESSGCRFLWSLRPPTEDAR
FPSNYENLEEILPEGFLERTKGIGKVIGWAPQLAILSHKSTGGFVSHCGWNSTLESTYFGVP
IA TWPMYAEQQANAFQLVKDLRMGVEIKMDYRKDMKVMGKEVIVKAEEIEKAIREIMDSESEIRVK
VKEMKEKSRAAQMEGGSSYTSIGGFIQIIMENSQ DNA UDP-glycosyltransferase 73C3

(NtGT4 - codon optimized for yeast expression) *Nicotiana tabacum* SEQ ID NO. 55

ATGGCTACTCAGGTGCATAAATTGCATTTCTATTCTGTTCCCACTGATGGCTCCCGGTCACATGA
TCCCTATGATAGACATCGCAAAACTATTGGCTAACCGTGGCGTGATACTACCATAATAACTAC
GCCCGTTAACGCCAATCGTTTTTCTCTACGATCACTAGGGCCATTAAATCAGGCCTAAGAATC
CAGATTTTAACCTTAAAATTCCCATCAGTTGAGGTAGGCCTGCCTGAAGGATGTGAAAACATCG
ACATGTTGCCATCTTTGGACTTAGCCTCTAAATTCTTTGCTGCTATTTCTATGCTTAAACAACA
AGTGGAGAACTTGCTAGAGGGTATTAACCCTAGTCCCTCATGCGTTATTTCTGACATGGGCTTC
CCATGGACGACACAGATCGCTCAAATTTCAATATTCCTCGTATCGTATTTTCATGGCACGTGTT
GCTTTTCTCTTCTTTGTTCTTACAAAATCCTGTCATCCAATATCTTAGAGAACATTACTAGTGA
CTCAGAGTATTTTGTCTGCGCCAGATCTGCCAGACCGTGTCTGAGCTAACTAAGGCCCAAGTCTCT
GGATCTACAAAGAATACTACATCAGTAAGTAGTTCAGTACTGAAGGAGGTTACAGAGCAGATCA
GGCTTGCAGAGGAATCATCCTACGGTGTGATAGTTAATTCCTTCGAAGAACTGGAACAGGTGTA
TGAAAAAGAGTACAGAAAAGCCAGGGGCAAAAAGGTCTGGTGCGTGGGTCCTGTCTCTTTGTGC
AACAAGGAGATTGAAGATCTTGTTACTAGAGGAAACAAAACCGCTATAGACAATCAGGATTGTC
TTAAGTGGTTAGACAACCTTCGAGACTGAATCCGTCGTCTATGCAAGTTTAGGCTCACTAAGTAG
GCTTACGTTACTGCAAATGGTTGAGCTGGGATTGGGACTGGAGGAGAGTAATAGGCCATTTGTA
TGGGTTCTGGGAGGAGGAGACAACTAAATGATCTTGAGAAATGGATATTGGAGAATGGCTTTG
AACAGCGTATAAAGGAGAGAGGTGTCCTGATACGTGGCTGGGCACCTCAAGTATTGATTTTAAG
TCACCCCGCAATTGGAGGAGTTTAAACGCATTGTGGATGGAACCTCTACATTAGAGGGCATTTC
GCCGACTACCCATGGTCACCTGGCCACTATTTGCCGAACAGTTCTGTAACGAAAAATTAGTAG
TGCAGGTTCTTAAAATCGGTGTCTCACTTGAGGTGAAGGTCCCTGTTAAGTGGGGTGACGAAGA
GAACGTAGGTGTCTTAGTGAAAAAGGATGACGTTAAAAAAGCACTGGATAAGCTAATGGATGAG
GGTGAGGAGGGCCAGGTTAGGAGGACCAAAGCCAAAGAGCTTGGTGAGTTAGCTAAAAAAGCCT
TTGGAGAGGGCGGATCATCCTACGTGAACCTAACGTCCCTAATTGAAGATATAATCGAGCAGCA
GACCATAAGGAGAAGTAG Amino Acid UDP-glycosyltransferase 73C3 (NtGT4 -

generated from codon optimized sequence for yeast expression) *Nicotiana tabacum* SEQ ID NO. 56

MATQVHKLHFILFPLMAPGHMIPMIDIAKLLANRGVITTIITTPVNANRFSSTITRAIKSGLRI
QILTLKFPSVEVGLPEGCENIDMLPSLDLASKFFAAISMLKQQVENLLEGINPSPSCVISDMGF
PWTTQIAQNFNIPRIVFHGTCCFSLCSYKILSSNILENITSDSEYFVVPDLPDRVELTKAQVS
GSTKNNTSVSSSVLKEVTEQIRLAESSYGVIVNSFEELEQVYEKEYRKARGKKVWCVGPVSLC
NKEIEDLVTRGNKTAIDNQDCLKWLDNFETESVYASLSLSRLTLLQMVELGLGLEESNRPFV
WVLGGGDKLNDLEKWILENGFEQRIKERVLRGWAPQVLILSHPAIGGVLTHCGWNSTLEGIS
AGLPMVTWPLFAEQFCNEKLVVQVLKIGVSLGVKVPVKWGDEENVGLVKKDDVKKALDKLMDE

GGGQVRRRTKAKELGELAKKAFGEAGSSYVNLTSIEDIEQQNHKEK DNA
Glycosyltransferase (NtGT5 - codon optimized for yeast expression) *Nicotiana tabacum*
SEQ ID NO. 57
ATGGGCTCTATCGGTGCAGAACTAACCAAGCCACACGCCGTATGCATTCCCTATCCCGCCCAGG
GACACATAAATCCTATGCTGAAGTTAGCTAAGATACTGCATCACAAGGGCTTCCATATAACCTT
CGTAAATACGGAATTTAATCACAGGCGTCTGCTGAAGTCCAGAGGTCCTGACTCCCTGAAAGGT
CTTTCAAGTTTCAGGTTTCGAGACGATACCTGACGGACTGCCCCCATGCGAAGCTGACGCTACAC
AGGACATTCCCTTCACTGTGTGAATCCACGACTAATACATGTCTAGCTCCTTTTAGAGACCTACT
TGCTAAGCTAAATGATACGAATACTTCTAACGTCCCTCCCGTAAGTTGTATTGTCAGTGACGGA
GTGATGTCATTTACCCTTGCAGCTGCACAGGAAGTGGGTGTCCCAGAGGTTTTATTTTGGACTA
CATCTGCTTGTGGATTCTTAGGITACATGCACTATTGCAAAGTCATTGAAAAAGGATATGCTCC
ATTAAAAGACGCATCAGACCTGACGAATGGCTATCTTGAGACAACCTTGGACTTCATCCCCGGC
ATGAAGGACGTCAGGCTGAGAGACTTACCTTCCTTTCTTAGGACCACCAATCCAGACGAATTTA
TGATTAAGTTTGTACTACAGGAACTGAGCGTGCTCGTAAGGCCAGTGCCATAATACTTAATAC
CTTTGAAACCTTAGAGGCAGAGGTATTAGAATCATTAAAGGAACCTTCTACCCCCCGTCTATCCA
ATCGGCCCTTGCATTTCTTGTCAAACACGTAGACGATGAGAACCTAAAAGGTCTACGTTCCCT
CACTTTGGAAGGAGGAACCTGAATGTATTCAATGGTTAGACACCAAAGAACCTAACTCTGTCTG
GTACGTGAATTTTCGGATCCATTACTGTGATGACTCCCAATCAATTAATAGAGTTCGCTTGGGGA
CTGGCAAACCTCTCAACAGACCTTCCTTTGGATCATAAGGCCTGACATCGTAAGTGGTGATGCTT
CCATATTACCTCCCGAGTTTGTGAGGAGACTAAGAACAGAGGCATGCTTGCCTCCTGGTGCTC
TCAGGAGGAGGTACTATCCCATCCCGCAATAGTGGGATTTTTGACGCACTCTGGTTGGAACCTCA
ACTTTAGAATCAATTTCTAGTGGCGTCCCCATGATCTGTTGGCCTTTCTTTGCTGAGCAGCAA
CGAACTGCTGGTTTTTCAGTGACGAAGTGGGACGTTGGAATGGAAATTGATTCAGATGTGAAGAG
AGATGAAGTAGAGAGTTTAGTAAGAGAGTTAATGGTGGGTGGTAAAGGCAAGAAGATGAAGAAG
AAGGCAATGGAGTGGAAAGGAAGTGGCCGAGGCTICAGCAAAGAACACTCTGGCTCCTCTTACG
TCAATATCGAGAAGTTGGTTAACGATATATTACTATCTAGTAAGCACTAA Amino Acid
Glycosyltransferase (NtGT5 - generated from codon optimized sequence for yeast
expression) *Nicotiana tabacum* SEQ ID NO. 58

MGSIGAEELTKPHAVCIPYPAQGHINPMLKLAKILHHKGFHITFVNTEFNHRRLKSRGPDSLKG
LSSFRFETIPDGLPPCEADATQDIPSLCESTTNTCLAPFRDLLAKLNDTNTSNVPPVSCIIVSDG
VMSFTLAAAQELGVPEVLFWTTTACGFLGYMHYCKVIEKGYAPLKDASDLTNGYLETTLDIFPG
MKDVRLRDLPSFLRRTNPDEFMIKFVLQETERARKASAILNTFETLEAEVLESLRNLLPPVYP
IGPLHFLVKHVDDENLKGLRSSLWKEEPECIQWLDTKEPNSVVYVNFSGSITVMTPNQLIEFAWG
LANSQQTFLWIIRPDIVSGDASILPPEFVEETKNRGM LASWCSQEEVLSHPAIVGFLTHSGWNS
TLESISSGVPMICWPFFAEQQTNCWFSVTKWDVGMEIDSDVKRDEVESLVRELMVGGKGKKMKK
KAMEWKELAEASAKEHSGSSYV NIEKL VNDILLSSKH DNA UDP glycosyltransferase 76G1
(UGT76G1 - codon optimized for yeast expression) *Stevia rebaudiana* SEQ ID NO. 59
ATGGAGAACAAAACCGAGACAACCGTTAGGCGTAGACGTAGGATAATATTGTTTCCCGTGCCCT
TTCAAGGCCATATAAACCAATCCTGCAGCTAGCCAACGTATTGTACTCAAAGGGCTTCAGTAT
AACGATCTTCCACACCAACTTTAATAAGCCAAAAACGTCTAATTATCCACACTTCACATTTAGA
TTTATACTTGATAACGACCCACAGGATGAAAGAATATCAAACCTTGCCCACGCACGGCCCACTAG
CCGGAATGAGAATACCAATAATCAATGAGCATGGCGCCGACGAGTTGCGTAGAGAGCTGGAATT
GTTGATGCTAGCCAGTGAGGAAGACGAAGAGGTGTCCTGCTTAATAACGGATGCACTTTGGTAT
TTTGCTCAATCTGTGGCCGACTCCCTTAACCTGAGGCGTCTTGTCCTTATGACCTCCAGTCTAT
TCAACTTTCATGCCCATGTCTCATTGCCCCAATTTGATGAGCTTGGCTATTTGGATCCTGATGA
CAAACCTAGGCTGGAGGAACAGGCTTCCGGTTTTCCCATGCTAAAGGTTAAGGACATCAAATCC
GCCTACTCAAACCTGGCAGATCCTTAAGGAAATTCTTGGCAAATGATCAAACAGACGAGGGCAT
CCAGTGGCGTCATCTGGAACCTCCTTTAAGGAACCTTGAAGAATCAGAACTTGAAACAGTAATCAG
AGAAATACCTGCCCAAGTTTCTTGATCCCTCTACCTAAGCACCTTACGGCTTCTAGTTCTTCT
TTGTTGGACCACGATCGTACTGTCTTTCAATGGTTAGATCAGCAACCCCCCTCATCAGTGCTAT
ATGTGTCATTCCGGTAGTACATCAGAAGTGGACGAAAAGGATTCCTTGAGATAGCCCGTGGATT
GGTGGACTCTAAACAGTCCTTTTTATGGGTTGTGAGACCTGGATTTGTAAAGGGATCCACGTGG
GTCGAACCTTGCCCGATGGTTTCCTGGGTGAAAGAGGAAGGATAGTGAAGTGGGTCCCTCAGC

AGAGGTACCTGGTGGCCCATGGCTATAGAGTGGCTTTCTGGACCCACTCCGGCTGGGAATAGTACACT
AGAATCCGTTTTCGAGGGGTGTCCCTATGATTTTTTCTGATTTTGGTTTAGATCAACCCCTGAAT
GCTAGGTACATGTCAGACGTCCTTAAAGTCGGCGTCTACCTAGAAAATGGCTGGGAGAGGGGTG
AGATAGCAAACGCTATCAGACGTGTTATGGTAGACGAAGAGGGAGAGTACATAAGGCAAAACGC
CAGGGTCCTGAAACAAAAAGCCGATGTGTCCTTGATGAAGGGCGGCTCTTCATACGAAAGTCTA
GAAAGTCTTGTTTCTTATATTTCTCTCACTATAA Amino Acid UDP glycosyltransferase 76G1
(UGT76G1 - generated from codon optimized sequence for yeast expression) *Stevia
rebaudiana* SEQ ID NO. 60

MENKTETTVRRRRRIILFPVPFQGHINPILQLANVLYSKGFSITIFHTNFNKPKTSNYPHFTFR
FILDNDPQDERISNLPHTHGLAGMRIPINEHGADELRRLELLMLASEEDEEVSLITDALWY
FAQSVADSLNLRRLVLMTSSLFNFHAHVSLPQFDELGYLDPDDKTRLEEQASGFPMMLKVVDIKS
AYSNWQILKEILGKMIKQTRASSGVIWNSFKELEESELETVIREIPAPSFLIPLPKHLTASSSS
LLDHDRTVFQWLDQQPPSSVLYVSFGSTSEVDEKDFLEIARGLVDSKQSFLWVVRPGFVKGSTW
VEPLPDGFLGERGRIVKWVPQQEVLAHGAIGAFWTHSGWNSTLESVCEGVPMIFSDFGLDQPLN
ARYMSDVLKVG VYLENGWERGEIANAIRVMVDEEGEYIRQNARVLKQKADVSLMKGGSSYESL
ESLVSYISSL DNA glycosyltransferase (UGT73A10) *Lycium barbarum* SEQ ID NO. 61
ATGGGTCAATTGCATTTTTTTTTTGTTCATGATGGCTCAAGGTCATATGATTCCAACCTTTGG
ATATGGCTAAGTTGATTGCTTCTAGAGGTGTTAAGGCTACTATTATTACTACTCCATTGAACGA
ATCTGTTTTTTCTAAGGCTATTCAAAGAAACAAGCAATTGGGTATTGAAATTGAAATTGAAATT
AGATTGATTAAAGTTTCCAGCTTTGGAAAACGATTTGCCAGAAGATTGTGAAAGATTGGATTGTA
TTCCAACCTGAAGCTCATTGCCAAACTTTTTTAAGGCTGCTGCTATGATGCAAGAACCATTGGA
ACAATTGATTCAAGAATGTAGACCAGATTGTTTGGTTTCTGATATGTTTTTGCCATGGACTACT
GATACTGCTGCTAAGTTTAACATTCCAAGAATTGTTTTTCATGGTACTAACTACTTTGCTTTGT
GTGTTGGTGATTCTATGAGAAGAAACAAGCCATTTAAGAACGTTTCTTCTGATTCTGAAACTTT
TGTTGTTCCAAACTTGCCACATGAAATTAAGTTGACTAGAACTCAAGTTTCTCCATTTGAACAA
TCTGATGAAGAATCTGTTATGTCTAGAGTTTTGAAGGAAGTTAGAGAATCTGATTGGAAGTCTT
ACGGTGTTATTTTTAACTCTTTTTACGAATTGGAACCAGATTACGTTGAACATTACACTAAGGT
TATGGGTAGAAAGTCTTGGGCTATTGGTCCATTGTCTTTGTGTAACAGAGATGTTGAAGATAAG
GCTGAAAGAGGTAAGAAGTCTTCTATTGATAAGCATGAATGTTTGGAATGGTTGGATTCTAAGA
AGCCATCTTCTATTGTTTACGTTTGTGTTTGGTTCTGTTGCTAACTTTACTGTTACTCAAATGAG
AGAATTGGCTTTGGGTTTGGAAAGCTTCTGGTTTGGATTTTATTTGGGCTGTTAGAGCTGATAAC
GAAGATTGGTTGCCAGAAGGTTTTGAAGAAAGAACTAAGGAAAAGGGTTTGATTATTAGAGGTT
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GAACTCTACTTTGGAAGGTATTTCTGCTGGTGTTCCAATGGTTACTTGGCCAGTTTTTGCTGAA
CAATTTTTTAACGAAAAGTTGGTTACTCAAGTTATGAGAACTGGTGCTGGTGTTGGTTCTGTTC
AATGGAAGAGATCTGCTTCTGAAGGTGTTGAAAAGGAAGCTATTGCTAAGGCTATTAAGAGAGT
TATGGTTTCTGAAGAAGCTGAAGGTTTTAGAAACAGAGCTAGAGCTTACAAGGAAATGGCTAGA
CAAGCTATTGAAGAAGGTGGTTCTTCTTACACTGGTTTGACTACTTTGTTGGAAGATATTTCTT
CTTACGAATCTTTGTCTTCTGATTAA Amino Acid Glycosyltransferase (UGT73A10) *Lycium
barbarum* SEQ ID NO. 62

MGQLHFFLFPMMAQGHMIPITLDMAKLIASRGVKATIITPLNESVFSKAIQRNKQLGIEIEIEI
RLIKFPALENDLPEDCERLDLIPTAEHLNPFKAAAMMQEPLEQLIQECRPDCLVSDMFLPWTT
DTAAKFNIPIRVFHTGNYFALCVGDSMRRNKPFPKNVSSDSETFVVPNLPHEIKLTRTQVSPFEQ
SDEESVMSRVLKEVRES DLKSYGVIFNSFYELEPDYVEHYTKVMGRKSWAIGPLSLCNRDVEDK
AERGKKSSIDKHECLEWLD SKKPSSIVYVCFGSVANFTVTQMRELALGLEASGLDFIWAVRADN
EDWLPEGFEERTKEKGLIIRGWAPQVLILDHESVGA FVTHCGWNSTLEGISAGVPMVTPVFAE
QFFNEKLVTQVMRTGAGVGSVQWKRSASEGVEKEAIAKAIKRV MVSEEAEGFRNRARAYKEMAR
QAIEEGGSSYTGLTTLLEDISSYESLSSD DNA Cytosolic-targeted UDP glycosyltransferase
76G1 (cytUTG) *Stevia rebaudiana* SEQ ID NO. 63
ATGGAAAATAAAACCGAAACCACCGTCCGCCGTCGTCGCCGTATCATTCTGTTCCCGGTCCCGT
TCCAGGGCCACATCAACCCGATTCTGCAACTGGCGAACGTGCTGTATTGAAAGGTTTCAGCAT
CACCATCTTCCATACGAACTTCAACAAGCCGAAGACCAGCAATTACCCGCACTTTACGTTCCGT
TTTATTCTGGATAACGACCCGCAGGATGAACGCATCTCTAATCTGCCGACCCACGGCCCGCTGG

CGGGTATCGATTATCCAGTATCAACCAACAGCGCGAGATGACGACGCGGAACTTGGAACT
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TTTGCCCAAGTCTGTTGCAGATAGTCTGAACCTGCGTCGCCTGGTCCTGATGACCAGCAGCCTGT
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GTTCCGGCGTCATCTGGAATAGTTTCAAAGAACTGGAAGAATCCGAACTGGAAACGGTGATTCTG
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GGAATCGGTTTTGTGAAGGTGTCCCGATGATTTTCTCAGATTTTGGCCTGGACCAGCCGCTGAAT
GCACGTTATATGTCGGATGTTCTGAAAGTCGGTGTGTACCTGGAAAACGGTTGGGAACGCGGCG
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TCGCGTCCTGAAACAAAAGGCGGACGTGAGCCTGATGAAAGGCGGTTTCATCGTATGAAAGTCTG
GAATCCCTGGTTTCATACATCAGCTCTCTGTAA Amino Acid Cytosolic-targeted UDP
glycosyltransferase 76G1 (cytUTG) *Stevia rebaudiana* SEQ ID NO. 64

MENKTETTVRRRRRIILFPVPFQGHINPILQLANVLYSKGFSITIFHTNFNPKPKTSNYPHFTFR
FILNDPQDERISNLPTHGPLAGMRIPIINEHGADELRLRELELLMLASEEDEEVSLITDALWY
FAQSVADSLNLRRLVLMTSSLFNFHAHVSLPQFDELGYLDPDDKTRLEEQASGFPMMLKVVDIKS
AYSNWQILKEILGKMIKQTKASSGVIWNSFKELEESELETVIREIPAPSFLIPLPKHLTASSSS
LLDHDRTVFQWLDQQPPSSVLYVSFGSTSEVDEKDFLEIARGLVDSKQSFLWVVRPGFVKGSTW
VEPLPDGFLGERGRIVKWVPQEVLAHGAIGAFWTHSGWNSTLESVCEGVPMIFSDFGLDQPLN
ARYMSDVLKVG VYLENGWERGEIANAIRVMVDEEGEYIRQNARVLKQKADVSLMKGGSSYESL
ESLVSYISSL Enhanced N-terminal chimera secretion signal with Ost1 signal sequence
S. cerevisiae SEQ ID NO. 65

MRQVWFSWIVGLFLCFFNVSSAAPVNTTTEDETAQIPAEAVIGYS DLEGDFDVAVLPFNSTNNG
LLFINTTIIASIAAKEEGVSLEKR Enhanced Ost1 secretion signal presequence *S. cerevisiae*
SEQ ID NO. 66 MRQVWFSWIVGLFLCFFNVSSA Amino Acid Enhanced Exportable
Hepatic Fatty acid-binding protein 1 *Mus musculus* SEQ ID NO. 67
MRQVWFSWIVGLFLCFFNVSSAAPVNTTTEDETAQIPAEAVIGYS DLEGDFDVAVLPFSNSTNN
GLLFINTTIIASIAAKEEGVSLEKRMNFSGKYQLQSQENFEPFMKAIGLPEDLIQKGKDIKGVSE
IVHEGKKIKLTITYGPKVVRNEFTLGEECELETMTGEKVKAVVKLEGDNKMVTTTFKGIKSVTEL
NGDTITNTMTLGDIVYKRVSKRI Amino Acid Enhanced Exportable Hepatic Fatty acid-
binding protein 1 *Bos taurus* SEQ ID NO. 68

MRQVWFSWIVGLFLCFFNVSSAAPVNTTTEDETAQIPAEAVIGYS DLEGDFDVAVLPFSNSTNN
GLLFINTTIIASIAAKEEGVSLEKRMNFSGKYQVQTQENYEAFMKAVGMPDDIIQKGKDIKGVSE
IVQNGKHFKFIITAGSKVIQNEFTLGEECEMEFMTGEKIKAVVQQEGDNKLVTTFKGIKSVTEF
NGDTVSTMTKGDVVFVKRVSKRI Amino Acid Sec signal peptide for *E coli* L-
asparaginase II *E. Coli* SEQ ID NO. 69 MEFFKKTALAALVMGFSGAALA Amino Acid
Tat signal peptide for *E coli* strain k12 periplasmic nitrate reductase *E. Coli* SEQ
ID NO. 70 MKLSRRSFMKANAVAAAAAAGLSVPGVARAVVGQQ Amino Acid secretion
signal from an extracellular protease Ara12 (At5g67360) *Arabidopsis thaliana* SEQ ID
NO. 71 MSSSFLSSTAFFLLCLGFCHVSSS Amino Acid secretion signal from a alpha
amylase barley (*Hordeum vulgare*) SEQ ID NO. 72 MGKKSHICCFSLLLLLFAGLASG
Amino Acid secretion signal from a a-Amylase rice SEQ ID NO. 73
MKNTSSLCLLLLVLCSLTCNSGQAAQV Amino Acid Chicken FABP7 *Gallus gallus*
SEQ ID NO. 74

MVEAFCATWKLADSHNFDEYMKALGVGFAMRQVGNVTKPTVIISSEGDKV VIRTQSTFKNTEIS
FKLGEEFDETTTPDDRNCSSVVTLDGDKLVHVQKWDGKETNFVREIKDGRMVMTLTFGDVVAVRH
YEKA Amino Acid Pig FABP1 WT *Sus scrofa* SEQ ID NO. 75

MNFSGKYQVQSQENFEAFMKAVGLPDLIQKGKDIKGTSEIVQNGKHFKLTITTTGSKVVQNEFT
 LGEECEMETLTGEKVKTVVQLEGDNKLVTTFKGIKSVTELNGDIITSTMTLGDIVFKRISKRI
 Amino Acid Horse FABP1 WT *Equus caballus* SEQ ID NO. 76
 MNFSGKYQLQSQENFEPFMKAVGMSDDLQKKGKDLKETSEIVHNGNHFKITITTTGPKVVHHEFT
 LGEEFELESFTGEKVKAITHMEGDKLVSTIKGIKSVTELKGDIIITNQLSVWHTPVVCATSALVP
 ACRPGPVLDLGTMTLGDIVFKRVSKRI Amino Acid Plant FABP WT *Rhodamnia argentea*
 SEQ ID NO. 77
 MTEFAGVYEFVKDDGKFEEILKAMDVNFLIRKVAGKMNPNTAIDVLDDGKMAFKTITPLKTVEI
 HFELGKEYENKRLDGTVVKGIVTRDGNKLIQEQMSEPKFKVIRELDGPEKLIVTWMCKDIVCVR
 EYKRIQT Amino Acid Mouse FABP1 WT *Mus musculus* SEQ ID NO. 78
 MNFSGKYQLQSQENFEPFMKAIGLPEDLIQKGKDIKGVSEIVHEGKKIKLTITYGPKVVRNEFT
 LGEECELETMTGEKVKAVVKLEGDNKMVTTTFKGIKSVTELNGDTITNTMTLGDIVYKRVSKRI
 Amino Acid Modified Chicken FABP7 mdl_r5_0225_ro *Gallus gallus* SEQ ID NO. 79
 MVEAFCATWKLADSHNMDEALKA~~W~~GVGFAIRQVANVTKPTVIISSEGDKVVIRM~~Q~~STFKNTEIS
 FKLGEEDFET~~T~~ADDRNCKSVVTLDGDKLVHVQKWDGKETN~~V~~VREIKDGRMVMT~~V~~TFGDVVAVRH
 YEKA Amino Acid Modified Chicken FABP7 mdl_r3_1403_sm *Gallus gallus* SEQ ID
 NO. 80
 MVEAFCATWKLADSHNQDEGLKALGVGFAMRQL~~F~~ENV~~D~~KPTVIISSEGDKVVIRIE~~G~~TVKNTEIS
 FKLGEEDFET~~T~~ADDRNCKSVVTLDGDKLVHVQKWDGKETN~~V~~VREIKDGRMV~~A~~T~~F~~TFGDVVAV~~A~~H
 YEKA Amino Acid Modified Pig FABP1 mdl_r5_1652_ro *Sus scrofa* SEQ ID NO. 81
 MNFSGKYQVQSQENFEAFMKAVGQ~~P~~DELIQKGKDIKGTSEIVQNGKHFK~~V~~TITTTGSKVVQ~~A~~EFT
 LGEECE~~A~~ET~~D~~TGEKVK~~V~~VVQLEGDNK~~A~~V~~V~~TAKGIK~~A~~V~~F~~EELNGDIIT~~A~~T~~A~~T~~L~~GDIVFKRISKRI
 Amino Acid Modified Pig FABP1 mdl_r4_0372_sm *Sus scrofa* SEQ ID NO. 82
 MNFSGKYQVQSQENFEAA~~A~~KA~~A~~G~~V~~PDEEIQKGKDIKGTLEIVQNGKHFKLT~~L~~TTGSKVVQNEFT
 LGEECEMETL~~G~~GEKA~~A~~KTVVQLEGDNK~~A~~V~~V~~WTFKGIK~~A~~V~~I~~ELNGDIIT~~L~~T~~V~~TLGDIVFKRISKRI
 Amino Acid Modified Horse FABP1 mdl_r5_0954_ro *Equus caballus* SEQ ID NO. 83
 MNFSGKYQLQSQEN~~A~~EP~~F~~FKAVGMSDDLQ~~L~~AKDLK~~V~~TGEIVHNGNHFKIT~~H~~TSG~~A~~AVVHHEFT
 LGEEFELES~~L~~TGEKVKAITHMEGDKLVSTIKGIKSVTELKGDIIITNQLSVWHTPVVCATSALVP
 ACRPGPVLDLGTMTLGDIV~~M~~K~~F~~VSKRI Amino Acid Modified Horse FABP1
 mdl_r5_1179_sm *Equus caballus* SEQ ID NO. 84
 MNFSGKYQLQSSEN~~V~~EPFNKAVGMSDDLQKA~~K~~DLK~~F~~ETGEIVHNGNHFKITITTTGPKVVHHEFT
 LGEEFELE~~H~~FTGEKVKAITHMEGDKLV~~M~~T~~V~~KGIKSV~~M~~V~~W~~KGDIIITNQLSVWHTPVVCATSALVP
 ACRPGPVLDLGT~~G~~V~~L~~GDIV~~M~~K~~H~~VSKRI Amino Acid Modified Plant FABP
 mdl_r3_0887_ro_sm *Rhodamnia argentea* SEQ ID NO. 85
 MTEFAGVYEFVKDDGKFEEILKAMDVNFLIRKV~~G~~MKMNA~~N~~V~~A~~IDVLDDGKMAIK~~V~~ITPLKTVEI
 HFELGKEYENK~~R~~EDGTVVKGIVTRDGNKLI~~E~~EQMSEPKFKVIA~~E~~ELDGPEKLI~~F~~ET~~E~~MCKDIVCVA
 EYKRIQT Amino Acid Modified Plant FABP mdl_r4_1489_ro *Rhodamnia argentea* SEQ
 ID NO. 86
 MTEFAGVYEFVKDDGKFEEILKAMDVNFLIRKVAGKMNPNI~~A~~IDVLDDGKMAIK~~M~~ITPLKTVEY
 HFELGKEYE~~A~~K~~A~~LDGTVVKA~~I~~VTRDGNKLI~~I~~ENMSEPKFT~~V~~IRELDGPEKLIVTWMCKDIVCVA
 EYKRIQT

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of vacuolar targeting, 1-12. [8] Barrero, J. J., Casler, J. C., Valero, F., Ferrer, P., & Glick, B. S. (2018). An improved secretion signal enhances the secretion of model proteins from *Pichia pastoris*. *Microbial Cell Factories*, 1-13. [9] Chen M H, Huang L F, Li H M, Chen Y R, Yu S M. Signal peptide-dependent targeting of a rice alpha-amylase and cargo proteins to plastids and extracellular compartments of plant cells. *Plant Physiol.* 2004; 135(3): 1367-77 [10] Castilho, A., Windwarder, M., Gatteringer, P., Mach, L., Strasser, R., Altmann, F., & Steinkellner, H. (2014). Proteolytic and N-Glycan Processing of Human 1-Antitrypsin Expressed in *Nicotiana benthamiana*. *Plant Physiology*, 166(4), 1839-1851 [11] Huang, L. F., Tan, C. C., Yeh, J. F., Liu, H. Y., Liu, Y. K., Ho, S. L., & Lu, C. A. (2015). Efficient secretion of recombinant proteins from rice suspension-cultured cells modulated by the choice of signal peptide. *PLoS ONE*, 10(10), 1-16 [12] Yu, X. C. & Strobel, H. W. Interactions of 8-anilino-1-naphthalenesulfonic acid (ANS) and cytochrome P450 2B1: Role of ANS as an effector as well as a reporter group. *Mol. Cell. Biochem.* 162, 89-95 (1996). [13] Chen M H, Huang L F, Li H M, Chen Y R, Yu S M. Signal peptide-dependent targeting of a rice alpha-amylase and cargo proteins to plastids and extracellular compartments of plant cells. *Plant Physiol.* 2004; 135(3): 1367-77

Claims

1. A method of solubilizing a cannabinoid comprising the steps of: modifying a Fatty-Acid Binding Protein (FABP)-carrier protein having affinity towards a cannabinoid to produce a recombinant FABP-carrier protein having enhanced affinity towards said cannabinoid; introducing said recombinant FABP-carrier protein to said cannabinoid, wherein said recombinant FABP-carrier protein binds said cannabinoid forming a water-soluble protein-cannabinoid complex; and wherein the recombinant FABP-carrier protein comprises a recombinant FABP-carrier protein having the amino acid sequence selected from the group consisting of: SEQ ID NOs. 79-86, and wherein said cannabinoid is cannabidiol (CBD), or Δ^9 -tetrahydrocannabinol (THC).
2. The method of claim 1, wherein said step of genetically modifying the FABP-carrier protein comprises at least one of the following: replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket with side chains orientated toward the binding cavity; replacing one or more amino acid residues of the FABP-carrier protein cannabinoid binding pocket having a hydrophilic side chain with amino acid residues having a hydrophobic side chain; and replacing one or more small hydrophobic amino acid residues of the FABP-carrier protein cannabinoid binding pocket with larger hydrophobic amino acid residues.
3. The method of claim 1, wherein the recombinant FABP-carrier protein is expressed in a protein production system selected from the group consisting of: a bacterial cell culture; a yeast cell culture; a plant cell culture; a fungi cell culture; an algae cell culture; a bioreactor production system; and a plant.
4. The method of claim 1, wherein the recombinant FABP-carrier protein is coupled with a secretion signal.
5. The method of claim 4, wherein said secretion signal is selected from the group consisting of: SEQ ID NO. 3, SEQ ID NO. 65, and SEQ ID NOs. 69-73.
6. The method of claim 1, wherein said recombinant FABP-carrier protein is further genetically modified to decrease aggregation propensity and potential antigenicity.
7. The method of claim 6, wherein said recombinant FABP-carrier protein comprises a recombinant FABP-carrier protein having one or both of the following mutations: V58E, and V42R.
8. A method of solubilizing a cannabinoid comprising the steps of: expressing in a protein production system a heterologous nucleotide sequence, operably linked to a promoter, encoding generating a Fatty-Acid Binding Protein (FABP)-carrier protein having affinity towards a cannabinoid; and introducing said FABP-carrier protein to said cannabinoid, wherein said FABP-carrier protein binds said cannabinoid to form a water-soluble protein-cannabinoid complex, wherein the FABP-carrier protein comprises a FABP-carrier protein having the amino acid sequence selected from the group of consisting of: SEQ ID NOs. 79-86, or a sequence having at least 95% or more sequence identity with any of the SEQ ID NOs. 79-86, and wherein said cannabinoid is cannabidiol (CBD), or Δ^9 -tetrahydrocannabinol (THC).
9. The method of claim 8, wherein the recombinant FABP-carrier protein is expressed in a protein

production system selected from the group consisting of: a bacterial cell culture; a yeast cell culture; a plant cell culture; a fungi cell culture; an algae cell culture; a bioreactor production system; and a plant.

10. The method of claim 8, wherein said recombinant FABP-carrier protein comprises a recombinant FABP-carrier protein having one or both of the following mutations: V58E, and V42R.
